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Multi-material topology optimization of strongly coupled nonlinear thermoelasticity considering temperature-dependent material properties

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ABSTRACT

The design of high-performance structures capable of enduring severe thermoelastic environments is of paramount importance in aerospace and mechanical engineering. However, multi-material topology optimization (TO) in such strongly coupled scenarios remains challenging due to mesh distortion-induced numerical singularities under large deformations and the intricate treatment of temperature-dependent material properties (TDMPs). To address these issues, this work proposes a strongly coupled thermoelastic TO framework based on a novel unified nonlinear multi-material (UNM) interpolation scheme. In contrast to conventional approaches, the UNM method employs a kinematic scaling technique, which effectively mitigates numerical instabilities in low-stiffness regions and allows the use of the considered constitutive models, including modified Saint Venant–Kirchhoff, neo-Hookean, and Ogden hyperelastic materials. A strongly coupled thermoelastic formulation is developed based on the multiplicative decomposition of the deformation gradient, thereby incorporating TDMPs and the influence of large deformation on heat conduction paths. Fully coupled adjoint sensitivities for end-compliance and stored-strain–energy objectives are derived. In the numerical cases investigated, kinematic scaling is shown to improve convergence under large deformation. For the prescribed material laws and loading conditions, accounting for deformation-dependent heat conduction and TDMPs leads to differences in the computed responses and optimized layouts. The numerical examples also exhibit structural bifurcation, boundary detachment, and load-direction-dependent material distributions. An additional exponential-integral thermal-expansion example demonstrates the applicability of the proposed framework to an alternative prescribed constitutive law. Overall, the proposed framework provides a computational basis for investigating material-specific thermoelastic designs.

1. Introduction

Structural design for extreme thermal environments, such as those encountered in gas turbines and aerospace applications, must account for both large deformations and temperature-dependent material properties (TDMPs) [1,2]. Under such conditions, the thermal and mechanical material responses are strongly coupled, forming a bidirectional interaction [3,4]. Elevated temperatures

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Table 1
Comparison between the proposed framework and representative finite strain thermoelastic TOs.

Reference	Design	Thermal kinematics	Temperature field	TDMPs	Coupling	Interpolated quantity
Zhu et al. [48]	MMTO	Additive	Solved (steady)	–	One-way ($T \rightarrow \boldsymbol{u}$)	Parameter
Chung et al. [49]	SMTO	Multiplicative	Prescribed (uniform)	–	–	Strain energy
Goh et al. [39]	SMTO	Multiplicative	Prescribed (uniform)	–	–	Strain energy
Yuan et al. [50]	SMTO	External load	Prescribed (uniform)	–	–	Parameter
Granlund et al. [40]	MMTO	Multiplicative	Solved (transient)	–	Two-way ($T \leftrightarrow \boldsymbol{u}$)	Parameter
This study	**MMTO**	Multiplicative	Solved (steady)	✓	Two-way ($T \leftrightarrow \boldsymbol{u}$)	**Strain**

induce thermal expansion and alter the stiffness of the constituent materials, which directly affects the stress distribution. Simultaneously, large structural deformations change the thermal pathways for heat conduction [5]. Accurate performance prediction in these environments requires capturing these coupled nonlinear effects [6].

Topology optimization (TO) is a computational framework for designing high-performance structures [7–9]. Most existing TO frameworks are based on simplified physical assumptions, primarily relying on single-material topology optimization (SMTO), linear elasticity, and constant material properties (CMPs) [10–13]. While these assumptions simplify numerical implementation and reduce computational cost, they limit the applicability of the optimized designs in practice. As a result, the conventional approaches relying on simplified physical assumptions often lead to performance degradation when structures are subjected to extreme thermal conditions.

From a thermal perspective, multi-material topology optimization (MMTO) offers the flexibility to address various design objectives, such as negative thermal expansion, thermal buckling, and heat dissipation [14–16]. Furthermore, incorporating TDMPs provides a more realistic description of material behavior than assuming constant properties. Since material parameters such as thermal conductivity, thermal expansion coefficient, and elastic properties vary with temperature, TDMPs enable the structural response under operating conditions to be captured more accurately [17–19]. In fact, Sun et al. [20] recently developed an MMTO framework incorporating TDMPs, enabling autonomous material switching based on local temperature fields for addressing both thermal compliance and peak temperature problems.

From a mechanical perspective, the use of multi-material structures can improve structural stiffness [21–25]. Recently, these MMTO approaches have been extended to the design of nonlinear multi-material structures undergoing large deformations [26–28], as well as the optimization of trusslike architectures under thermoelastic environments [29]. However, performing MMTO under a finite strain assumption for various types of materials introduces theoretical challenges; different constituent materials are often governed by distinct constitutive laws [30–32], such as different hyperelastic models with their own strain energy density functions. Thereby, traditional material interpolations [33–35], which typically interpolate material parameters (e.g., Lamé constants), become insufficient and may lead to physical inconsistency; refer to Han et al. [36] for an elastoplastic example. Hence, a unified nonlinear multi-material interpolation framework is required to achieve consistent MMTO under finite strain conditions, similar to the extended discrete material optimization (XDMO) [37] that allows distinct constitutive laws to be considered in one optimization problem.

For MMTO targeting extreme thermal environments, it is necessary to account for both thermal and mechanical effects. Integrating MMTO, TDMPs, and finite strain kinematics leads to a strongly coupled nonlinear thermoelastic system. Despite its engineering relevance, MMTO under such conditions has not been sufficiently studied in the literature [38–40]. Table 1 lists representative previous studies at finite strain, including both SMTO and MMTO approaches. Among these studies, Granlund et al. [40] considered the effect of structural deformation on heat-conduction paths in compliant-mechanism design and found only minor differences between weakly and strongly coupled formulations. However, their study assumed temperature-independent material properties. When TDMPs are included under finite strains, deformation-induced changes in the temperature field also modify the material properties and, consequently, the mechanical response. A bottleneck to this integration is the numerical instability encountered during optimization. Extreme thermo-mechanical loads and significant order differences in material parameters can cause excessive mesh distortion in low- and intermediate-density regions in finite element analyses. Standard numerical treatments, such as element removal [41], additive hyperelasticity [42–44], and energy interpolation [45–47], often fail to maintain stability in multi-material settings, as observed in our numerical experiments.

Motivated by these challenges, this study establishes a multi-material topology optimization (MMTO) framework for finite-strain thermoelasticity with TDMPs. Although the underlying thermoelastic formulations are well established, their integration into topology optimization requires a fully coupled treatment of strong thermo-mechanical interactions and poses severe computational challenges. To address these issues, a finite-strain thermoelastic MMTO framework with a unified nonlinear multi-material (UNM) interpolation scheme is proposed. The main contributions of this work are twofold: (i) a kinematically scaled interpolation scheme applied directly to the strain, rather than to material parameters, which improves the stability of intermediate-density elements in the investigated large-deformation cases while allowing distinct hyperelastic models within a single optimization problem; and (ii) an MMTO framework for finite-strain thermoelasticity with TDMPs, including adjoint sensitivity analyses for different objective functions. Combined with the discrete material optimization (DMO) [37,51] and material field series expansion (MFSE) [52] methods, this framework is evaluated using prescribed constitutive laws and loading conditions.

The remainder of this work is organized as follows. Section 2 presents the finite element formulation for strongly coupled thermoelasticity, including finite strain kinematics and TDMPs. Section 3 introduces the kinematically scaled material interpolation

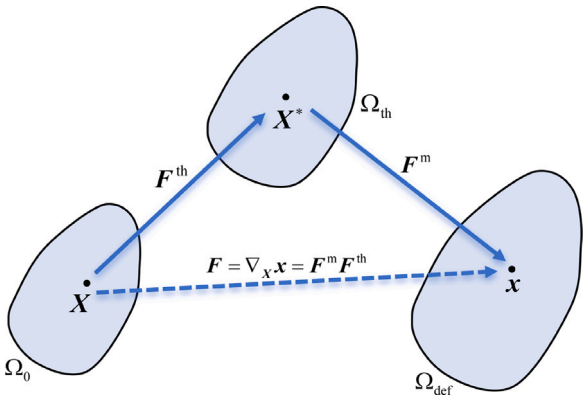


Fig. 1. Schematic illustration of the kinematics of nonlinear thermoelasticity.

scheme for large-deformation problems, followed by the proposed UNM interpolation method. The integration of the UNM method with the DMO-based MFSE framework for multi-material representation is also described. Several numerical examples are presented in Section 4 to demonstrate the effects of large deformation and multi-physics coupling on the optimized results. Finally, Section 5 concludes this work and outlines directions for future research.

2. Finite strain nonlinear thermoelasticity

This section presents the finite strain formulation adopted in this work. First, the kinematics based on the multiplicative decomposition rule [53] are introduced, and then the governing equations, finite element discretization, and constitutive models are described.

2.1. Kinematics

The deformation state of a thermoelastic structure under the finite strain framework can be decomposed into elastic (or mechanical) and thermal parts based on the multiplicative decomposition rule [53], as shown in Fig. 1. Suppose the reference and current configurations are denoted as $\Omega_0(\boldsymbol{X})$ and $\Omega_{\text{def}}(\boldsymbol{x})$, respectively. The intermediate configuration caused by thermal stress can be introduced as $\Omega_{\text{th}}(\boldsymbol{X}^*)$. Consequently, the deformation gradient tensor $\boldsymbol{F}$, which maps a material point in the reference configuration $\boldsymbol{X} \in \Omega_0$ to the position in the current configuration $\boldsymbol{x} \in \Omega_{\text{def}}$, can be decoupled as

$$\boldsymbol{F} = \nabla_{\boldsymbol{X}} \boldsymbol{x} = \boldsymbol{F}^{\text{m}} \boldsymbol{F}^{\text{th}}, \tag{1}$$

where $\boldsymbol{F}^{\text{m}}$ and $\boldsymbol{F}^{\text{th}}$ denote the elastic and thermal parts, respectively. Correspondingly, the elastic deformation gradient tensor is defined as $\boldsymbol{F}^{\text{m}} = \nabla_{\boldsymbol{X}^*} \boldsymbol{x}$, which maps material point positions from $\Omega_{\text{th}}(\boldsymbol{X}^*)$ to $\Omega_{\text{def}}(\boldsymbol{x})$, and the thermal deformation gradient tensor is defined as $\boldsymbol{F}^{\text{th}} = \nabla_{\boldsymbol{X}} \boldsymbol{X}^*$, which maps material point positions between $\Omega_0(\boldsymbol{X})$ and $\Omega_{\text{th}}(\boldsymbol{X}^*)$, respectively. The total displacement vector yields $\boldsymbol{u} = \boldsymbol{x} - \boldsymbol{X}$, and the subscript of the gradient operator (e.g., $\nabla_{\boldsymbol{X}}$) specifies the configuration with respect to which the gradient is taken.

Building upon the kinematic description in Eq. (1), the elastic Green–Lagrange strain $\boldsymbol{E}^{\text{m}}$ can be derived as

$$\boldsymbol{E}^{\text{m}} = \frac{1}{2}\left(\left(\boldsymbol{F}^{\text{m}}\right)^{\text{T}} \boldsymbol{F}^{\text{m}} - \boldsymbol{I}\right) = \frac{1}{2}\left(\left(\boldsymbol{F}^{\text{th}}\right)^{-\text{T}} \left(\boldsymbol{F}^{\text{T}} \boldsymbol{F}\right) \left(\boldsymbol{F}^{\text{th}}\right)^{-1} - \boldsymbol{I}\right), \tag{2}$$

where $\boldsymbol{I}$ represents the second-order identity tensor. Here, the thermal deformation gradient $\boldsymbol{F}^{\text{th}}$ is assumed to be isotropic and dependent on the temperature T and temperature-dependent expansion coefficient $\alpha_{\text{th}}(T)$, which can be written as

$$\boldsymbol{F}^{\text{th}} = \left(1 + \alpha_{\text{th}}(T)(T - T_{\text{ref}})\right) \boldsymbol{I} = \vartheta \boldsymbol{I}. \tag{3}$$

Here, $\alpha_{\text{th}}(T)$ denotes the conventional secant coefficient of thermal expansion, defined with respect to the engineering strain measured from the reference temperature T_{ref}. Thus, the engineering thermal strain and the corresponding thermal stretch are given by $\varepsilon_{\text{th}} = \alpha_{\text{th}}(T)(T - T_{\text{ref}})$ and $\vartheta = 1 + \varepsilon_{\text{th}} = 1 + \alpha_{\text{th}}(T)(T - T_{\text{ref}})$. This definition directly yields Eq. (3). This linear form is commonly used in engineering thermal analysis [54] and has also been adopted in previous finite strain thermoelastic topology optimization studies [39,49].

Alternatively, the instantaneous logarithmic coefficient is defined as $\alpha_{\text{th}}^{\text{ins}} = (1/L)\mathrm{d}_T L$, where L denotes the current length of the material at temperature T. This definition leads to the exponential thermal stretch $\vartheta = \exp\left(\int_{T_{\text{ref}}}^{T} \alpha_{\text{th}}^{\text{ins}}\, \mathrm{d}\tau\right)$. The secant and instantaneous coefficients are related by $\alpha_{\text{th}} = \left[\exp\left(\int_{T_{\text{ref}}}^{T} \alpha_{\text{th}}^{\text{ins}}\, \mathrm{d}\tau\right) - 1\right] / \Delta T$. Thus, the two expressions for the thermal stretch correspond to different definitions of the thermal expansion coefficient, and the present work adopts the secant definition. Because

each expression for the thermal stretch is derived from a distinct definition of α_{th}, they do not represent approximations of one another. Rather, adopting a specific form defines the thermal expansion behavior of the material. The proposed framework allows the exponential form to be adopted without any modification to the methodology, as demonstrated in Section 4.1.3.

Substituting Eq. (3) into Eq. (2) yields the following expression:

$$\boldsymbol{E}^{\text{m}} = \frac{1}{2}\left((\vartheta \boldsymbol{I})^{-\text{T}}\left(\boldsymbol{F}^{\text{T}}\boldsymbol{F}\right)(\vartheta \boldsymbol{I})^{-1} - \boldsymbol{I}\right) = \frac{1}{2}\left(\vartheta^{-2}\boldsymbol{C} - \boldsymbol{I}\right), \tag{4}$$

where $\boldsymbol{C} = \boldsymbol{F}^{\text{T}}\boldsymbol{F}$ is the right Cauchy–Green tensor. Hence, the following relationship holds:

$$\boldsymbol{C}^{\text{m}} = \left(\boldsymbol{F}^{\text{m}}\right)^{\text{T}}\boldsymbol{F}^{\text{m}} = \vartheta^{-2}\boldsymbol{C}, \tag{5}$$

in which $\boldsymbol{C}^{\text{m}}$ denotes the elastic right Cauchy–Green tensor.

2.2. Governing equations

This work considers steady-state thermoelasticity, and "strong coupling" refers to the two-way interaction between the thermal and mechanical fields. Specifically, structural deformation modifies the heat conduction path through deformation-dependent thermal conductivity, while the resulting temperature field governs the mechanical response through thermal expansion and TDMPs. The thermoelastic constitutive interaction considered here is assumed to be reversible, with no mechanically induced dissipation included. In contrast, steady-state Fourier heat conduction remains an irreversible, entropy-producing process in the presence of a temperature gradient. More general formulations involving mechanically induced heat generation are beyond the scope of this study.

2.2.1. Thermal field

In this work, the thermal conductivity is assumed to be isotropic and temperature-dependent. Since nonlinear heat conduction occurs in the current configuration $\Omega_{\text{def}}(\boldsymbol{x})$, the governing equations are written as

$$\begin{cases} \nabla_{\boldsymbol{x}} \cdot \boldsymbol{q} = Q & \text{in } \Omega_{\text{def}}, \\ T = \bar{T} & \text{on } \partial\Omega_{\text{def}}^{DT}, \\ \boldsymbol{q} \cdot \boldsymbol{n}_{\text{def}} = \bar{q} & \text{on } \partial\Omega_{\text{def}}^{NT}, \end{cases} \tag{6}$$

where $\boldsymbol{q} := -k(T)\nabla_{\boldsymbol{x}} T$ denotes the heat flux vector with $k(T)$ being the temperature-dependent thermal conductivity, Q is the volumetric heat source, $\bar{T}$ is the prescribed temperature on the current Dirichlet boundary $\partial\Omega_{\text{def}}^{DT}$, $\bar{q}$ is the prescribed normal heat flux on the current Neumann boundary $\partial\Omega_{\text{def}}^{NT}$, and $\boldsymbol{n}_{\text{def}}$ represents the outward unit normal vector to the current boundary $\partial\Omega_{\text{def}}$, respectively.

Eq. (6) is pulled back to the reference configuration $\Omega_0(\boldsymbol{X})$ as follows:

$$\begin{cases} \nabla_{\boldsymbol{X}} \cdot \boldsymbol{q}_0 = Q_0 & \text{in } \Omega_0, \\ T = \bar{T} & \text{on } \partial\Omega_0^{DT}, \\ \boldsymbol{q}_0 \cdot \boldsymbol{N}_0 = \bar{q}_0 & \text{on } \partial\Omega_0^{NT}, \end{cases} \tag{7}$$

where $\boldsymbol{q}_0 := -k(T)J\boldsymbol{C}^{-1}\nabla_{\boldsymbol{X}} T$ denotes the reference heat flux vector, $J = \det(\boldsymbol{F}) = \sqrt{\det(\boldsymbol{C})}$ is the volume ratio (or determinant), $Q_0 := JQ$ denotes the reference volumetric heat source, $\bar{q}_0$ denotes the prescribed normal heat flux on the reference Neumann boundary, and $\boldsymbol{N}_0$ is the outward unit normal vector to the reference boundary $\partial\Omega_0$. Notably, by defining

$$\boldsymbol{k}_{\text{def}}(T, \boldsymbol{F}) := k(T)J\boldsymbol{C}^{-1} \quad \text{with} \quad \boldsymbol{q}_0 := -\boldsymbol{k}_{\text{def}}\nabla_{\boldsymbol{X}} T, \tag{8}$$

the nonlinear heat conduction can be interpreted as being governed by an effective thermal conductivity tensor $\boldsymbol{k}_{\text{def}}$ that depends on both the temperature and the deformation. In fact, under the finite strain framework, structural deformation modifies the heat conduction paths and alters the effective area through which the thermal flux passes. Specifically, the inverse right Cauchy–Green tensor $\boldsymbol{C}^{-1}$ captures the directional modification of the conduction paths, while the determinant J accounts for volumetric changes. Although Eq. (8) follows the standard pull-back operation, its incorporation into TO departs from the conventional weakly coupled treatment. As detailed in Appendices A–C, this integration requires a monolithic solution scheme and a fully coupled adjoint sensitivity formulation.

Remark 1. $\bar{q}_0$ can be derived from $\bar{q}$ using Nanson's formula; however, in most cases of numerical simulations, $\bar{q}_0$ is simply treated as an input parameter.

2.2.2. Mechanical field

The elastic strain energy density function of a hyperelastic material is independent of strain rate and deformation history. Based on the kinematics in Section 2.1 and TDMPs assumption, the strain energy density function yields

$$\phi = \phi(\boldsymbol{E}^{\text{m}}) = \phi(\boldsymbol{F}, T). \tag{9}$$

Using Eqs. (4) and (9), the first Piola–Kirchhoff stress tensor (PK1) is derived as

$$\boldsymbol{P} = \frac{\partial \phi(\boldsymbol{E}^{\mathrm{m}})}{\partial \boldsymbol{E}^{\mathrm{m}}} : \frac{\partial \boldsymbol{E}^{\mathrm{m}}}{\partial \boldsymbol{C}} : \frac{\partial \boldsymbol{C}}{\partial \boldsymbol{F}} = \boldsymbol{S}^{\mathrm{m}} : \frac{1}{2}\vartheta^{-2}\mathbb{I}_{\mathrm{sym}} : \left(\boldsymbol{F}^{\mathrm{T}} \overline{\otimes} \boldsymbol{I} + \boldsymbol{I} \underline{\otimes} \boldsymbol{F}^{\mathrm{T}}\right) = \boldsymbol{F}\left(\vartheta^{-2}\boldsymbol{S}^{\mathrm{m}}\right) = \boldsymbol{F}\boldsymbol{S}, \tag{10}$$

where $\boldsymbol{S}$ and $\boldsymbol{S}^{\mathrm{m}}$ denote the second Piola–Kirchhoff stress tensors (PK2 and PK2i) with respect to the reference and intermediate configurations, $\mathbb{I}_{\mathrm{sym}}$ is the fourth-order symmetric identity tensor, and $\overline{\otimes}$ and $\underline{\otimes}$ are the dyadic-up and dyadic-down operators (i.e., $(\boldsymbol{I} \overline{\otimes} \boldsymbol{I})_{ijkl} = \delta_{ik}\delta_{jl}$ and $(\boldsymbol{I} \underline{\otimes} \boldsymbol{I})_{ijkl} = \delta_{il}\delta_{jk}$), respectively. Recalling Eq. (3), it turns out that the following relationship holds:

$$\boldsymbol{S} = \vartheta^{-2}\boldsymbol{S}^{\mathrm{m}} = \left(\vartheta^{-1}\boldsymbol{I}\right)\boldsymbol{S}^{\mathrm{m}}\left(\vartheta^{-1}\boldsymbol{I}\right) = \left(\boldsymbol{F}^{\mathrm{th}}\right)^{-1}\boldsymbol{S}^{\mathrm{m}}\left(\boldsymbol{F}^{\mathrm{th}}\right)^{-\mathrm{T}}, \tag{11}$$

which indicates that $\boldsymbol{S}$ is obtained by pulling back $\boldsymbol{S}^{\mathrm{m}}$ from the intermediate configuration via $\boldsymbol{F}^{\mathrm{th}}$.

Following the standard equilibrium formulation under the finite strain framework, the governing equations of the mechanical field can be written with respect to the reference configuration $\Omega_0(\boldsymbol{X})$ as follows:

$$\begin{cases} \nabla_{\boldsymbol{X}} \cdot \boldsymbol{P} + \boldsymbol{b} = \boldsymbol{0} & \text{in } \Omega_0, \\ \boldsymbol{u} = \bar{\boldsymbol{u}} & \text{on } \partial\Omega_0^{Du}, \\ \boldsymbol{P} \cdot \boldsymbol{N}_0 = \bar{\boldsymbol{t}} & \text{on } \partial\Omega_0^{Nu}, \end{cases} \tag{12}$$

where $\boldsymbol{b}$ is the body force vector, $\bar{\boldsymbol{u}}$ denotes the prescribed displacement on the Dirichlet boundary $\partial\Omega_0^{Du}$, and $\bar{\boldsymbol{t}}$ is the prescribed traction on the Neumann boundary $\partial\Omega_0^{Nu}$, respectively. Notably, the geometric dependence of the effective thermal conductivity $\boldsymbol{k}_{\mathrm{def}}$ on $\boldsymbol{F}$, together with the temperature dependence of the PK1 stress $\boldsymbol{P}$, results in a fully coupled nonlinear system in which Eqs. (7) and (12) must be solved simultaneously. Also, it should be noted that the TDMPs are considered in this study; in addition to the temperature-dependent thermal conductivity k in Eq. (8), all mechanical parameters, including the thermal expansion coefficient α_{th}, are also temperature-dependent. Detailed parameters will be discussed in Section 2.4.

2.3. Finite element discretization

To solve Eqs. (7) and (12) in a strongly coupled manner, the following weak forms are considered:

$$\begin{aligned} \mathcal{R}^{\mathrm{m}}(\boldsymbol{u}, T; \delta\boldsymbol{u}) &= \int_{\Omega_0} \boldsymbol{S} : \delta\boldsymbol{E}\,\mathrm{d}V - \int_{\Omega_0} \boldsymbol{b}\cdot\delta\boldsymbol{u}\,\mathrm{d}V - \int_{\partial\Omega_0^{Nu}} \bar{\boldsymbol{t}}\cdot\delta\boldsymbol{u}\,\mathrm{d}A = 0, \\ \mathcal{R}^{\mathrm{th}}(\boldsymbol{u}, T; \delta T) &= -\int_{\Omega_0} \boldsymbol{q}_0\cdot\nabla_{\boldsymbol{X}}\delta T\,\mathrm{d}V - \int_{\Omega_0} Q_0\delta T\,\mathrm{d}V + \int_{\partial\Omega_0^{NT}} \bar{q}_0\delta T\,\mathrm{d}A = 0, \end{aligned} \tag{13}$$

where $\delta\boldsymbol{u}$ and δT are the test functions. Here, the relationship $\boldsymbol{P} : \delta\boldsymbol{F} = \boldsymbol{S} : \delta\boldsymbol{E}$ has been adopted, and $\delta\boldsymbol{E}$ is expressed by

$$\delta\boldsymbol{E} = \delta\left(\frac{1}{2}\left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} - \boldsymbol{I}\right)\right) = \frac{1}{2}\left((\nabla_{\boldsymbol{X}}\delta\boldsymbol{u})^{\mathrm{T}}\boldsymbol{F} + \boldsymbol{F}^{\mathrm{T}}\nabla_{\boldsymbol{X}}\delta\boldsymbol{u}\right) = (\boldsymbol{F}^{\mathrm{T}}\nabla_{\boldsymbol{X}}\delta\boldsymbol{u})_{\mathrm{sym}}. \tag{14}$$

Following the standard Galerkin finite element procedure, the continuous domain Ω_0 is discretized into n_{e} finite elements to solve Eq. (13). Within one finite element, the material coordinate $\boldsymbol{X}$, displacement vector $\boldsymbol{u}$, temperature T, and their variations $\delta\boldsymbol{u}$ and δT are approximated using the shape functions N^I as follows:

$$\boldsymbol{X} \approx \sum_{I=1}^{n_{\mathrm{d}}} N^I\boldsymbol{X}^I, \quad \boldsymbol{u} \approx \sum_{I=1}^{n_{\mathrm{d}}} N^I\boldsymbol{u}^I, \quad T \approx \sum_{I=1}^{n_{\mathrm{d}}} N^I T^I, \quad \delta\boldsymbol{u} \approx \sum_{I=1}^{n_{\mathrm{d}}} N^I\delta\boldsymbol{u}^I, \quad \delta T \approx \sum_{I=1}^{n_{\mathrm{d}}} N^I\delta T^I, \tag{15}$$

in which n_{d} denotes the number of nodes per element, and $\boldsymbol{X}^I$, $\boldsymbol{u}^I$, T^I, $\delta\boldsymbol{u}^I$, and δT^I represent the corresponding nodal quantities, respectively. The substitution of Eq. (15) into Eq. (13) leads to the following global residual vectors as

$$\begin{aligned} \boldsymbol{\mathcal{R}}^{\mathrm{m}}(\boldsymbol{U}, \boldsymbol{T}) &= \bigcup_{e=1}^{n_{\mathrm{e}}}\left[\int_{\Omega_0^e} (\boldsymbol{B}^{\mathrm{m}})^{\mathrm{T}}\boldsymbol{S}\,\mathrm{d}V - \int_{\Omega_0^e} \left(\boldsymbol{N}^{\mathrm{m}}\right)^{\mathrm{T}}\boldsymbol{b}\,\mathrm{d}V - \int_{\partial\Omega_0^{Nu,e}} \left(\boldsymbol{N}^{\mathrm{m}}\right)^{\mathrm{T}}\bar{\boldsymbol{t}}\,\mathrm{d}A\right], \\ \boldsymbol{\mathcal{R}}^{\mathrm{th}}(\boldsymbol{U}, \boldsymbol{T}) &= \bigcup_{e=1}^{n_{\mathrm{e}}}\left[-\int_{\Omega_0^e} (\boldsymbol{B}^{\mathrm{th}})^{\mathrm{T}}\boldsymbol{q}_0\,\mathrm{d}V - \int_{\Omega_0^e} \left(\boldsymbol{N}^{\mathrm{th}}\right)^{\mathrm{T}}Q_0\,\mathrm{d}V + \int_{\partial\Omega_0^{NT,e}} \left(\boldsymbol{N}^{\mathrm{th}}\right)^{\mathrm{T}}\bar{q}_0\,\mathrm{d}A\right], \end{aligned} \tag{16}$$

where $\bigcup$ is the finite element assembly operator, $\boldsymbol{U}$ and $\boldsymbol{T}$ denote the global displacement and temperature vectors, $\boldsymbol{B}^{\mathrm{m}}$ and $\boldsymbol{B}^{\mathrm{th}}$ are the displacement and thermal gradient matrices, and $\boldsymbol{N}^{\mathrm{m}}$ and $\boldsymbol{N}^{\mathrm{th}}$ represent the displacement and thermal shape function matrices, respectively. Note that $\boldsymbol{B}^{\mathrm{th}} = \nabla_{\boldsymbol{X}}\boldsymbol{N}^{\mathrm{th}}$ is constant, whereas $\boldsymbol{B}^{\mathrm{m}} = (\boldsymbol{F}^{\mathrm{T}}\nabla_{\boldsymbol{X}}\boldsymbol{N}^{\mathrm{m}})_{\mathrm{sym}}$ is deformation-dependent. For clarity in the subsequent numerical implementation parts, the residual vectors can be partitioned into internal and external components, i.e., $\boldsymbol{\mathcal{R}}^{\mathrm{m}} = \boldsymbol{F}_{\mathrm{int}} - \boldsymbol{F}_{\mathrm{ext}}$ and $\boldsymbol{\mathcal{R}}^{\mathrm{th}} = \boldsymbol{Q}_{\mathrm{int}} - \boldsymbol{Q}_{\mathrm{ext}}$. Specifically, the global external mechanical loading vector $\boldsymbol{F}_{\mathrm{ext}}$ and the global external thermal loading vector $\boldsymbol{Q}_{\mathrm{ext}}$ are assembled as

$$\begin{aligned} \boldsymbol{F}_{\mathrm{ext}} &= \bigcup_{e=1}^{n_{\mathrm{e}}}\left[\int_{\Omega_0^e} \left(\boldsymbol{N}^{\mathrm{m}}\right)^{\mathrm{T}}\boldsymbol{b}\,\mathrm{d}V + \int_{\partial\Omega_0^{Nu,e}} \left(\boldsymbol{N}^{\mathrm{m}}\right)^{\mathrm{T}}\bar{\boldsymbol{t}}\,\mathrm{d}A\right], \\ \boldsymbol{Q}_{\mathrm{ext}} &= \bigcup_{e=1}^{n_{\mathrm{e}}}\left[\int_{\Omega_0^e} \left(\boldsymbol{N}^{\mathrm{th}}\right)^{\mathrm{T}}Q_0\,\mathrm{d}V - \int_{\partial\Omega_0^{NT,e}} \left(\boldsymbol{N}^{\mathrm{th}}\right)^{\mathrm{T}}\bar{q}_0\,\mathrm{d}A\right]. \end{aligned} \tag{17}$$

Solving Eq. (16) via the Newton–Raphson iterative procedure requires consistent linearization. This is achieved by taking the Gâteaux derivative of Eq. (13) for $(\boldsymbol{u}, T)$ in the direction of their increments $(\Delta\boldsymbol{u}, \Delta T)$, yielding the linearized system of equations:

$$\begin{cases} \mathcal{R}^{\mathrm{m}} + D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{m}} + D_{\Delta T}\mathcal{R}^{\mathrm{m}} = 0, \\ \mathcal{R}^{\mathrm{th}} + D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{th}} + D_{\Delta T}\mathcal{R}^{\mathrm{th}} = 0. \end{cases} \tag{18}$$

Specifically, the directional derivatives associated with the mechanical residual yield

$$\begin{aligned} D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{m}} &= \int_{\Omega_0} \Big(\delta\boldsymbol{E} : \underbrace{\mathbb{C}_{\tan}}_{\text{Eq. (A.1)}} : \Delta\boldsymbol{E} + \boldsymbol{S} : \frac{\partial\delta\boldsymbol{E}}{\partial\nabla_{\boldsymbol{X}}\boldsymbol{u}} : \nabla_{\boldsymbol{X}}\Delta\boldsymbol{u}\Big)\,\mathrm{d}V, \\ D_{\Delta T}\mathcal{R}^{\mathrm{m}} &= \int_{\Omega_0} \delta\boldsymbol{E} : \underbrace{\frac{\partial\boldsymbol{S}}{\partial T}}_{\text{Eq. (A.2)}}\Delta T\,\mathrm{d}V, \end{aligned} \tag{19}$$

where $\Delta\boldsymbol{E} = (\boldsymbol{F}^{\mathrm{T}}\nabla_{\boldsymbol{X}}\Delta\boldsymbol{u})_{\mathrm{sym}}$ denotes the incremental Green–Lagrange strain tensor, $\mathbb{C}_{\tan}$ is the fourth-order material tangent tensor, and $\boldsymbol{S} : \big((\nabla_{\boldsymbol{X}}\delta\boldsymbol{u})^{\mathrm{T}}\nabla_{\boldsymbol{X}}\Delta\boldsymbol{u}\big)$ accounts for the initial stress stiffness (or geometric stiffness), respectively. Also, the directional derivatives of the thermal residual yield

$$\begin{aligned} D_{\Delta T}\mathcal{R}^{\mathrm{th}} &= -\int_{\Omega_0} \nabla_{\boldsymbol{X}}\delta T \cdot \Bigg(\underbrace{\frac{\partial\boldsymbol{q}_0}{\partial T}}_{\text{Eq. (A.3)}_1}\Delta T + \underbrace{\frac{\partial\boldsymbol{q}_0}{\partial\nabla_{\boldsymbol{X}}T}}_{\text{Eq. (A.3)}_2}\cdot\nabla_{\boldsymbol{X}}\Delta T\Bigg)\mathrm{d}V, \\ D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{th}} &= -\int_{\Omega_0} \nabla_{\boldsymbol{X}}\delta T \cdot \underbrace{\frac{\partial\boldsymbol{q}_0}{\partial\boldsymbol{F}}}_{\text{Eq. (A.4)}} : \nabla_{\boldsymbol{X}}\Delta\boldsymbol{u}\,\mathrm{d}V. \end{aligned} \tag{20}$$

Since both terms $D_{\Delta T}\mathcal{R}^{\mathrm{m}}$ and $D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{th}}$ are non-zero, Eq. (16) should be solved in a strongly coupled manner. That is, the spatial discretization maps these analytical directional derivatives to the global monolithic tangent system as

$$\underbrace{\begin{bmatrix} \boldsymbol{\mathcal{K}}_{uu} & \boldsymbol{\mathcal{K}}_{uT} \\ \boldsymbol{\mathcal{K}}_{Tu} & \boldsymbol{\mathcal{K}}_{TT} \end{bmatrix}}_{\boldsymbol{\mathcal{K}}_{\tan}} \begin{bmatrix} \Delta\boldsymbol{U} \\ \Delta\boldsymbol{T} \end{bmatrix} = -\begin{bmatrix} \boldsymbol{\mathcal{R}}^{\mathrm{m}} \\ \boldsymbol{\mathcal{R}}^{\mathrm{th}} \end{bmatrix}, \tag{21}$$

in which $\Delta\boldsymbol{U}$ and $\Delta\boldsymbol{T}$ denote the global displacement and temperature increment vectors. Here, the diagonal sub-matrices $\boldsymbol{\mathcal{K}}_{uu}$ and $\boldsymbol{\mathcal{K}}_{TT}$ (derived from $D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{m}}$ and $D_{\Delta T}\mathcal{R}^{\mathrm{th}}$) represent the mechanical tangent stiffness and thermal tangent conductivity. The off-diagonal coupling matrices $\boldsymbol{\mathcal{K}}_{uT}$ and $\boldsymbol{\mathcal{K}}_{Tu}$ (derived from $D_{\Delta T}\mathcal{R}^{\mathrm{m}}$ and $D_{\Delta\boldsymbol{u}}\mathcal{R}^{\mathrm{th}}$) account for the thermally induced internal force variation and the geometrically modified heat conduction path, respectively. The derivation result for these sub-matrices is provided in Appendix A. Furthermore, to verify the theoretical consistency and implementation correctness of these formulations, numerical verification against analytical solutions is presented in Appendix B.1. Also, a comparison of different nonlinear thermoelasticity computational methods in terms of efficiency and correctness is provided in Appendix B.2.

2.4. Constitutive models

2.4.1. Modified Saint Venant–Kirchhoff model

The elastic strain energy density function for the modified Saint Venant–Kirchhoff (SVK) model [53] can be expressed as

$$\phi = \mu\boldsymbol{E}^{\mathrm{m}} : \boldsymbol{E}^{\mathrm{m}} + \frac{\lambda}{2}(J^{\mathrm{m}} - 1)^2, \tag{22}$$

where $J^{\mathrm{m}} := \det(\boldsymbol{F}^{\mathrm{m}}) = \sqrt{\det(2\boldsymbol{E}^{\mathrm{m}} + \boldsymbol{I})}$ denotes the elastic volume ratio, and μ and λ are the temperature-dependent Lamé parameters. Recalling Eq. (10), the derivative of Eq. (22) with respect to $\boldsymbol{E}^{\mathrm{m}}$ leads to the following intermediate PK2 stress [55]:

$$\boldsymbol{S}^{\mathrm{m}} = \frac{\partial\phi}{\partial\boldsymbol{E}^{\mathrm{m}}} = 2\mu\boldsymbol{E}^{\mathrm{m}} + \lambda J^{\mathrm{m}}(J^{\mathrm{m}} - 1)(\boldsymbol{C}^{\mathrm{m}})^{-1}. \tag{23}$$

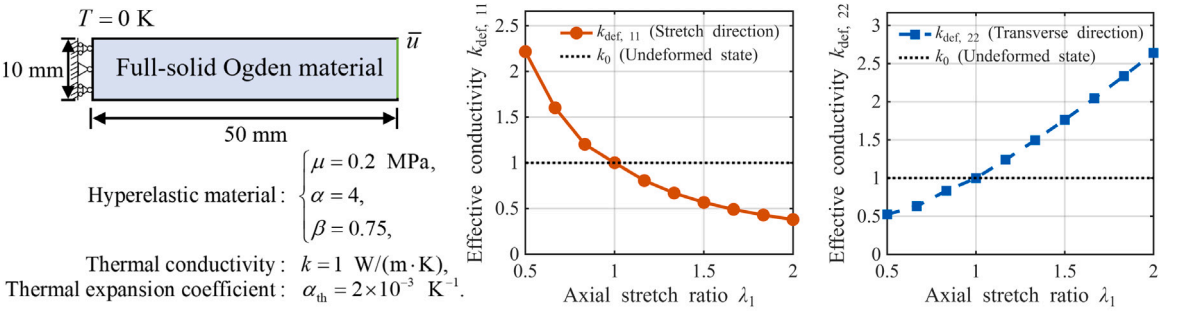
Also, from Eq. (19) (or Eq. (A.1)), the intermediate material tangent tensor is derived as follows:

$$\begin{aligned} \mathbb{C}^{\mathrm{m}} = \frac{\partial\boldsymbol{S}^{\mathrm{m}}}{\partial\boldsymbol{E}^{\mathrm{m}}} &= 2\mu\mathbb{I}_{\mathrm{sym}} + \lambda J^{\mathrm{m}}(2J^{\mathrm{m}} - 1)(\boldsymbol{C}^{\mathrm{m}})^{-1} \otimes (\boldsymbol{C}^{\mathrm{m}})^{-1} \\ &\quad - \lambda J^{\mathrm{m}}(J^{\mathrm{m}} - 1)\Big((\boldsymbol{C}^{\mathrm{m}})^{-1} \overline{\otimes} (\boldsymbol{C}^{\mathrm{m}})^{-1} + (\boldsymbol{C}^{\mathrm{m}})^{-1} \underline{\otimes} (\boldsymbol{C}^{\mathrm{m}})^{-1}\Big). \end{aligned} \tag{24}$$

2.4.2. Compressible neo-Hookean model

The elastic strain energy density function for the compressible neo-Hookean model [31] can be formulated as:

$$\phi = \frac{1}{2}K\left(\frac{(J^{\mathrm{m}})^2 - 1}{2} - \ln(J^{\mathrm{m}})\right) + \frac{1}{2}G\left((J^{\mathrm{m}})^{-2/3}\mathrm{tr}(\boldsymbol{C}^{\mathrm{m}}) - 3\right), \tag{25}$$



(a) Problem setup (Ogden model (Eq. (28))) (b) Effective thermal conductivity $k_{\text{def},11}$ (c) Effective thermal conductivity $k_{\text{def},22}$

Fig. 2. Evolution of the components in the effective thermal conductivity tensor $\boldsymbol{k}_{\text{def}}$ under uniaxial deformation.

where K and G denote the temperature-dependent bulk and shear moduli, respectively. By taking the derivative of Eq. (25) with respect to $\boldsymbol{E}^{\text{m}}$ yields

$$\boldsymbol{S}^{\text{m}} = \frac{\partial \phi}{\partial \boldsymbol{E}^{\text{m}}} = \frac{K}{2}((J^{\text{m}})^2 - 1)(\boldsymbol{C}^{\text{m}})^{-1} + G(J^{\text{m}})^{-2/3}\left(\boldsymbol{I} - \frac{1}{3}\text{tr}(\boldsymbol{C}^{\text{m}})(\boldsymbol{C}^{\text{m}})^{-1}\right), \tag{26}$$

and the derivative of Eq. (26) with respect to $\boldsymbol{E}^{\text{m}}$ becomes

$$\begin{aligned}\mathbb{C}^{\text{m}} = \frac{\partial \boldsymbol{S}^{\text{m}}}{\partial \boldsymbol{E}^{\text{m}}} &= K(J^{\text{m}})^2(\boldsymbol{C}^{\text{m}})^{-1} \otimes (\boldsymbol{C}^{\text{m}})^{-1} - \frac{K}{2}((J^{\text{m}})^2 - 1)\left((\boldsymbol{C}^{\text{m}})^{-1} \overline{\otimes} (\boldsymbol{C}^{\text{m}})^{-1} + (\boldsymbol{C}^{\text{m}})^{-1} \underline{\otimes} (\boldsymbol{C}^{\text{m}})^{-1}\right) \\ &\quad - \frac{2}{3}G(J^{\text{m}})^{-2/3}\left(\boldsymbol{I} \otimes (\boldsymbol{C}^{\text{m}})^{-1} + (\boldsymbol{C}^{\text{m}})^{-1} \otimes \boldsymbol{I}\right) + \frac{2}{9}G(J^{\text{m}})^{-2/3}\text{tr}(\boldsymbol{C}^{\text{m}})(\boldsymbol{C}^{\text{m}})^{-1} \otimes (\boldsymbol{C}^{\text{m}})^{-1} \\ &\quad + \frac{1}{3}G(J^{\text{m}})^{-2/3}\text{tr}(\boldsymbol{C}^{\text{m}})\left((\boldsymbol{C}^{\text{m}})^{-1} \overline{\otimes} (\boldsymbol{C}^{\text{m}})^{-1} + (\boldsymbol{C}^{\text{m}})^{-1} \underline{\otimes} (\boldsymbol{C}^{\text{m}})^{-1}\right).\end{aligned} \tag{27}$$

2.4.3. Compressible Ogden model

The elastic strain energy density function for the compressible Ogden model [30] is defined as

$$\phi(\lambda_1, \lambda_2, \lambda_3, J^{\text{m}}) = \frac{\mu}{\alpha}(\lambda_1^{\alpha} + \lambda_2^{\alpha} + \lambda_3^{\alpha} - 3) + \frac{\mu}{\alpha\beta}((J^{\text{m}})^{-\alpha\beta} - 1), \tag{28}$$

where the principal stretches λ_a are obtained via the spectral decomposition of $\boldsymbol{C}^{\text{m}} = \sum_{a=1}^{3}\lambda_a^2 \boldsymbol{N}_a \otimes \boldsymbol{N}_a$, with $\boldsymbol{N}_a$ being the orthonormal eigenvectors. Here, μ, α, and β represent the initial shear modulus, strain-hardening exponent, and compressibility parameters, and they are temperature-dependent. The power α enables the Ogden model to capture the characteristic "S-shaped" stress–strain behavior [32]. Since $\boldsymbol{S}^{\text{m}}$ is coaxial with $\boldsymbol{C}^{\text{m}}$ for isotropic materials, $\boldsymbol{S}^{\text{m}}$ can be rewritten as $\boldsymbol{S}^{\text{m}} = \sum_{a=1}^{3} S_a \boldsymbol{N}_a \otimes \boldsymbol{N}_a$. For the Ogden model, the principal stresses $S_a = 1/\lambda_a \partial_{\lambda_a}\phi$ are derived as

$$S_a = \mu(\lambda_a^{\alpha-2} - (J^{\text{m}})^{-\alpha\beta}\lambda_a^{-2}) \quad (a = 1, 2, 3). \tag{29}$$

Additionally, $\mathbb{C}^{\text{m}}$ can be derived as the form of an eigenvalue decomposition as follows:

$$\mathbb{C}^{\text{m}} = \sum_{a=1}^{3}\sum_{b=1}^{3} C_{ab}(\boldsymbol{M}_a \otimes \boldsymbol{M}_b) + \sum_{a=1}^{3}\sum_{b\neq a} \mathcal{G}_{ab}(\boldsymbol{N}_a \otimes \boldsymbol{N}_b \otimes \boldsymbol{N}_a \otimes \boldsymbol{N}_b + \boldsymbol{N}_a \otimes \boldsymbol{N}_b \otimes \boldsymbol{N}_b \otimes \boldsymbol{N}_a), \tag{30}$$

where $\boldsymbol{M}_a = \boldsymbol{N}_a \otimes \boldsymbol{N}_a$ denotes the eigen-projections. Here, the principal stiffness components C_{ab} for the compressible Ogden model are

$$C_{ab} = \frac{1}{\lambda_b}\frac{\partial S_a}{\partial \lambda_b} = \begin{cases} \mu\left((\alpha-2)\lambda_a^{\alpha} + (\alpha\beta+2)(J^{\text{m}})^{-\alpha\beta}\right)\lambda_a^{-4} & \text{if } a = b, \\ \mu\alpha\beta(J^{\text{m}})^{-\alpha\beta}(\lambda_a\lambda_b)^{-2} & \text{if } a \neq b, \end{cases} \tag{31}$$

and the shear stiffness coefficients $\mathcal{G}_{ab}$ are

$$\mathcal{G}_{ab} = \begin{cases} \frac{1}{2}(C_{aa} - C_{ab}) & \text{if } \lambda_a = \lambda_b, \\ \frac{S_a - S_b}{\lambda_a^2 - \lambda_b^2} & \text{if } \lambda_a \neq \lambda_b, \end{cases} \tag{32}$$

in which L'Hôpital's rule has been applied when $\lambda_a = \lambda_b$.

2.5. Benchmark showing strongly coupled thermoelastic responses

When thermo-mechanical coupling from mechanics to heat is considered, mechanical deformation alters both the heat conduction paths and the effective areas through which the thermal flux passes. This effect is described by the deformation-dependent

conductivity tensor $\boldsymbol{k}_{\text{def}}(T, \boldsymbol{F}) := k(T)J\boldsymbol{C}^{-1}$ in Eq. (8), where volumetric change J and directional stretches of $\boldsymbol{C}$ modify heat transfer. This coupling appears in the off-diagonal tangent term $D_{\Delta\boldsymbol{u}}\mathcal{R}^{\text{th}}$ in Eq. (21), indicating that the temperature field depends not only on thermal boundary conditions but also on the deformed configuration. To provide an intuitive understanding, the effective thermal conductivity tensor $\boldsymbol{k}_{\text{def}}$ is examined under uniaxial deformation. As shown in Fig. 2(a), a 2D domain composed of the Ogden model (Section 2.4.3) is subjected to a prescribed displacement $\bar{u}$ on the right edge.

The resulting components of $\boldsymbol{k}_{\text{def}}$ are directly extracted at the integration points, and the variations of the axial and transverse components, $k_{\text{def},11}$ and $k_{\text{def},22}$, with respect to the stretch ratio λ_1 are shown in Fig. 2(b) and (c). In the undeformed state ($\lambda_1 = 1$), the material is thermally isotropic. As λ_1 increases (or λ_2 decreases), $k_{\text{def},11}$ decreases due to elongation of the conduction path ($\mathrm{d}\lambda_1/\mathrm{d}\bar{u} > 0$) and reduction of the cross-sectional area ($\mathrm{d}\lambda_2/\mathrm{d}\bar{u} < 0$), while $k_{\text{def},22}$ increases as lateral contraction shortens the transverse path. This demonstrates that finite strain theory induces thermal anisotropy, highlighting the necessity of incorporating $\boldsymbol{k}_{\text{def}}$ in Eq. (8) for reliable thermoelastic material responses.

Meanwhile, large deformations in finite strain TO often lead to finite element distortion and Newton–Raphson divergence, preventing access to regimes with pronounced deformation-dependent heat conduction. Therefore, a robust material interpolation strategy capable of sustaining large deformations is required, which will be discussed in the next section.

3. Thermoelastic multi-material topology optimization

This section presents a thermoelastic multi-material topology optimization (MMTO) framework for finite strain problems. A material-field series expansion (MFSE) method [52] combined with a DMO-based interpolation is employed to represent multi-material distributions with reduced design variables. To ensure numerical robustness under large deformations, a kinematically scaled unified nonlinear multi-material (UNM) interpolation scheme is introduced, followed by the formulation of the optimization problem and its solution procedure.

3.1. Material-field series expansion

In this work, the material-field series expansion (MFSE) method [52] is adopted to reduce the computational cost by decreasing the number of design variables. To describe n_{m} types of material distributions within the design domain Ω_{Des}, n_{m} material fields $\varphi_i(\boldsymbol{X})$ ($i = 1, 2, \dots, n_{\text{m}}$) are introduced. The introduced material field is treated as an uncertain field with spatially correlated dependency, and the correlation between any two points in the design domain is calculated by a correlation function $f_{\text{cor}}(\boldsymbol{X}_i, \boldsymbol{X}_j)$ as follows:

$$\begin{cases} f_{\text{cor}}(\boldsymbol{X}_i, \boldsymbol{X}_j) = \exp\left(-\dfrac{(X_i - X_j)^2}{l_{cx}^2} - \dfrac{(Y_i - Y_j)^2}{l_{cy}^2}\right) & \text{for 2D cases,} \\ f_{\text{cor}}(\boldsymbol{X}_i, \boldsymbol{X}_j) = \exp\left(-\dfrac{(X_i - X_j)^2}{l_{cx}^2} - \dfrac{(Y_i - Y_j)^2}{l_{cy}^2} - \dfrac{(Z_i - Z_j)^2}{l_{cz}^2}\right) & \text{for 3D cases,} \end{cases} \tag{33}$$

where l_{cx}, l_{cy}, and l_{cz} are correlation lengths in the X, Y, and Z directions of the reference configuration $\Omega_0(\boldsymbol{X})$. Similar to the role of spatial filters in density-based methods, the correlation length implicitly controls the minimum feature size of the optimized design. To maintain consistency, the correlation lengths are uniformly set to 10% of the corresponding edge lengths of the design domain in all numerical examples. In the MFSE method, there are many observation points $\boldsymbol{X}_i (i = 1, 2, \dots, n_{\text{p}})$ distributed in the design domain Ω_{Des}. The correlation matrix $\boldsymbol{M}_{\text{cor}}$ of the material field $\varphi_i(\boldsymbol{X})$ can be constructed by the spatial correlation of any two observation points using Eq. (33) as

$$\boldsymbol{M}_{\text{cor}} = \begin{bmatrix} 1 & f_{\text{cor}}(\boldsymbol{X}_1, \boldsymbol{X}_2) & \cdots & f_{\text{cor}}(\boldsymbol{X}_1, \boldsymbol{X}_{n_{\text{p}}}) \\ f_{\text{cor}}(\boldsymbol{X}_2, \boldsymbol{X}_1) & 1 & \cdots & f_{\text{cor}}(\boldsymbol{X}_2, \boldsymbol{X}_{n_{\text{p}}}) \\ \vdots & \vdots & \ddots & \vdots \\ f_{\text{cor}}(\boldsymbol{X}_{n_{\text{p}}}, \boldsymbol{X}_1) & f_{\text{cor}}(\boldsymbol{X}_{n_{\text{p}}}, \boldsymbol{X}_2) & \cdots & 1 \end{bmatrix}. \tag{34}$$

Benefiting from the orthogonal symmetry of the correlation matrix $\boldsymbol{M}_{\text{cor}}$, the eigenvalue decomposition is adopted to extract its eigenvalues ξ_k and normalized eigenvectors $\boldsymbol{m}_k$. Thus, the material field can be expressed using the Expansion Optimal Linear Estimator (EOLE) method [56] as

$$\varphi_i(\boldsymbol{X}) = \sum_{k=1}^{n_{\text{p}}} \eta_{i,k} \frac{1}{\sqrt{\xi_k}} \boldsymbol{m}_k^{\text{T}} \boldsymbol{m}_{\text{div}}(\boldsymbol{X}), \quad \boldsymbol{X} \in \Omega_{\text{Des}}, \tag{35}$$

where $\eta_{i,k}$ is the kth coefficient with respect to the basic mode of the ith material field $\varphi_i(\boldsymbol{X})$, and $\boldsymbol{m}_{\text{div}}(\boldsymbol{X}) = (f_{\text{cor}}(\boldsymbol{X}, \boldsymbol{X}_1), \cdots, f_{\text{cor}}(\boldsymbol{X}, \boldsymbol{X}_{n_{\text{p}}}))^{\text{T}}$ is the corresponding column of $\boldsymbol{M}_{\text{cor}}$ evaluated at $\boldsymbol{X}$. Notably, eigenmodes associated with small eigenvalues can be neglected when constructing the material field. Therefore, by truncating Eq. (35) based on the specified truncation error $\epsilon > 0$, the material field can be rewritten in the truncated form as

$$\varphi_i(\boldsymbol{X}) \approx \sum_{k=1}^{\tilde{n}_{\text{p}}} \eta_{i,k} \frac{1}{\sqrt{\xi_k}} \boldsymbol{m}_k^{\text{T}} \boldsymbol{m}_{\text{div}}(\boldsymbol{X}) = \boldsymbol{\eta}_i^{\text{T}} \boldsymbol{\Xi}^{-1/2} \boldsymbol{\Phi}^{\text{T}} \boldsymbol{m}_{\text{div}}(\boldsymbol{X}), \quad \boldsymbol{X} \in \Omega_{\text{Des}}, \tag{36}$$

where $\tilde{n}_{\text{p}} \le n_{\text{p}}$ is determined by a specified truncation error ϵ, defined as $\epsilon = 1 - \sum_{k=1}^{\tilde{n}_{\text{p}}} \xi_k / \sum_{k=1}^{n_{\text{p}}} \xi_k$, which is set to 0.01% in this work. $\boldsymbol{\eta}_i = (\eta_{i,1}, \dots, \eta_{i,\tilde{n}_{\text{p}}})^{\text{T}}$ is the vector of undetermined design coefficients for the ith material field, $\boldsymbol{\Xi} = \text{diag}(\xi_1, \dots, \xi_{\tilde{n}_{\text{p}}})$ is an

$\tilde{n}_p \times \tilde{n}_p$ diagonal matrix storing the dominant eigenvalues, and $\boldsymbol{\Phi} = (\boldsymbol{m}_1, \boldsymbol{m}_2, \dots, \boldsymbol{m}_{\tilde{n}_p})$ is an $n_p \times \tilde{n}_p$ matrix collecting the corresponding normalized eigenvectors, respectively. Based on the expansion and truncation of the material field, the number of design variables can be reduced while preserving the capability to represent the continuum configuration.

The material field in Eq. (36) is subsequently substituted into the following sigmoid-based function to confine its values within the range of $[0, 1]$ to get the physical variable ρ_i as

$$\rho_i(\boldsymbol{X}) = \frac{1}{1 + \exp(-\beta_{\text{sig}} \varphi_i(\boldsymbol{X}))}. \tag{37}$$

Here, β_{sig} is the sharpness parameter for the sigmoid-based function, which is increased from 1 to 40 during the optimization process to ensure stability. It is worth noting that in the absence of the MFSE method, the physical variable ρ_i can be directly treated as the design variable.

3.2. Kinematically scaled material interpolation scheme

Large deformations can occur in extreme thermal environments, such as gas turbines and aerospace applications. Thus, when considering the optimization problem, it is necessary to employ a material interpolation scheme suitable for large deformation problems. In geometrically nonlinear TO, excessive distortion of low-stiffness regions (e.g., void elements) would induce negative element Jacobians and singular tangent stiffness matrices, leading to numerical divergence. To improve the numerical stability, Wang et al. [45] proposed an energy interpolation scheme that decouples the kinematic assumptions of solid and void regions. The corresponding strain energy density function $\phi_{\text{wang}}(\rho, \boldsymbol{u})$ is defined as

$$\begin{aligned} \phi_{\text{wang}}(E(\rho), \boldsymbol{u}) &= \phi_{\text{NL}}(E(\rho), \boldsymbol{F}_{\text{wang},\rho}, T) - \phi_{\text{L}}(E(\rho), \boldsymbol{\varepsilon}_{\text{wang},\rho}) + \phi_{\text{L}}(E(\rho), \boldsymbol{\varepsilon}) \\ &= \phi_{\text{NL}}(E(\rho), \boldsymbol{F}_{\text{wang},\rho}, T) + (1 - \gamma^2)\phi_{\text{L}}(E(\rho), \boldsymbol{\varepsilon}), \end{aligned} \tag{38}$$

where ϕ_{NL} and ϕ_{L} represent the nonlinear hyperelastic and linear elastic energy densities, $\boldsymbol{F}_{\text{wang},\rho} = \boldsymbol{I} + \gamma \nabla_{\boldsymbol{X}} \boldsymbol{u}$ and $\boldsymbol{\varepsilon}_{\text{wang},\rho} = \gamma \left(\nabla_{\boldsymbol{X}} \boldsymbol{u}\right)_{\text{sym}}$ denote the design variable-dependent deformation gradient tensor and small strain tensor with $\boldsymbol{\varepsilon} = \left(\nabla_{\boldsymbol{X}} \boldsymbol{u}\right)_{\text{sym}}$ being the original small strain tensor, and $E(\rho)$ denotes the interpolated Young's modulus, respectively. Notably, the factor $\gamma \in [0, 1]$ serves as a kinematic switch parameter; $\gamma = 1$ and $\gamma = 0$ leads to the hyperelastic and linear elastic kinematics, respectively. The computation stability can be maintained even if mesh distortion occurs in the void elements; refer to Han et al. [57] for comprehensive discussions. To ensure differentiability, the switching factor γ is parameterized via a tangent-based Heaviside projection:

$$\gamma = \frac{\tanh(\beta_{\text{tang}}\zeta) + \tanh(\beta_{\text{tang}}(\rho^p - \zeta))}{\tanh(\beta_{\text{tang}}\zeta) + \tanh(\beta_{\text{tang}}(1 - \zeta))}, \tag{39}$$

where the penalization power p is used to penalize the intermediate value of ρ and push it to binary values. The sharpness parameter β_{tang} and threshold value ζ are empirically set to 500 and 0.02, respectively.

Although the interpolation scheme in Eq. (38) is widely known to be effective for large deformation TO, the computational stability may be impaired under the level of extreme large deformation considered in this study, particularly in intermediate regions. To maintain computational stability in intermediate regions, an alternative material interpolation scheme by directly scaling the kinematic field is proposed:

$$\begin{aligned} \phi_{\text{Scale}}(\rho, \boldsymbol{u}) &= \phi_{\text{NL}}(E_{\text{NL}}, \boldsymbol{F}_\rho, T) - \phi_{\text{L}}(E_{\text{L}}, \boldsymbol{\varepsilon}_\rho) + \phi_{\text{L}}(E_{\text{L}}, \boldsymbol{\varepsilon}_\rho/\gamma) \\ &= \phi_{\text{NL}}(E_{\text{NL}}, \boldsymbol{F}_\rho, T) + (1 - \gamma^2)\rho^{2p}\phi_{\text{L}}(E_{\text{L}}, \boldsymbol{\varepsilon}), \end{aligned} \tag{40}$$

where $\boldsymbol{F}_\rho = \boldsymbol{I} + \gamma\rho^p \nabla_{\boldsymbol{X}} \boldsymbol{u}$ and $\boldsymbol{\varepsilon}_\rho = \gamma\rho^p \left(\nabla_{\boldsymbol{X}} \boldsymbol{u}\right)_{\text{sym}}$ denote the design variable-dependent deformation gradient and small strain tensors employed in this study. Also, E_{NL} and E_{L} are the prescribed Young's moduli for the material and void phases. It turns out from Eq. (40) that ρ^p has been moved to the structural deformation gradient rather than the material parameters, and it acts as a kinematic scaling. While this approach improves the FE convergence in intermediate regions, it should be acknowledged that the direct kinematic scaling of the deformation gradient in intermediate regions compromises the strict physical consistency of finite strain mechanics, in particular the principle of objectivity (frame indifference). This modification is therefore introduced as a pragmatic numerical compromise: without such a relaxation, the large deformation TO problem considered in this study becomes extremely difficult to solve due to severe distortion-induced instabilities in intermediate regions. The underlying mathematical mechanism of this interpolation scheme to ensure numerical stability is detailed in Appendix D. Since the optimizer ultimately drives the design variables toward binary values, the non-physical artifacts vanish in the final design, thereby preserving the physical consistency of the optimized structural configuration.

3.3. Kinematically scaled material interpolation scheme for MMTO

As described in Section 2.4, nonlinear constitutive laws are characterized by their own strain energy density functions and associated sets of material parameters. Thus, unlike MMTO for linear elastic materials, interpolation based on material parameters becomes highly challenging or physically unreasonable [36]. This is because, even among the afore-introduced three hyperelastic materials, each model is defined by a different elastic strain energy density function, and the Ogden model in particular possesses its own unique material parameters, namely α and β.

To enable flexible selection of material models in MMTO, the concept of extended discrete material optimization (XDMO) [37] is incorporated into Eq. (40). Specifically, the hybrid strain energy density function for n_{m} types of candidate materials is reformulated as follows:

$$\begin{aligned}\phi_{\mathrm{UNM}}(w_1, w_2, \ldots, w_{n_{\mathrm{m}}}, \boldsymbol{u}) &= \sum_{i=1}^{n_{\mathrm{m}}} \left(\phi_{\mathrm{NL},i}(\boldsymbol{\kappa}_{\mathrm{NL},i}, \boldsymbol{F}_{w,i}, T) - \phi_{\mathrm{L},i}(\boldsymbol{\kappa}_{\mathrm{L},i}, \boldsymbol{\varepsilon}_{w,i}) + \phi_{\mathrm{L},i}(\boldsymbol{\kappa}_{\mathrm{L},i}, \boldsymbol{\varepsilon}_{w,i}/\gamma_w)\right) \\ &= \sum_{i=1}^{n_{\mathrm{m}}} \left(\phi_{\mathrm{NL},i}(\boldsymbol{\kappa}_{\mathrm{NL},i}, \boldsymbol{F}_{w,i}, T) + \left(1-\gamma_w^2\right) w_i^2 \phi_{\mathrm{L},i}(\boldsymbol{\kappa}_{\mathrm{L},i}, \boldsymbol{\varepsilon})\right),\end{aligned} \tag{41}$$

where γ_w is the tangent-based Heaviside projection for MMTO, $\phi_{\mathrm{NL},i}$ and $\phi_{\mathrm{L},i}$ represent the elastic strain energy density functions of the ith hyperelastic and linear elastic materials, and $\boldsymbol{\kappa}_{\mathrm{NL},i}$ and $\boldsymbol{\kappa}_{\mathrm{L},i}$ denote the corresponding material parameter sets for the ith hyperelastic and linear elastic materials, respectively. Also, $\boldsymbol{F}_{w,i}$ and $\boldsymbol{\varepsilon}_{w,i}$ are the ith weight-dependent deformation gradient and small strain tensors defined as

$$\begin{aligned}&\boldsymbol{F}_{w,i} = \boldsymbol{I} + \gamma_w w_i \nabla_{\boldsymbol{X}} \boldsymbol{u}, \\ &\boldsymbol{\varepsilon}_{w,i} = \gamma_w w_i \left(\nabla_{\boldsymbol{X}} \boldsymbol{u}\right)_{\mathrm{sym}}, \\ &\gamma_w = \frac{\tanh(\beta_{\mathrm{tang}}\zeta) + \tanh(\beta_{\mathrm{tang}}(\sum_{i=1}^{n_{\mathrm{m}}} w_i - \zeta))}{\tanh(\beta_{\mathrm{tang}}\zeta) + \tanh(\beta_{\mathrm{tang}}(1-\zeta))}.\end{aligned} \tag{42}$$

Additionally, $w_i(\rho_1, \rho_2, \ldots, \rho_{n_{\mathrm{m}}})$ denotes the ith weight function, and is interpolated through the DMO-type interpolation function [51] as

$$w_i(\boldsymbol{X}) = \left(\rho_i(\boldsymbol{X})\right)^p \prod_{j=1, j\neq i}^{n_{\mathrm{m}}} \left(1 - \left(\rho_j(\boldsymbol{X})\right)^p\right). \tag{43}$$

For convenience, the proposed material interpolation scheme is referred to as the "unified nonlinear multi-material (UNM) interpolation scheme" in this study.

As a particular case, let us consider an MMTO problem involving the previously introduced three hyperelastic materials as follows:

$$\begin{aligned}&\sum_{i=1}^{n_{\mathrm{m}}} \phi_{\mathrm{NL},i}(\boldsymbol{\kappa}_{\mathrm{NL},i}, \boldsymbol{F}_{w,i}, T) \\ &\quad = \phi_1^{\mathrm{SVK}}(\boldsymbol{\kappa}_{\mathrm{NL},1}, \boldsymbol{F}_{w,1}, T) + \phi_2^{\text{neo-Hookean}}(\boldsymbol{\kappa}_{\mathrm{NL},2}, \boldsymbol{F}_{w,2}, T) + \cdots + \phi_{n_{\mathrm{m}}}^{\mathrm{Ogden}}(\boldsymbol{\kappa}_{\mathrm{NL},n_{\mathrm{m}}}, \boldsymbol{F}_{w,n_{\mathrm{m}}}, T), \\ &\text{with } \boldsymbol{\kappa}_{\mathrm{NL},1} = \left(\lambda_1, \mu_1, \alpha_{\mathrm{th},1}, k_1\right), \ \boldsymbol{\kappa}_{\mathrm{NL},2} = \left(K_2, G_2, \alpha_{\mathrm{th},2}, k_2\right), \ldots, \ \boldsymbol{\kappa}_{\mathrm{NL},n_{\mathrm{m}}} = \left(\mu_{n_{\mathrm{m}}}, \alpha_{n_{\mathrm{m}}}, \beta_{n_{\mathrm{m}}}, \alpha_{\mathrm{th},n_{\mathrm{m}}}, k_{n_{\mathrm{m}}}\right), \\ &\sum_{i=1}^{n_{\mathrm{m}}} \left(1-\gamma_w^2\right) w_i^2 \phi_{\mathrm{L},i}(\boldsymbol{\kappa}_{\mathrm{L},i}, \boldsymbol{\varepsilon}) \\ &\quad = \left(1-\gamma_w^2\right) w_1^2 \phi_{\mathrm{L},1}(\boldsymbol{\kappa}_{\mathrm{L},1}, \boldsymbol{\varepsilon}) + \left(1-\gamma_w^2\right) w_2^2 \phi_{\mathrm{L},2}(\boldsymbol{\kappa}_{\mathrm{L},2}, \boldsymbol{\varepsilon}) + \cdots + \left(1-\gamma_w^2\right) w_{n_{\mathrm{m}}}^2 \phi_{\mathrm{L},n_{\mathrm{m}}}(\boldsymbol{\kappa}_{\mathrm{L},n_{\mathrm{m}}}, \boldsymbol{\varepsilon}) \\ &\text{with } \phi_{\mathrm{L},1} = \phi_{\mathrm{L},2} = \cdots = \phi_{\mathrm{L},n_{\mathrm{m}}} = \frac{\lambda_{\mathrm{L}}}{2}(\mathrm{tr}(\boldsymbol{\varepsilon}))^2 + \mu_{\mathrm{L}} \boldsymbol{\varepsilon} : \boldsymbol{\varepsilon}, \quad \boldsymbol{\kappa}_{\mathrm{L},1} = \boldsymbol{\kappa}_{\mathrm{L},2} = \cdots = \boldsymbol{\kappa}_{\mathrm{L},n_{\mathrm{m}}} = \left(\lambda_{\mathrm{L}}, \mu_{\mathrm{L}}\right),\end{aligned} \tag{44}$$

which is formulated in terms of the strain energy density function; thus, the functional forms and material parameter sets of the individual base materials can be chosen arbitrarily. Thereby, Eq. (41) can be interpreted as follows:

1. The formulation of Wang et al. [45] is adopted to distinguish the material and void regions.
2. Within the material region, the XDMO concept [37] is employed to allow flexible selection of material models.

Once the proposed UNM interpolation scheme is adopted, the strain measurements in Eq. (42) require the derivative with respect to the displacement gradient tensor $\nabla_{\boldsymbol{X}} \boldsymbol{u}$ instead of the small and finite strain tensors. For instance, the PK1 stress in Eq. (11) and the stiffness tangent tensor in Eq. (19) are modified to be

$$\begin{aligned}\text{Eq. (10)} \Rightarrow \boldsymbol{P} &= \sum_{i=1}^{n_{\mathrm{m}}} \left(\frac{\partial \phi_{\mathrm{NL},i}}{\partial \boldsymbol{E}_{w,i}^{\mathrm{m}}} : \frac{\partial \boldsymbol{E}_{w,i}^{\mathrm{m}}}{\partial \boldsymbol{C}_{w,i}} : \frac{\partial \boldsymbol{C}_{w,i}}{\partial \boldsymbol{F}_{w,i}} : \frac{\partial \boldsymbol{F}_{w,i}}{\partial \nabla \boldsymbol{u}} + \left(1-\gamma_w^2\right) w_i^2 \frac{\partial \phi_{\mathrm{L},i}}{\partial \nabla \boldsymbol{u}}\right) \\ &= \sum_{i=1}^{n_{\mathrm{m}}} \left(\gamma_w w_i \boldsymbol{F}_{w,i} \boldsymbol{S}_{w,i} + \left(1-\gamma_w^2\right) w_i^2 \boldsymbol{\sigma}_{\mathrm{L},i}\right) = \boldsymbol{P}_{\mathrm{NL}} + \boldsymbol{P}_{\mathrm{L}} \quad \text{with} \quad \boldsymbol{\sigma}_{\mathrm{L},i} = \lambda_{\mathrm{L},i} \mathrm{tr}(\boldsymbol{\varepsilon}) \boldsymbol{I} + 2\mu_{\mathrm{L},i} \boldsymbol{\varepsilon},\end{aligned} \tag{45}$$

and

$$\begin{aligned}\text{Eq. (19)} \Rightarrow \frac{\partial \boldsymbol{P}}{\partial \nabla \boldsymbol{u}} &= \sum_{i=1}^{n_{\mathrm{m}}} \left(\gamma_w w_i \frac{\partial \boldsymbol{F}_{w,i} \boldsymbol{S}_{w,i}}{\partial \nabla \boldsymbol{u}} + \left(1-\gamma_w^2\right) w_i^2 \frac{\partial \boldsymbol{\sigma}_{\mathrm{L},i}}{\partial \nabla \boldsymbol{u}}\right) \\ &= \sum_{i=1}^{n_{\mathrm{m}}} \left(\gamma_w^2 w_i^2 \left(\boldsymbol{I} \overline{\otimes} \boldsymbol{S}_{w,i} + \boldsymbol{F}_{w,i} \left(\vartheta_i^{-4} \mathbb{C}_{w,i}^{\mathrm{m}}\right) : \frac{1}{2}\left(\boldsymbol{F}_{w,i}^{\mathrm{T}} \overline{\otimes} \boldsymbol{I} + \boldsymbol{I} \underline{\otimes} \boldsymbol{F}_{w,i}^{\mathrm{T}}\right)\right) \right. \\ &\qquad \left. + \left(1-\gamma_w^2\right) w_i^2 \mathbb{C}_{\mathrm{L},i} \right) = \frac{\partial \boldsymbol{P}_{\mathrm{NL}}}{\partial \nabla \boldsymbol{u}} + \frac{\partial \boldsymbol{P}_{\mathrm{L}}}{\partial \nabla \boldsymbol{u}} \quad \text{with} \quad \mathbb{C}_{\mathrm{L},i} = \lambda_{\mathrm{L},i} \boldsymbol{I} \otimes \boldsymbol{I} + 2\mu_{\mathrm{L},i} \mathbb{I}_{\mathrm{sym}},\end{aligned} \tag{46}$$

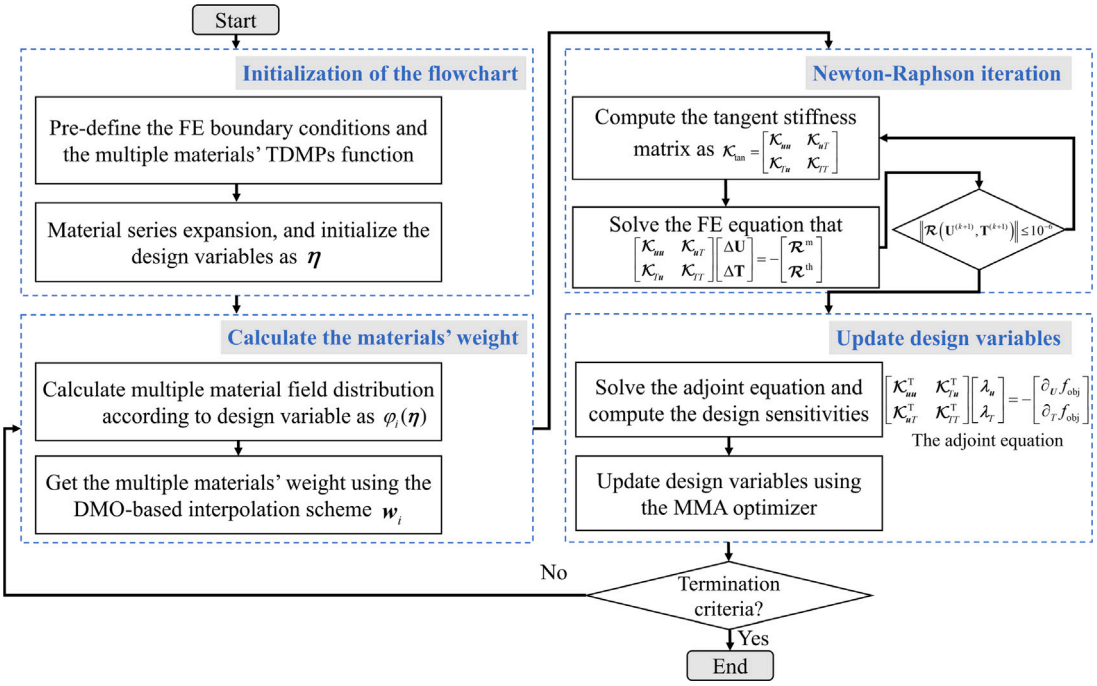


Fig. 3. Optimization algorithm.

both of which have the nonlinear and linear contributions. Here, the subscript $\bullet_w$ represents that the design variable-dependent (or weight function) strain measurements have been used. It is important to distinguish the mathematical consistency of this weighted interpolation from its physical interpretability. Because each base material is formulated as a hyperelastic potential (i.e., a strain energy density function), any non-negative weighted superposition of these functions guarantees a mathematically well-posed hyperelastic potential [37]. However, this does not imply that the interpolated intermediate-density material possesses a physically realizable constitutive response. From an optimization perspective, the intermediate-density regions serve primarily as a mathematical relaxation. Through the use of penalization and projection schemes, the optimization process ultimately eliminates these intermediate mixtures upon convergence, as demonstrated in Appendix D. Consequently, the final optimized structure is composed exclusively of distinct, physically realizable material phases (e.g., pure modified SVK, neo-Hookean, or Ogden) and the void phase.

Additionally, when the proposed material interpolation scheme is adopted, the effective thermal conductivity tensor in Eq. (8) can be rewritten as follows:

$$\boldsymbol{k}_{\mathrm{def}}=\sum_{i=1}^{n_{\mathrm{m}}} w_i\, k_i(T)\, J_{w,i}\, \boldsymbol{C}_{w,i}^{-1}. \tag{47}$$

Note that both $J_{w,i}$ and $\boldsymbol{C}_{w,i}$ are design variable-dependent, i.e., $J_{w,i}=\det\left(\boldsymbol{F}_{w,i}\right)$ and $\boldsymbol{C}_{w,i}=\boldsymbol{F}_{w,i}^{\mathrm{T}}\boldsymbol{F}_{w,i}$.

3.4. Optimization problem

A general optimization problem of the thermoelastic MMTO considering the TDMPs can be given as

$$\begin{aligned}&\text{Find:}\quad \boldsymbol{\eta}=(\boldsymbol{\eta}_1,\boldsymbol{\eta}_2,\ldots,\boldsymbol{\eta}_{n_m})^{\mathrm{T}},\\ &\text{Minimize:}\quad f_{\mathrm{obj}}(\boldsymbol{\eta},\boldsymbol{U},\boldsymbol{T}),\\ &\text{Subject to:}\quad \boldsymbol{\mathcal{R}}(\boldsymbol{\eta})=\begin{bmatrix}\boldsymbol{\mathcal{R}}^{\mathrm{m}}(\boldsymbol{\eta},\boldsymbol{U},\boldsymbol{T})\\ \boldsymbol{\mathcal{R}}^{\mathrm{th}}(\boldsymbol{\eta},\boldsymbol{U},\boldsymbol{T})\end{bmatrix}=\boldsymbol{0}\ \text{ and }\ M(\boldsymbol{\eta})=\frac{\sum_{e=1}^{n_{\mathrm{e}}}\sum_{i=1}^{n_{\mathrm{m}}} M_i\rho_i(\boldsymbol{X}_e)v(\boldsymbol{X}_e)}{M_0}\le f_{\max},\end{aligned} \tag{48}$$

where f_{obj} denotes the general thermo-mechanical objective function, the first constraint denotes the residual vectors in Eq. (16), and the second constraint denotes the total mass constraint. Here, M_0 is the reference total mass of the full-solid design domain, M_i denotes the mass factor for the ith material, v_e denotes the elemental volume of the eth element, and $f_{\max}$ is the prescribed upper limit of the mass fraction, respectively. In this study, the mass factors for all materials are set as $M_i=1$ to enable unbiased material distribution. It should be noted that the MMTO problem is inherently non-convex, and this characteristic is further intensified by the strongly nonlinear coupled thermoelastic behavior and the presence of TDMPs. To minimize artificial bias in the optimization process, the initial guess for all numerical examples is set to a uniform distribution, following common practice in DMO-based

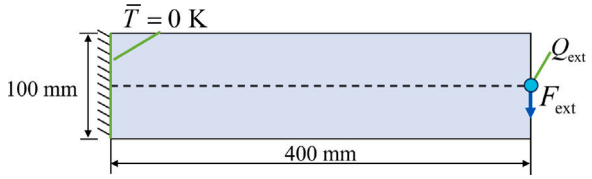


Fig. 4. Problem setup for Example 1: 2D cantilever beam.

approaches [21,51]. Specifically, the MFSE design variables are initialized by assigning each element an equal volume fraction of all candidate materials, i.e., $\rho_{e,i}^{(0)} = f_{\max}/n_{\mathrm{m}}$.

In finite strain thermoelastic TO, the choice of objective function influences the resulting thermo-mechanical response. As demonstrated by Goh et al. [39], minimizing the strain energy tends to suppress thermally induced deformation by limiting the overall thermal stress, whereas minimizing the end compliance allows the structure to actively exploit thermal expansion to counteract external mechanical loads. Since the primary objective of this work is to investigate geometric self-adaptation induced by TDMPs and thermal expansion, the structural end compliance is adopted as the primary objective function (i.e., $f_{\mathrm{obj}} = c_{\mathrm{obj}}$):

$$c_{\mathrm{obj}}(\boldsymbol{\eta}, \boldsymbol{U}, \boldsymbol{T}) = \boldsymbol{F}_{\mathrm{ext}}^{\mathrm{T}} \boldsymbol{U}, \tag{49}$$

where the external force $\boldsymbol{F}_{\mathrm{ext}}$ in Eq. $(17)_1$ is assumed to be design independent.

To discuss the characteristics of the optimized results under different objective functions, the total stored strain energy is employed as an alternative objective functional (i.e., $f_{\mathrm{obj}} = \Pi_{\mathrm{obj}}$):

$$\Pi_{\mathrm{obj}}(\boldsymbol{\eta}, \boldsymbol{U}, \boldsymbol{T}) = \int_{\Omega_0} \psi \, \mathrm{d}V, \tag{50}$$

where ψ denotes the elastic strain–energy density of the constituent hyperelastic models defined in Eq. (44), and Ω_0 is the reference configuration. Unlike end compliance, the stored-energy objective depends on the temperature field through both thermal deformation and the TDMPs. Its minimization therefore suppresses thermally induced stresses rather than exploiting them, as discussed by Goh et al. [39]. For the sake of conciseness, the detailed derivation and verification of the design sensitivities are provided in Appendix C.

The overall optimization algorithm is illustrated in Fig. 3. An optimization begins with the initialization of boundary conditions, design variables $\boldsymbol{\eta}$, and mechanical and thermal parameters, respectively. In each optimization iteration, the design variables are first converted into material weights through the DMO-based MFSE interpolation scheme. Next, the state variables, $\boldsymbol{U}$ and $\boldsymbol{T}$, are solved via the Newton–Raphson procedure. Following the nonlinear analysis, the design sensitivities are calculated by solving the adjoint equation. Finally, the design variables are updated using the MMA optimizer. This optimization loop repeats until the convergence criterion is satisfied, such that the relative change in the objective function in Eq. (49) is below 0.5% and the projection parameter β_{sig} in Eq. (37) reaches 40.

To balance computational efficiency and convergence stability, a two-stage strategy is adopted within the Newton–Raphson solver. The full thermo-mechanical load is first applied using the converged solution from the previous design iteration as the initial guess (warm-start). If convergence is not achieved, the solver automatically switches to an adaptive load-stepping scheme with Armijo line search [58]. The load increment is halved upon divergence and increased by a factor of 1.5 after successful convergence. The warm-start strategy succeeds in more than 60% of the optimization iterations, resulting in an approximately 40%–50% reduction in the overall computational time. The adaptive load-stepping scheme serves as a robust fallback mechanism for the remaining cases.

4. Numerical examples

To demonstrate the effectiveness of the proposed MMTO framework for strongly coupled thermoelasticity with the TDMPs, this section presents five numerical examples. We begin with two classical examples to assess numerical stability and physical consistency. Subsequently, irregular design domains are employed to demonstrate the potential for practical engineering design. For all numerical examples, the reference temperature T_{ref} is set to 0 K. Following Eq. (17), external loads are assigned to the corresponding degrees of freedom of the global loading vectors. Here, F_{ext} and Q_{ext} denote the prescribed scalar mechanical and thermal nodal loads, respectively. The TDMP functions in the following examples are prescribed to investigate the computational treatment of different constitutive responses and their crossover behavior, and they are not fitted to specific engineering materials. In practical applications, thermal and mechanical constitutive functions require material-specific characterization, for which polynomial functions are commonly used to approximate measured temperature dependence [17–19].

4.1. Example 1: 2D cantilever beam

As the first example, a TO problem of the 2D cantilever beam in Fig. 4 with dimensions $L \times H = 400 \times 100$ mm^2 is investigated. The left edge of the design domain is fully clamped ($\bar{\boldsymbol{u}} = \boldsymbol{0}$) and acts as a heat sink ($\bar{T} = 0$). At the midpoint of the right free end, a concentrated external mechanical loading and an external thermal loading are applied simultaneously. The design domain is also discretized into 200 × 50 elements, and the total mass constraint $f_{\max}$ is prescribed as 0.50.

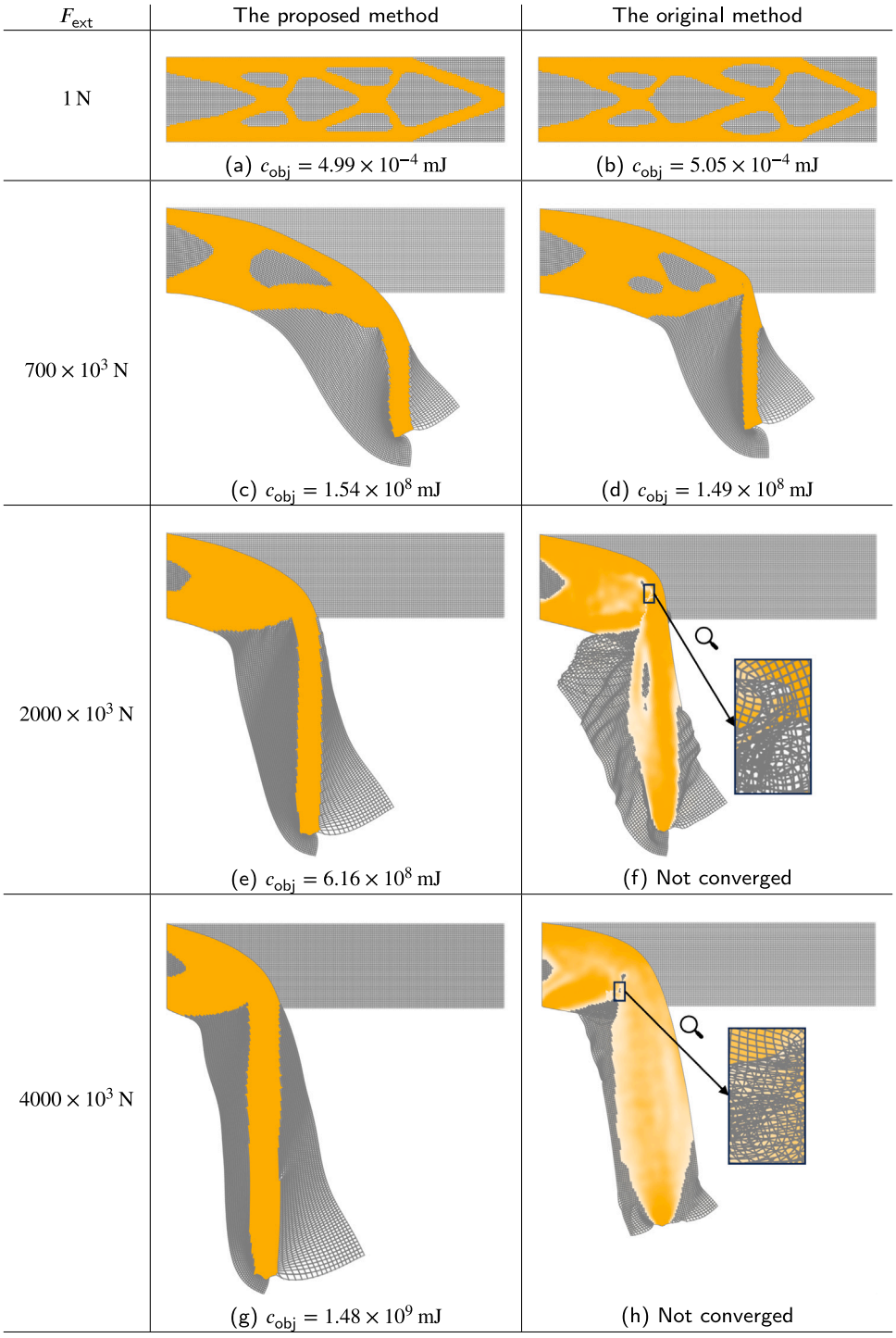


Fig. 5. Comparison of optimized results by the proposed and original interpolation methods for Example 1.

4.1.1. Verification of the kinematically scaled material interpolation method in Section 3.2

Before exploring MMTO, we briefly compare the proposed kinematically scaled interpolation in Eq. (40) with the original energy interpolation in Eq. (38). While both ensure nonlinear and linear consistencies in solid and void regions, they differ in the treatment of intermediate density regions. In the original method, the displacement gradient is scaled as $\gamma \nabla_{\boldsymbol{X}} \boldsymbol{u}$, allowing large deformation gradients even in intermediate density regions, which may cause excessive distortion and numerical singularities. In contrast, the

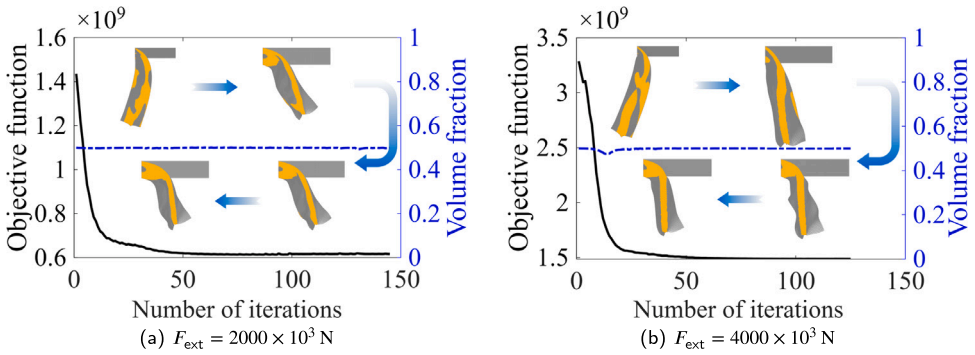


Fig. 6. Optimization histories of the proposed interpolation method for Example 1.

Table 2
Comparison of the objective function values (c_{obj}) and reevaluated values ($\hat{c}_{obj}$) under different external thermal loads (Q_{ext}) for Example 1.

Q_{ext} (W)	Analysis method	c_{obj} (mJ)	$\hat{c}_{obj}$ (mJ, Reevaluated)
50	Strongly coupled method	1.06×10^9	–
	Weakly coupled method	1.03×10^9	1.08×10^9
100	Strongly coupled method	1.50×10^9	–
	Weakly coupled method	1.42×10^9	1.56×10^9
150	Strongly coupled method	2.01×10^9	–
	Weakly coupled method	1.87×10^9	2.08×10^9

proposed method scales the displacement gradient as $\gamma \rho^p \nabla_{\boldsymbol{X}} \boldsymbol{u}$, so that the deformation is reduced in accordance with the material density. This kinematic scaling suppresses nonlinear distortion in intermediate regions and improves numerical stability under large deformations.

To demonstrate this mechanism, the Ogden model with $\mu = 5.36 \times 10^5$ MPa, $\alpha = 4$, and $\beta = 2$ is considered without the TDMPs, and the external thermal loading is prescribed as zero. By employing the MFSE method, the number of design variables in this example is reduced to $\tilde{n}_p = 168$. This is considerably smaller than the number of finite elements ($n_e = 10000$), thereby accelerating the variable update process within the optimizer. However, because the dominant computational cost of the present framework arises from the strongly coupled nonlinear analysis, the reduction in design variables contributes only marginally to the overall computational efficiency. The corresponding optimized results, along with their deformed configurations, are given in Fig. 5.

For small deformations (e.g., Fig. 5(a) and (b)), both interpolation methods yield comparable performance despite minor differences in configuration. Notably, the proposed method yields a simpler optimized configuration than the original method, particularly in the highly compressed region near the lower-left support (see Fig. 5(c) and (d)). This behavior can be explained by the stabilization mechanism described in Appendix D. The proposed method introduces fictitious strain energy that acts as a penalty against local distortions in intermediate-density regions. As a consequence, the optimizer favors simpler configurations that suppress distorted intermediate-density regions, thereby preventing premature mesh distortion during the optimization process. Meanwhile, as deformation increases, the original method suffers from numerical instability due to excessive distortion in intermediate density regions, leading to non-convergence for N–R iterations; see Fig. 5(f) and (h). In contrast, the proposed method maintains stability by scaling the displacement gradient $\nabla_{\boldsymbol{X}} \boldsymbol{u}$ in the intermediate density regions via Eq. (40), enabling convergence even under large deformations; refer to Fig. 5(e) and (g).

Furthermore, as the mechanical loading increases, the optimized structure changes from slender trusses to a solid vertical arm. This thicker design provides sufficient stiffness to resist large deformations and reduce local buckling. Fig. 6 shows the iteration histories for $F_{ext} = 2000 \times 10^3$ N and $F_{ext} = 4000 \times 10^3$ N cases. With kinematic scaling, the proposed method converges in fewer than 150 iterations. This indicates reasonable computational efficiency, with the largest mechanical loading case ($F_{ext} = 4000 \times 10^3$ N) completed in approximately 20 min on a standard desktop computer.

4.1.2. Verification of the thermoelastic TO considering the effective thermal conductivity tensor $\boldsymbol{k}_{def}$

Subsequently, we study the effect of including the effective thermal conductivity tensor $\boldsymbol{k}_{def}$ in Eq. (8). The thermal conductivity is set to $k = 25$ W/(m K), and the thermal expansion coefficient in Eq. (3) is set to $\alpha_{th} = 5 \times 10^{-2}$ K^{-1}. The external thermal loading is varied, while the mechanical loading is fixed at $F_{ext} = 2000 \times 10^3$ N.

The optimized configurations and their corresponding temperature fields are compared in Fig. 7. A distinct difference appears in the temperature distributions. In the classical weakly coupled approach, heat transfer is calculated based on the undeformed configuration, assuming a constant thermal conductivity. However, in the proposed strongly coupled framework, large deformations modify the effective thermal conductivity $\boldsymbol{k}_{def}$. This deformation-induced effective thermal conductivity leads to a noticeably higher temperature distribution due to the elongation of the heat conduction path, as shown in Fig. 7(c)–(f).

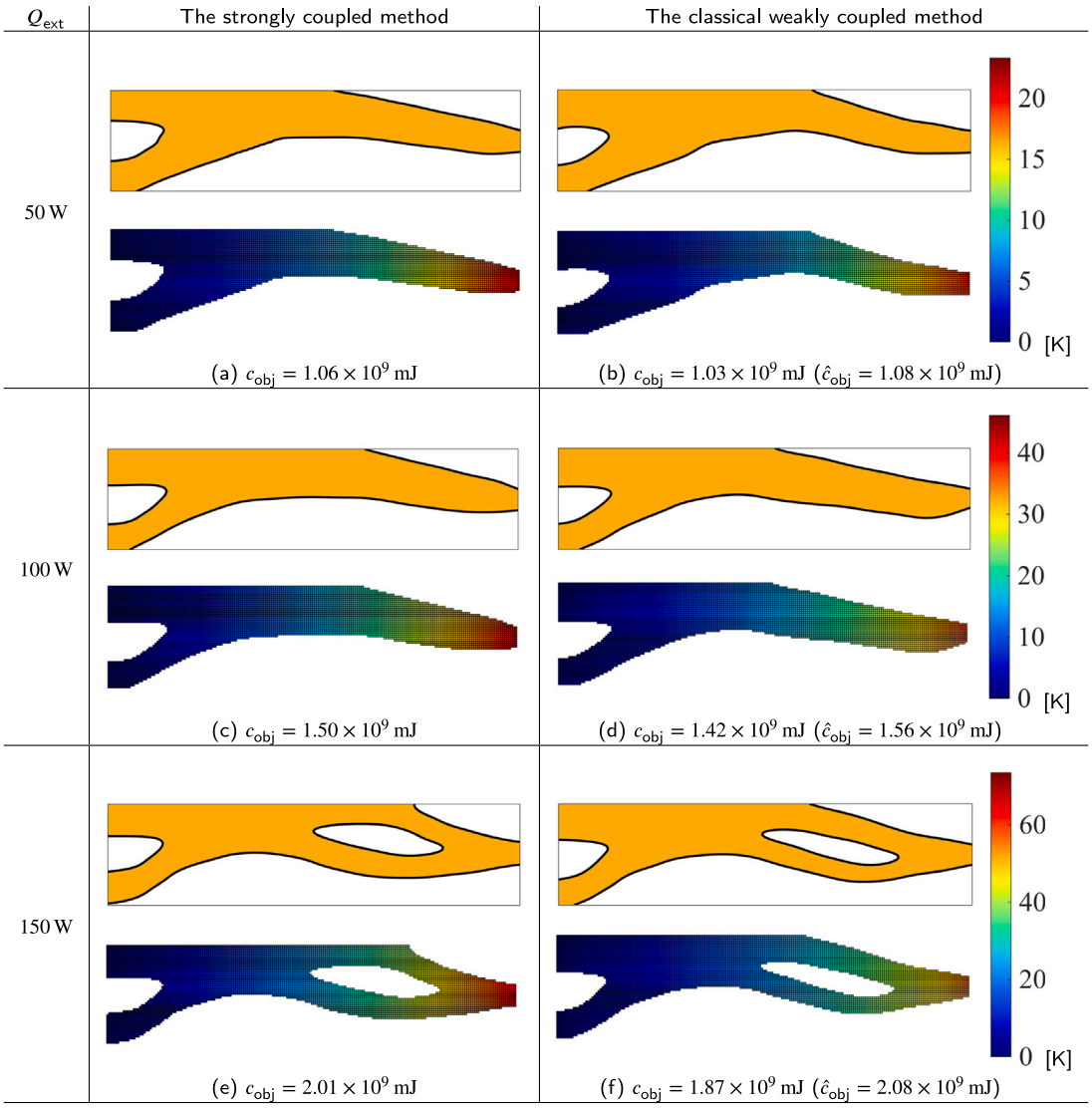


Fig. 7. Comparison of optimized results by the effective and constant thermal conductivity for Example 1. (The top and bottom images show the material and temperature distributions [K], respectively. $\hat{c}_{obj}$ is the objective function that is reevaluated using the strongly coupled method.)

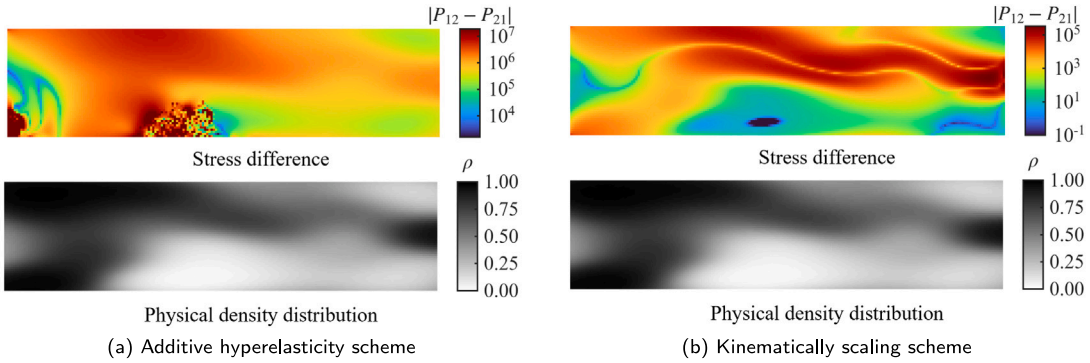


Fig. 8. Element-level stress and physical density distributions for Example 1: comparison with the additive hyperelasticity scheme under strong thermo-mechanical coupling.

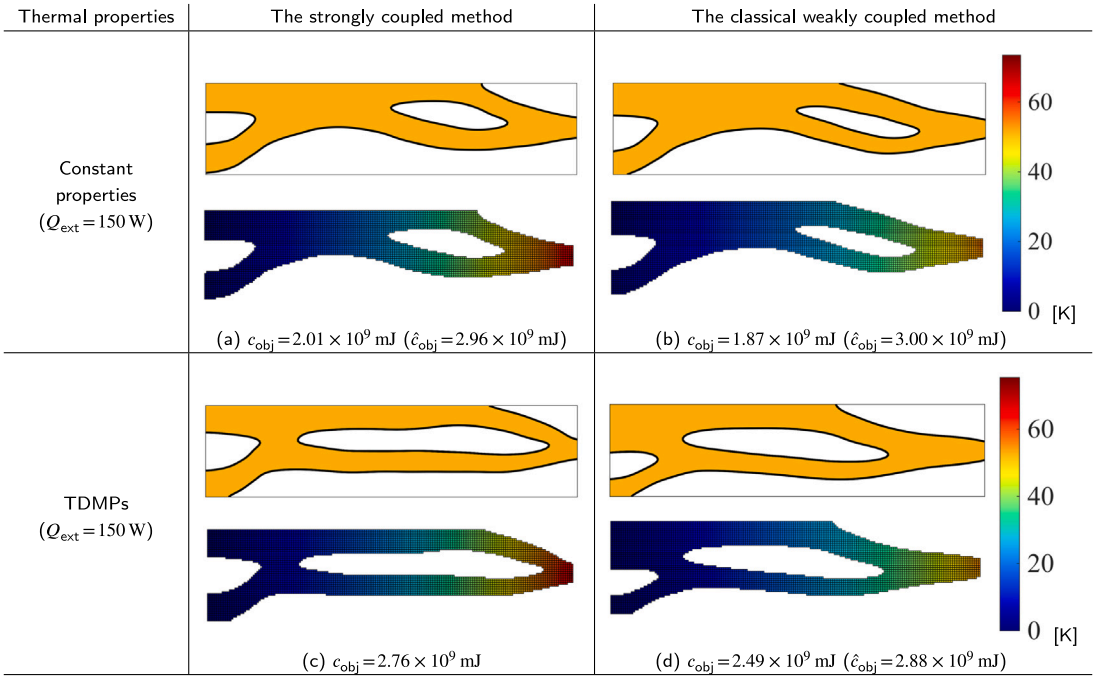


Fig. 9. Comparisons of optimized designs with constant material properties and temperature-dependent material properties (TDMPs) for Example 1. (The top and bottom images show the material and temperature distributions [K], respectively. $\hat{c}_{obj}$ is the objective function reevaluated using the strongly coupled method with TDMPs)

For the prescribed secant coefficient and loads, the strongly coupled calculation produces a higher temperature field and a different optimized geometry, including enlarged internal voids in Fig. 7(e). The corresponding weakly coupled design in Fig. 7(f) has smaller internal voids. When the weakly coupled designs are reevaluated using the same strongly coupled secant model, their end compliances are higher than those of the corresponding strongly coupled designs, as reported in Table 2. These results quantify the consequences of neglecting deformation-dependent heat conduction for this specified material law.

To further demonstrate the effectiveness of the kinematically scaled scheme, a comparison with the additive hyperelasticity scheme [42–44] is performed. The additive hyperelasticity scheme employs a shear-resistant artificial strain energy density function (e.g., the Yeoh model) to alleviate mesh distortion. However, under large deformations, it requires careful parameter tuning and may still develop numerical instabilities in intermediate-density regions. For the additive hyperelasticity scheme under $Q_{ext} = 100$ W, the non-linear solver diverges at a specific increment step within the 7th optimization iteration. Fig. 8 compares the element-level stress and physical density distributions of both methods at this identical increment step and iteration. The off-diagonal nominal stress difference, $|P_{12} - P_{21}|$, is used to visualize localized mesh distortions [57]. Severe stress concentrations and numerical instabilities are observed in the intermediate-density regions for the additive hyperelasticity scheme, whereas the proposed method remains stable and exhibits no excessive values of $|P_{12} - P_{21}|$.

4.1.3. *Verification of the thermoelastic TO considering the TDMPs assumption*

Building upon the previous section, we now incorporate temperature-dependent material properties (TDMPs) to demonstrate the role of TDMPs in thermoelastic coupling effects. For this cantilever beam example, the following material parameters are assumed to be temperature-dependent:

$$\begin{aligned} \text{Shear modulus:}\quad & \mu(T) = 5.36\times10^5 + 1\times10^3 T, \\ \text{Thermal conductivity:}\quad & k(T) = 25 + 0.01T, \\ \text{Thermal expansion coefficient:}\quad & \alpha_{th}(T) = \left(5 + 0.1\,T - 0.001\,T^2\right)\times10^{-2}. \end{aligned} \tag{51}$$

Here, the units for the shear modulus μ, thermal conductivity k, and thermal expansion coefficient α_{th} are given in MPa, W/(m K), and K^{-1}, respectively.

The results in Fig. 9 compare the selected constant-property and TDMP models and the corresponding strongly and weakly coupled analyses. For the prescribed secant TDMP functions, the strongly coupled design develops branched bars and enlarged internal voids. Its end compliance is 2.76×10^9 mJ, whereas the weakly coupled design, when reevaluated using the same strongly coupled TDMP model, gives 2.88×10^9 mJ. Similarly, the constant-property strongly coupled design gives 2.96×10^9 mJ when reevaluated using the same TDMP model. Thus, within this prescribed constitutive setting, neglecting either the deformation

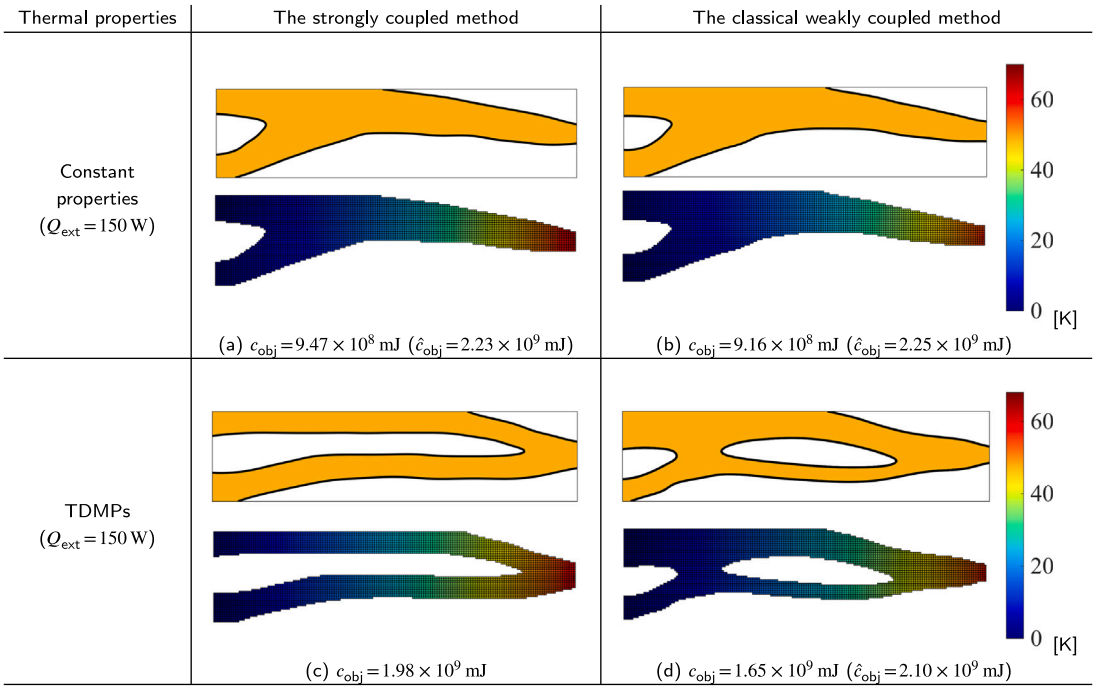


Fig. 10. Comparisons of optimized designs with constant material properties and with TDMPs for Example 1, under the independently prescribed exponential-integral thermal stretch in Eq. (53). (The top and bottom images show the material and temperature distributions [K], respectively. $\hat{c}_{obj}$ is the objective function reevaluated using the strongly coupled method with TDMPs.)

dependence of heat conduction or the selected temperature dependence of the material properties leads to differences in both the optimized layout and its reevaluated performance.

As an additional numerical illustration, the cantilever problem is solved using an independently prescribed instantaneous logarithmic expansion coefficient and the corresponding thermal stretch obtained by exponential integration. All other material parameters and the problem settings are kept unchanged. This case is intended to demonstrate the applicability of the proposed framework to this alternative thermal-expansion law. Because both the magnitude and the temperature dependence of the expansion coefficient are altered, the results should be interpreted as an algorithmic demonstration rather than as a matched-material comparison with Eq. (51). The prescribed instantaneous coefficient is

$$\alpha_{\mathrm{th}}^{\mathrm{ins}}(T)=\left(1+0.07\,T-1\times10^{-4}\,T^{2}\right)\times10^{-2}\,\mathrm{K}^{-1}, \tag{52}$$

and the associated thermal stretch follows

$$\vartheta=\exp\left(\int_{T_{\mathrm{ref}}}^{T}\alpha_{\mathrm{th}}^{\mathrm{ins}}(\tau)\,\mathrm{d}\tau\right)=\exp\left[\left(T+0.035\,T^{2}-\frac{10^{-4}}{3}\,T^{3}\right)\times10^{-2}\right]. \tag{53}$$

The instantaneous coefficient in Eq. (52) is an independently chosen illustrative function, which is not a conversion of the secant coefficient in Eq. (51). For the corresponding constant-property calculation, the value at the reference temperature is adopted, $\alpha_{\mathrm{th}}^{\mathrm{ins}}=1\times10^{-2}\,\mathrm{K}^{-1}$. At the comparison temperature change $\Delta T=70$ K, the constant-property exponential stretch is 2.014, compared with 4.500 for the constant-property secant law. For the TDMP case at the same temperature change, the exponential stretch in Eq. (53) is 9.981, compared with 5.970 for the secant law in Eq. (51). These values are obtained by evaluating the prescribed functions at the same temperature and confirm that the two cases employ different thermal-expansion laws.

The designs in Fig. 10 exhibit differences between strongly and weakly coupled optimization and between the selected constant-property and TDMP models. Similar qualitative differences are also observed in the secant example in Fig. 9. This agreement of trends is an observation for two prescribed constitutive settings. Because the thermal stretch affects deformation, deformation-dependent conductivity, and the temperature field, the resulting responses and layouts may depend on the selected thermal-expansion law.

All reevaluated objective values $\hat{c}_{obj}$ in Fig. 10 are computed using the strongly coupled exponential model with TDMPs. Within this constitutive setting, the weakly coupled TDMP design has an end compliance that is 6.1% higher than that of the strongly coupled TDMP design (2.10×10^{9} versus 1.98×10^{9} mJ). Similarly, the constant-property strongly coupled design has an end compliance 12.6% higher than that of the strongly coupled TDMP design when reevaluated using the same TDMP model (2.23×10^{9} versus 1.98×10^{9} mJ). The corresponding differences for the secant example in Fig. 9 are 4.3% and 7.2%, respectively. These percentages quantify differences among designs within each constitutive setting and should not be interpreted as measures of the sensitivity of

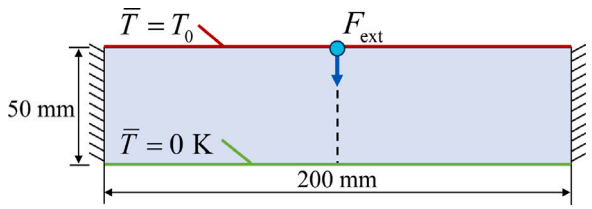


Fig. 11. Problem setup for Example 2: 2D clamped beam.

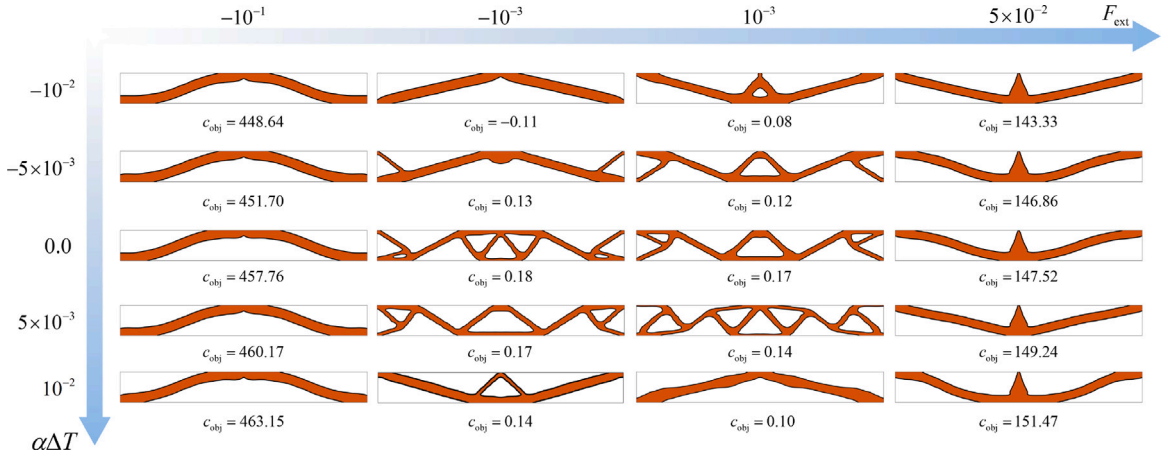


Fig. 12. Optimized results without TDMPs for Example 2: under a uniform temperature distribution.

the temperature fields, objective values, or optimized layouts to the choice between the secant and exponential thermal-expansion laws. The secant definition is retained throughout the remainder of this work.

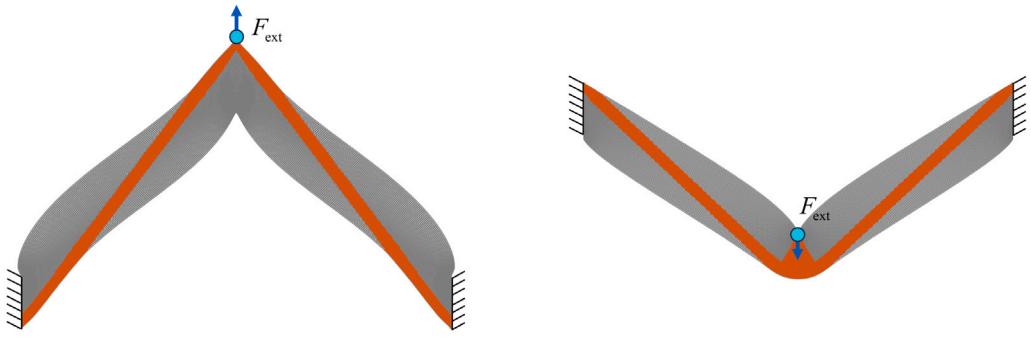
4.2. Example 2: 2D clamped beam

Following the initial investigation, the 2D clamped beam example in Fig. 11 is used to demonstrate the capability of the proposed method in handling multi-material design with TDMPs. As depicted in the figure, the design domain is a $200\times50\,\text{mm}^2$ rectangle, which is fully clamped at the two ends ($\bar{\boldsymbol{u}} = \boldsymbol{0}$). A downward external loading $\boldsymbol{F}_{\text{ext}}$ is applied at the center of the top edge. Simultaneously, a prescribed thermal distribution is imposed; the top surface is maintained at a constant temperature $\bar{T} = T_0$, and the bottom surface is fixed at $\bar{T} = 0\,\text{K}$, creating a vertical temperature gradient across the design domain. For all sub-cases in this example, the design domain is also discretized into 200×50 elements. The total mass constraint $f_{\max}$ is set to 0.3. The investigation in this example is structured into four parts as follows:

1. A one-material optimization under a uniform temperature distribution is performed to compare with the reference solution [49];
2. A two-material optimization without the TDMPs assumption is conducted to establish a baseline configuration;
3. The TDMPs assumption is introduced into the two-material design to highlight its importance;
4. The optimization problem is extended to a three-material design under the TDMPs assumption.

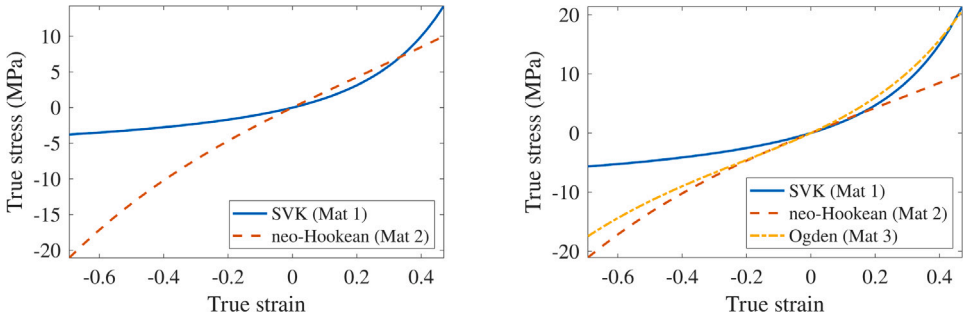
4.2.1. One-material optimized result with uniform temperature distribution

Before exploring the optimization with TDMPs and multi-material layouts, we verify that the proposed method reproduces results comparable to those reported in the existing literature, for which the neo-Hookean constitutive model in Eq. (25) is adopted. Following the dimensionless setting of Chung et al. [49], the Lamé parameters are taken as $\mu = 0.4$ and $\lambda = 1$, and a slender bi-clamped beam of dimensionless size 8×1 is considered. Accordingly, the external load F_{ext} and the end compliance c_{obj} in this comparison case are reported as dimensionless quantities, consistent with Chung et al. [49]. As shown in Fig. 11, a downward concentrated mechanical load F_{ext} is applied at the center of the bottom edge, and a uniform thermal stretch $\alpha\Delta T$ is prescribed across the entire domain. For this prescribed-temperature case, the heat conduction solver and TDMPs are deactivated. By exploiting symmetry, only half of the design domain is considered, discretized using 200×50 elements, and the total mass constraint is set to 0.3.



(a) $F_{ext} = -10^{-1}$, and $\alpha\Delta T = -10^{-2}$ (b) $F_{ext} = 5 \times 10^{-2}$, and $\alpha\Delta T = -10^{-2}$

Fig. 13. Deformed shapes for Example 2: under a uniform temperature distribution and larger external forces.



(a) Two models: SVK and neo-Hookean (b) Three models: SVK, neo-Hookean, and Ogden

Fig. 14. Nonlinear material responses for Example 2: true stress-true strain relationships of the constituent materials under uniaxial tension.

The optimized configurations under pure mechanical loading and combined thermoelastic loading are compared in Fig. 12. Consistent with the results reported in [49], under a purely mechanical downward load ($F_{ext} = 10^{-3}$ and $\alpha\Delta T = 0$), the optimized structure forms a truss-like layout that maximizes resistance against bending. When a uniform temperature increase ($\alpha\Delta T > 0$) is applied simultaneously, the optimized configuration adapts markedly, evolving toward an inverted V-shape. This geometric adaptation uses thermally induced upward deformation to counteract the downward mechanical force. Reproducing this classical thermo-mechanical counteraction confirms that the proposed formulation, together with the kinematically scaled interpolation scheme, produces physically reasonable designs.

Furthermore, the deformed configurations of the optimized structures under $F_{ext} = -10^{-1}$ and 5×10^{-2} are shown in Fig. 13. These results demonstrate that the proposed interpolation scheme remains stable under large structural deformations. Since the pull-back operation of the effective thermal conductivity tensor in Eq. (8) depends on the deformation magnitude, the thermo-mechanical coupling becomes increasingly pronounced as the deformation increases. Consequently, larger deformations lead to more noticeable differences between the strongly and weakly coupled analyses (refer to Section 4.1.3), which in turn alter the temperature-dependent material properties and the resulting optimized layouts.

4.2.2. *Two-material optimized result without the TDMPs assumption*

Subsequently, the modified SVK and compressible neo-Hookean constitutive models are considered for the design domain in Fig. 11. Their material parameters are assigned as Lamé parameters $\mu = 5.77$ MPa and $\lambda = 8.65$ MPa for SVK, and bulk modulus $K = 16.67$ MPa and shear modulus $G = 7.69$ MPa for neo-Hookean. The thermal conductivities are set to $k_1 = 1$ W/(m K) for SVK and $k_2 = 1$ W/(m K) for neo-Hookean, and the thermal expansion coefficients are set to $\alpha_{th,1} = 2\times10^{-4}$ K^{-1} for SVK and $\alpha_{th,2} = 1\times10^{-4}$ K^{-1} for neo-Hookean. To elucidate the distinct material responses, the true stress-true strain relationships of the two materials are shown in Fig. 14(a). The comparison reveals a stiffness crossover: the neo-Hookean material is stiffer under compression and small tensile strains, whereas the SVK material exhibits pronounced strain hardening at large strains. This suggests that the optimizer preferentially assigns the SVK material to regions experiencing large tensile strains.

The optimized results under different external mechanical loads F_{ext} and thermal conditions T_0 are presented in Fig. 15. It is evident that at higher temperatures, the structure undergoes significant thermal expansion. Furthermore, as the loading increases, the topological configuration evolves from two bars to a sharp-angle frame. To verify the physics-driven material layout, the spatial distributions of the equivalent stress and the mean stress for the optimized structures in Fig. 15(d) and (g) are provided in Fig. 16. The equivalent stress field demonstrates a smooth and efficient load-transfer configuration. The mean stress field with the material

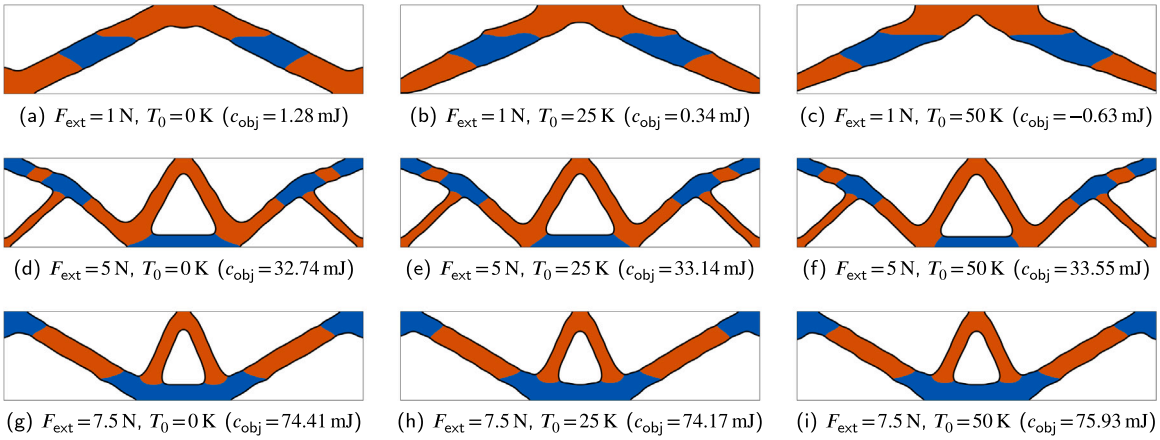


Fig. 15. Optimized results without TDMPs for Example 2. (The blue- and orange-colored materials correspond to the SVK and neo-Hookean.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

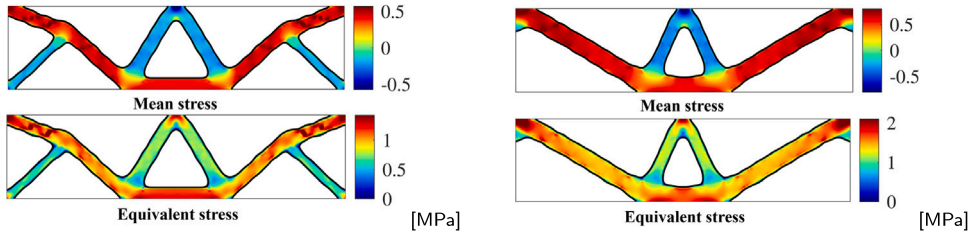


Fig. 16. Spatial distribution of the mean stress and equivalent stress for the optimized structure.

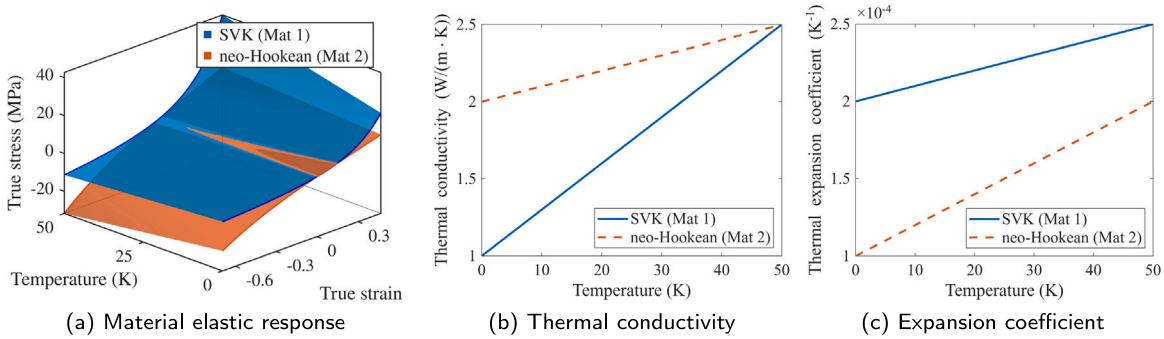


Fig. 17. TDMPs functions for Example 2: two-material optimization.

distribution reveals that the SVK material (marked in blue in Fig. 15) is allocated to the regions subjected to severe tension. This evidence confirms that the proposed method exploits the pronounced strain-stiffening property of the SVK material under large tension to maximize the overall structural load-bearing capacity. On the other hand, since the TDMPs are disregarded in this case, the material parameters remain constant. Consequently, under low mechanical loading conditions ($F_{ext}=1$ N; refer to Fig. 15(a–c)), the structural response is mainly dominated by thermal expansion. Notably, at $T_0=50$ K, thermal expansion significantly counteracts the external mechanical load, resulting in a negative objective function ($c_{obj}=-0.63$ mJ). This indicates that the thermally induced deformation acts in the opposite direction to the applied downward force F_{ext}. Conversely, under high loading conditions ($F_{ext}=5$ N and 7.5 N; refer to Fig. 15(d–i)), the structure becomes mechanically dominated. Hence, the differences in both the objective function and topological configuration across different temperatures become negligible.

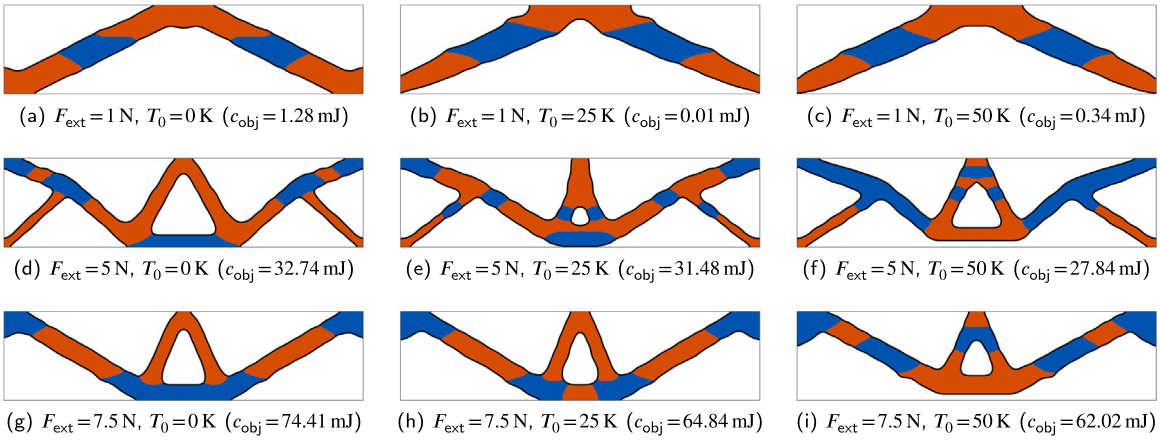


Fig. 18. Optimized results with TDMPs for Example 2: two-material optimization. (The blue- and orange-colored materials correspond to the SVK and neo-Hookean.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

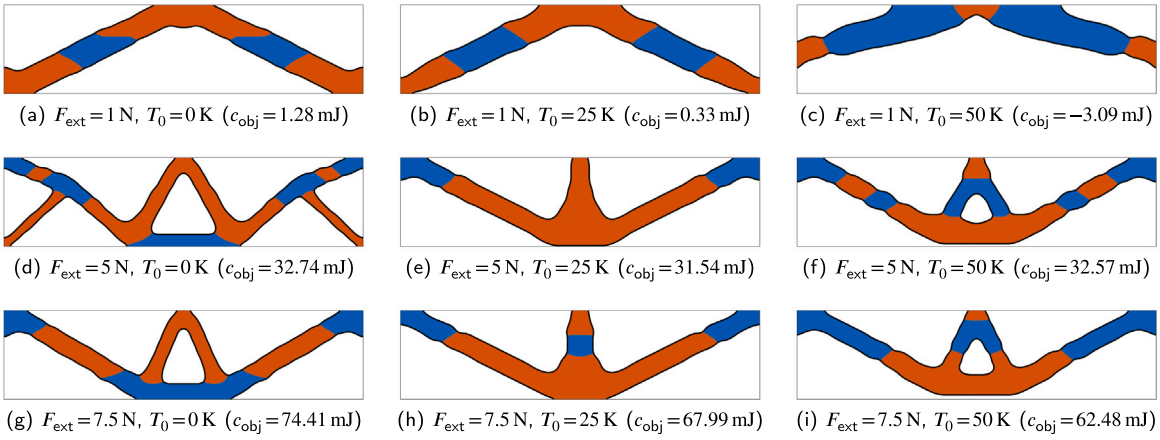


Fig. 19. Optimized results with different TDMPs for Example 2: two-material optimization. (The blue- and orange-colored materials correspond to the SVK and neo-Hookean.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2.3. *Two-material optimized result under the TDMPs assumption*

Incorporating temperature-dependent constitutive behavior leads to a more comprehensive optimization framework that better reflects realistic physical conditions. Accordingly, the TDMP functions for the different material models are implemented as shown in Fig. 17 and are defined as follows:

$$\begin{aligned}
&\text{Hyperelastic material parameter:} \begin{cases} \mu_1(T) = \dfrac{3}{26}T + \dfrac{75}{13}, \quad \lambda_1(T) = \dfrac{9}{52}T + \dfrac{225}{26}, \\ K_2(T) = \dfrac{1}{6}T + \dfrac{50}{3}, \quad G_2(T) = \dfrac{1}{13}T + \dfrac{100}{13}, \end{cases} \\
&\text{Thermal conductivity}: \begin{cases} k_1(T) = 0.03T + 1, \\ k_2(T) = 0.01T + 2, \end{cases} \\
&\text{Thermal expansion coefficient:} \begin{cases} \alpha_{th,1}(T) = (0.01T + 2) \times 10^{-4}, \\ \alpha_{th,2}(T) = (0.02T + 1) \times 10^{-4}. \end{cases}
\end{aligned} \tag{54}$$

The hyperelastic material parameters $(\mu_1, \lambda_1, K_2, G_2)$ are expressed in MPa, the thermal conductivities (k_1, k_2) in W/(m K), and the thermal expansion coefficients $(\alpha_{th,1}, \alpha_{th,2})$ in K^{-1}. At the reference temperature ($T_{ref} = 0\,K$), the material properties recover the baseline values used in the temperature-independent case (Section 4.2.2). As the temperature increases, the material parameters evolve according to the TDMP functions shown in Fig. 17. Consequently, the true stress becomes a function of both the true strain

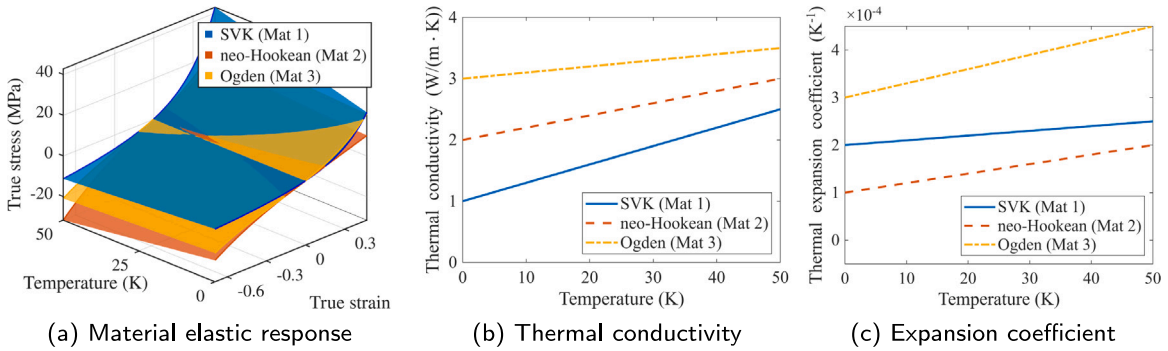


Fig. 20. TDMPs functions for Example 2: three-material optimization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

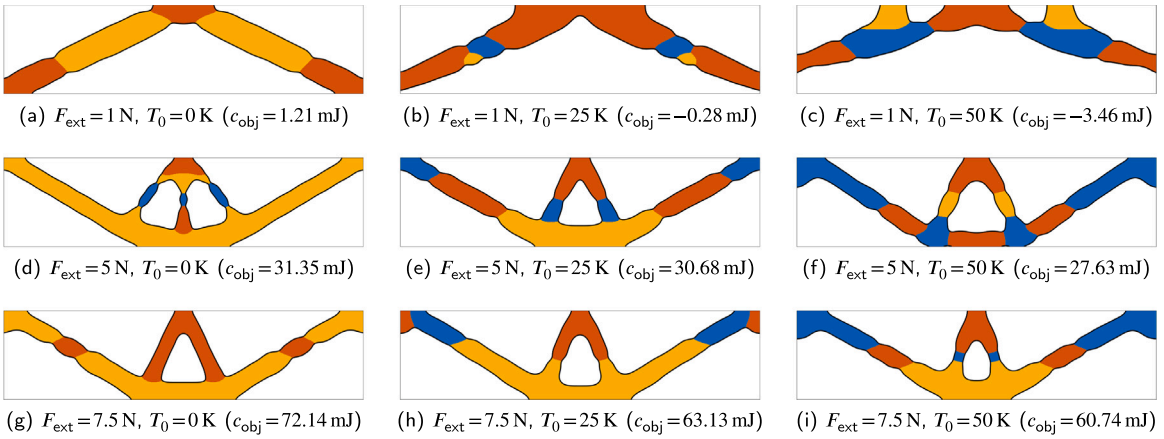


Fig. 21. Optimized results with TDMPs for Example 2: three-material optimization. (The blue-, orange-, and yellow-colored materials correspond to the SVK, neo-Hookean, and Ogden.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and the temperature. To illustrate this dependency, the three-dimensional constitutive surfaces of the two candidate materials are presented in Fig. 17(a). Notably, under these constitutive descriptions, the materials exhibit an intricate stiffness crossover behavior.

The optimized results under different loads and temperatures are shown in Fig. 18. For the baseline thermal condition ($T_0 = 0$ K), the optimized results in Fig. 18(a), (d), and (g) coincide with those in Section 4.2.2. As the applied temperature increases, the structural configuration tends to detach from the bottom edge; since the bottom edge acts as a heat sink ($\bar{T} = 0$ K), this detachment effectively provides thermal insulation. By reducing heat dissipation, the structure retains thermal energy from the top surface, maintaining a higher internal temperature and thereby enhancing TDMP-driven thermal expansion against the downward mechanical load. However, when both mechanical loading and temperature increase, as shown in Fig. 18(h), the material is reallocated to the bottom edge to improve structural stiffness against pronounced bending. This trade-off between thermal insulation and mechanical support highlights the importance of incorporating TDMPs in strongly coupled nonlinear thermoelastic design.

To evaluate the sensitivity of the optimized configurations to the specific TDMP functions, an additional numerical study was conducted by modifying the thermal conductivity function of the neo-Hookean material. Specifically, the slope of $k_2(T)$ is doubled from 0.01 to 0.02, yielding $k_2(T) = 0.02T + 2$, thereby enhancing its heat transfer capability at elevated temperatures. The optimized results are shown in Fig. 19. Compared with the original TDMP setting, this modification of $k_2(T)$ changes both the objective function value and the distribution of the neo-Hookean material, highlighting the importance of the TDMP functions in strongly coupled thermoelastic TO. Meanwhile, despite the significant change in thermal conductivity, the geometric self-adaptation mechanism remains active in the optimized designs, as evidenced by the boundary detachment associated with thermal insulation observed in Fig. 19(f) and (i).

4.2.4. Three-material optimized result under the TDMPs assumption

Subsequently, all three constitutive models (the SVK, neo-Hookean, and Ogden materials) are incorporated into the optimization framework. The TDMP functions associated with the Ogden constitutive model (Mat 3, marked in yellow) are shown in Fig. 20 and

Table 3
Comparison of objective function values (c_{obj} mJ) between two-material and three-material optimizations for Example 2.

Load F_{ext} (N)	Temperature T_0 (K)	Two-material (Fig. 19)	Three-material (Fig. 21)
1	0	1.28	1.21
	25	0.33	−0.28
	50	−3.09	−3.46
5	0	32.74	31.35
	25	31.54	30.68
	50	32.57	27.63
7.5	0	74.41	72.14
	25	67.99	63.13
	50	62.48	60.74

Table 4
End-compliance reevaluation (c_{obj} in mJ) of the fixed three-material designs in Example 2 using the prescribed secant law and an alternative exponential-integral law with the same numerical coefficient functions.

Load F_{ext} (N)	Temperature T_0 (K)	Linear stretch	Exponential-integral stretch	Performance deviation
1	0	1.2124	1.2124	–
	25	−0.2787	−0.2722	2.332%
	50	−3.4606	−3.4430	0.509%
5	0	31.3510	31.3510	–
	25	30.6802	30.6163	0.208%
	50	27.6316	27.5031	0.465%
7.5	0	72.1433	72.1433	–
	25	63.1343	63.0453	0.141%
	50	60.7438	60.4836	0.428%

are defined as follows:

$$\begin{aligned} \text{Hyperelastic material parameter:}\quad & \mu_3(T) = 0.033T + 4.81, \\ \text{Thermal conductivity:}\quad & k_3(T) = 0.01T + 3, \\ \text{Thermal expansion coefficient:}\quad & \alpha_{\text{th},3}(T) = (0.03T + 3) \times 10^{-4}. \end{aligned} \tag{55}$$

Similar to the previous definitions, the units of μ_3, k_3, and $\alpha_{\text{th},3}$ are MPa, W/(m K), and K^{-1}, respectively, whereas the Ogden parameters α_3 and β_3 are dimensionless.

Introducing a third material expands the design space while also increasing the complexity of the optimization problem; the optimized results are shown in Fig. 21. The TDMP functions for the other two materials (the SVK and neo-Hookean) are the same as in Fig. 19. To enable a clearer comparison of structural performance, the objective function values (c_{obj}) for the two-material (Fig. 19) and three-material (Fig. 21) cases are summarized in Table 3. As observed, the three-material design exhibits improved structural performance compared to the two-material case. This enhancement is primarily attributed to the expanded design space provided by the newly introduced third material, whose specific parameters prove favorable under the given thermo-mechanical loading conditions. Furthermore, the optimized results exhibit a distinct material distribution, reflecting the ability of the proposed method to exploit the specific advantages of each material.

To elucidate the underlying physical mechanism for the material selection, Fig. 14(b) illustrates the non-linear constitutive responses of the three candidate materials at the reference temperature ($T = 0\,\text{K}$), plotted as true stress versus true strain. As observed, at $T = 0\,\text{K}$, the Ogden model exhibits higher stiffness than the other materials under tension strain. Hence, the Ogden material is largely distributed in the design domain at the lower temperature, as shown in Fig. 21(a), (d), and (g). As the temperature and strain increase, the other two materials gradually appear in the optimized results. This observation further highlights the necessity of tailoring structural configurations to specific application scenarios, as simplistic assumptions may compromise structural performance.

For the three-material designs in Fig. 21, the end compliance is reevaluated using $\vartheta = \exp\left(\int_{T_{\text{ref}}}^{T} \alpha_{\text{th}}^{\text{ins}}(\tau)\,\mathrm{d}\tau\right)$, with the same temperature-dependent coefficient functions prescribed as $\alpha_{\text{th}}^{\text{ins}} = \alpha_{\text{th}}$ for the purpose of this comparison. The designs are held fixed during the reevaluation. Under the thermo-mechanical loading conditions in Table 4, the largest relative difference in end compliance is 2.332%. This maximum occurs at $F_{\text{ext}} = 1\,\text{N}$ and $T_0 = 25\,\text{K}$, with an absolute difference of 0.0065 mJ (−0.2787 versus −0.2722 mJ). The relatively large percentage difference results from the small magnitude of the reference end compliance. This comparison is limited to the end-compliance differences of the fixed designs under the tested loading conditions and does not assess the influence of alternative thermal-expansion laws on the resulting optimized designs.

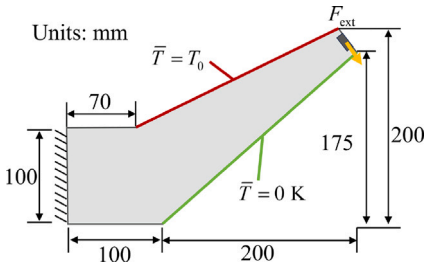


Fig. 22. Problem setup for Example 3: 2D antenna support bracket. (The arrow represents the "downward" mechanical loading.)

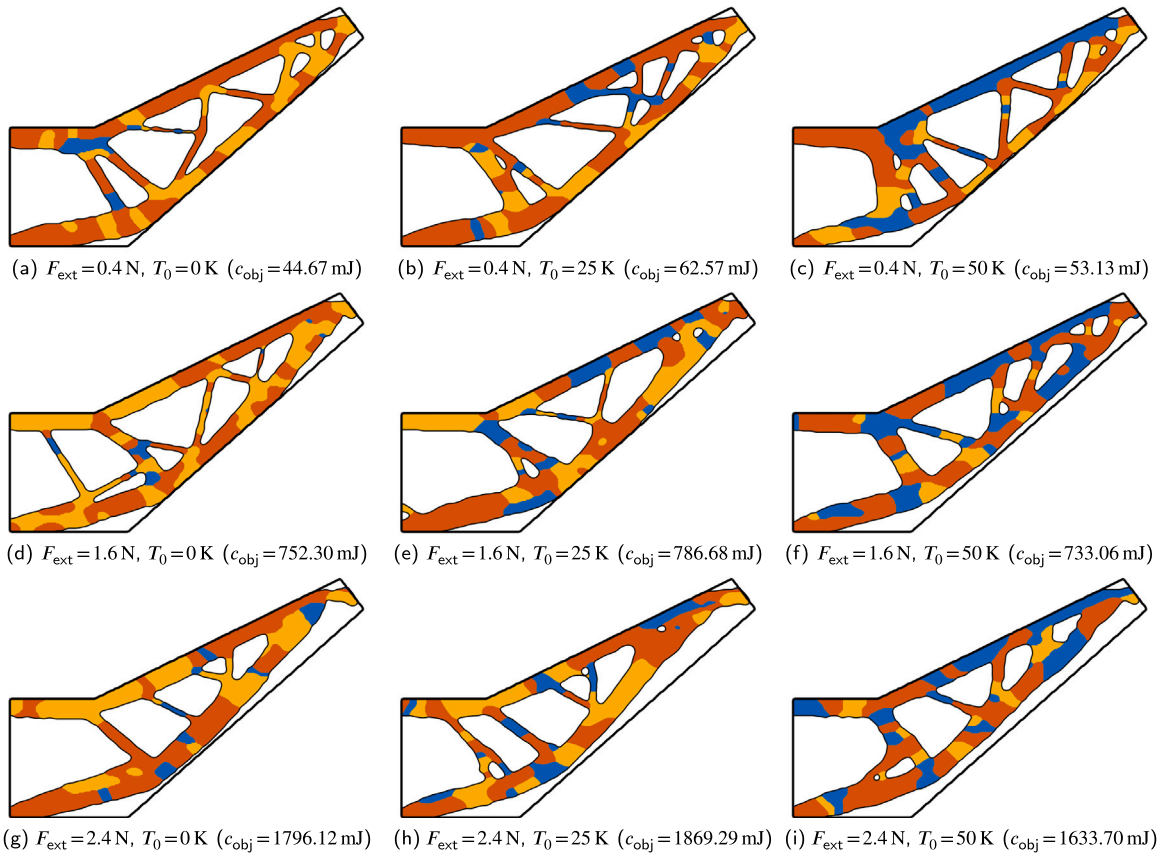


Fig. 23. Optimized results with TDMPs for Example 3: downward mechanical loading. (The blue-, orange-, and yellow-colored materials correspond to the SVK, neo-Hookean, and Ogden.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. Example 3: 2D antenna support bracket

To demonstrate the effectiveness of the proposed method in solving practical engineering problems, the third numerical example considers the design of a 2D antenna support bracket [58], as illustrated in Fig. 22. The design domain has an irregular geometry with overall dimensions of $300\times200\,\text{mm}^2$, consisting of a fixed rectangular base and a tapered cantilever arm. The left vertical edge is fully clamped ($\bar{\boldsymbol{u}} = \boldsymbol{0}$), and a concentrated external loading $\boldsymbol{F}_{\text{ext}}$ is applied at the upper-right tip, acting parallel to the loading surface to induce rotation. Also, a Dirichlet boundary condition T_0 is prescribed on the upper inclined edge to simulate external radiation, while the lower inclined edge is maintained at $\bar{T} = 0\,\text{K}$, thereby creating a thermal gradient. The design domain is discretized using a structured mesh of 25,016 elements, and the total mass constraint $f_{\max}$ is set to 0.5. In this example, the candidate materials and their TDMP functions are consistent with those considered in Example 2. Additionally, the total stored strain energy Π_{obj} defined in Eq. (50) is considered for optimization.

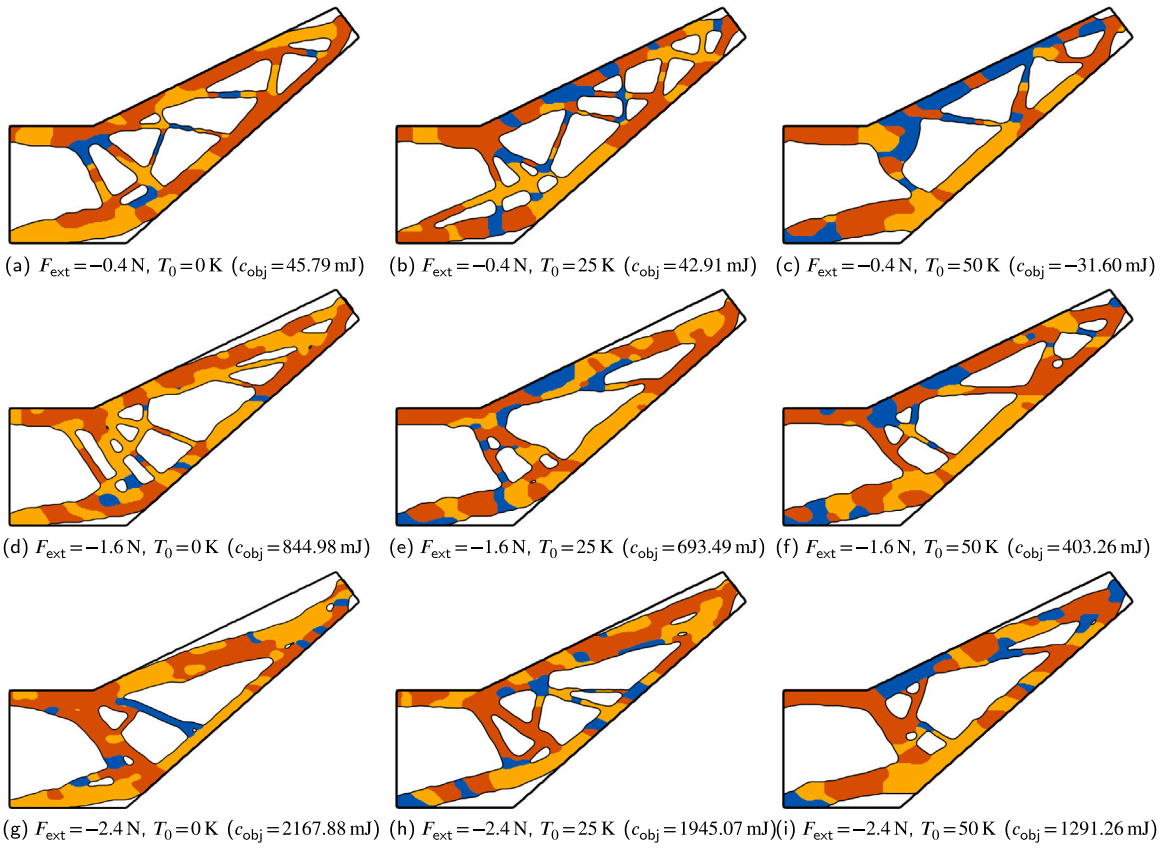


Fig. 24. Optimized results with TDMPs for Example 3: upward mechanical loading. (The blue-, orange-, and yellow-colored materials correspond to the SVK, neo-Hookean, and Ogden.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3.1. Optimized result for the downward mechanical loading

Unlike linear elasticity, where reversing the loading direction merely changes the sign of the displacement field without altering the optimized topology, the finite strain kinematics considered in this work introduce a clear dependency on the loading direction. To investigate this asymmetry, this example first examines the design under a downward mechanical loading ($F_{ext} = 0.4$ N, 1.6 N, and 2.4 N). The structural performance is evaluated under three distinct thermal gradients, with the optimized results summarized in Fig. 23.

As observed in the figure, the configurations develop downward-curved profiles to efficiently counteract the bending moment. Consistent with the physical insights gained from Example 2 (2D clamped beam in Section 4.2), a geometric self-adaptation characterized by a boundary detachment is observed under high thermal gradient (Fig. 23(f) and (i) at $T_0 = 50$ K). To fully leverage thermal expansion against the mechanical downward load, the structural configuration is detached from the lower heat sink. This boundary detachment not only allows the configuration to fully leverage thermal expansion against the downward mechanical load, but also triggers the stiffness enhancement provided by the TDMPs function. Furthermore, the three constituent materials are distributed according to their thermo-mechanical properties, demonstrating the capability of the proposed method to generate complex multi-material layouts in irregular design domains.

Notably, under severe thermo-mechanical loading, the optimized layouts exhibit isolated material islands. Since the temperature field is not predefined but evolves in response to structural deformation and material redistribution, local material selection is influenced by the complex crossover behaviors of the TDMPs (as previously shown in Fig. 20). Consequently, the optimizer assigns specific materials to localized regions to adapt to the evolving thermo-mechanical environment, leading to these complex configurations. As evidenced by the results under smaller thermal gradients and external loads in Fig. 23(a), the extent of this material fragmentation is substantially reduced, indicating that these isolated islands are physically driven responses to severe thermo-mechanical conditions.

4.3.2. Optimized result for the upward mechanical loading

Subsequently, the loading direction is reversed upward while all other optimization parameters are kept identical; the optimized configurations are presented in Fig. 24. It is worth noting that hyperelastic materials exhibit different material responses in tension

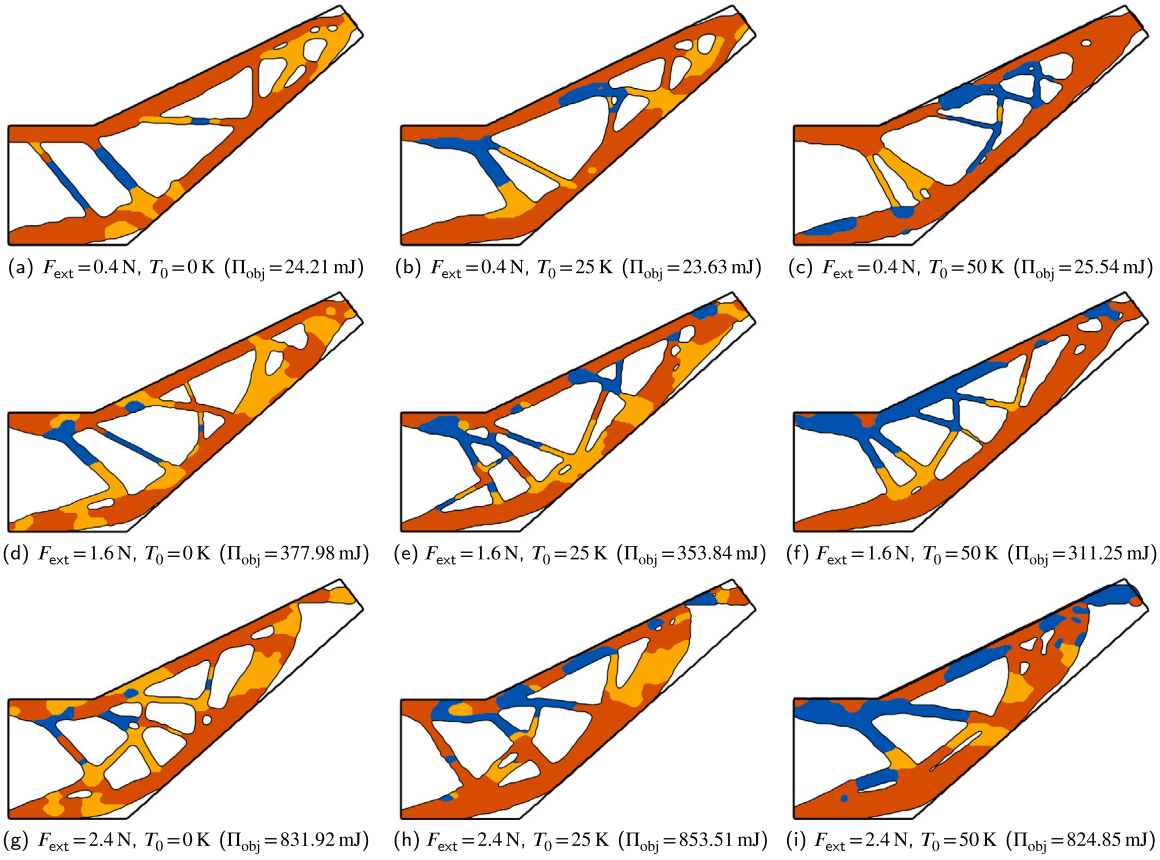


Fig. 25. Optimized results minimizing the total stored strain energy (Π_{obj}) for Example 3: downward mechanical loading. (The blue-, orange-, and yellow-colored materials correspond to the SVK, neo-Hookean, and Ogden.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and compression. Specifically, a downward loading induces severe compression near the bottom root of the bracket, whereas an upward loading subjects the same region to significant tension.

Consequently, this fundamental kinematic asymmetry, coupled with the TDMPs, leads to topological configurations that differ entirely from those in the downward loading cases; compare Fig. 24 with Fig. 23. Driven by the imposed thermal gradient and the upward force, the optimized structures fully conform to the upper thermal boundary. These configurations exploit the elevated temperature to activate the TDMPs locally, generating a pronounced upward-curved shape to minimize the objective function. This geometric self-adaptation and material redistribution demonstrates that the proposed framework can harness complex nonlinear thermoelastic behavior to achieve superior structural designs that linear TO would fail to predict.

4.3.3. Optimized result under the total stored strain energy minimization

To demonstrate the versatility of the proposed framework and the distinct physical responses induced by different thermo-mechanical objective functions, we additionally solve optimization problems to minimize the total stored strain energy. Here, the structure is subjected to the same downward mechanical loads and thermal gradients as in Section 4.3.1. The preceding examples minimized the structural end compliance to exploit thermally induced deformation against the external load. In this case, the total stored strain energy Π_{obj} in Eq. (50) is minimized instead to suppress the internal elastic energy generated by the combined thermo-mechanical loading.

The resulting topologies obtained by minimizing the total stored strain energy are shown in Fig. 25. The optimized layouts exhibit distinct geometric adaptations under the same thermo-mechanical conditions. Under pure mechanical loading ($T_0 = 0\,K$), the layouts in Fig. 25(a), (d), and (g) resemble their end-compliance counterparts in Fig. 23 because both objectives favor mechanically stiff designs. As T_0 increases, however, the two objectives produce distinct responses. End-compliance minimization promotes thermal expansion that counteracts the mechanical load, as shown in Fig. 23(f) and (i), whereas stored-energy minimization reduces the thermally induced internal stress. Consequently, the designs in Fig. 25(c), (f), and (i) connect to the thermal boundary to limit strain–energy accumulation and exhibit different material distributions. These results demonstrate the applicability of the proposed MMTO framework to different thermo-mechanical objective functions.

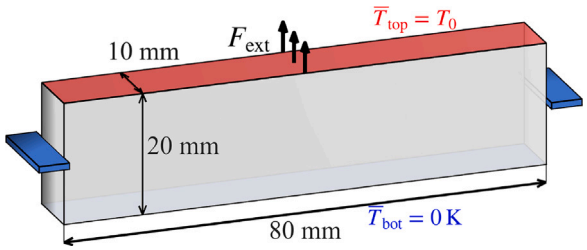


Fig. 26. Problem setup for Example 4: 3D clamped beam.

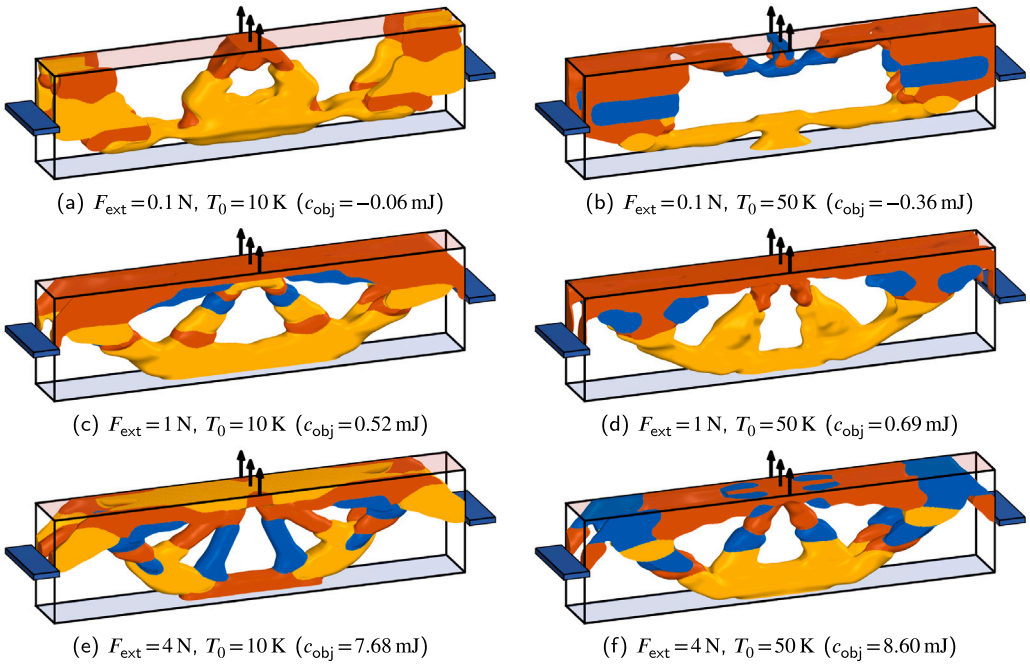


Fig. 27. Optimized results with TDMPs for Example 4. (The blue-, orange-, and yellow-colored materials correspond to the SVK, neo-Hookean, and Ogden.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

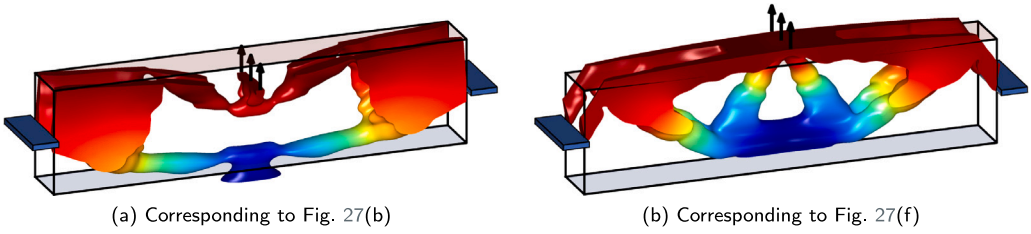


Fig. 28. Deformed configurations along with temperature distribution for Example 4.

It is worth noting that, when the external mechanical load is small, minimizing the end compliance $c_{\text{obj}} = \boldsymbol{F}_{\text{ext}}^{\text{T}}\boldsymbol{U}$ in a thermoelastic setting produces a thermally driven actuator-like design [39]. In TO of compliant mechanisms and multiphysics actuators [59–61], an artificial output spring or dummy load is commonly introduced to maintain structural connectivity and define the actuation direction. Here, the small design-independent mechanical load serves the same purpose. Minimizing the end compliance therefore drives the structure to exploit thermal expansion against the applied load, maximizing the output displacement opposite to $\boldsymbol{F}_{\text{ext}}$ and yielding negative compliance, as also reported by Goh et al. [39].

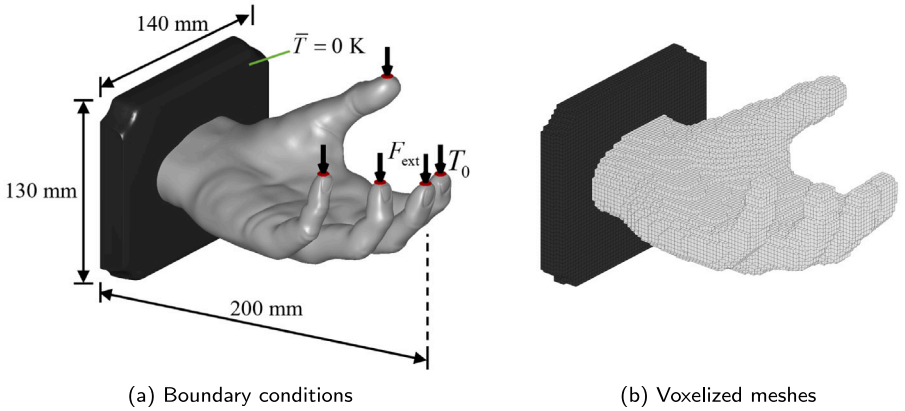


Fig. 29. Problem setup for Example 5: 3D hand gripper. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.4. Example 4: 3D clamped beam

The fourth numerical example considers the design of a 3D clamped beam, as depicted in Fig. 26. The design domain is defined as a rectangular block with dimensions of $80\times20\times10\,\text{mm}^3$, and is discretized into a structured hexahedral mesh of $96 \times 24 \times 12$ elements. The employed material properties are consistent with those used in Example 2 and Example 3. This structure is fully fixed along the central lines at both the left and right ends ($\bar{\boldsymbol{u}} = \mathbf{0}$), and an external vertically distributed loading F_{ext} is applied along the longitudinal centerline of the top surface. Also, Dirichlet boundary conditions are imposed to establish a vertical temperature gradient: the top surface is prescribed a constant temperature $\bar{T}_{\text{top}} = T_0$, while the bottom surface is maintained at a lower temperature $\bar{T}_{\text{bot}} = 0\,\text{K}$, with all remaining surfaces assumed to be adiabatic. The total mass constraint $f_{\max}$ is set to 0.25.

The optimized configurations are illustrated in Fig. 27. As shown in Fig. 27(a) and (b), in the low mechanical loading case ($F_{\text{ext}} = 0.1\,\text{N}$), the optimization yields a bottom-heavy structure featuring four auxiliary pillars that connect the clamped supports to the top surface. These pillars effectively work as built-in thermal actuators. By harnessing thermal expansion induced by the elevated top temperature ($\bar{T}_{\text{top}}$), they drive a downward structural deformation that actively counteracts the external mechanical loading. Consequently, the resulting negative compliance highlights the dominant actuating role of thermal expansion. Conversely, as the external mechanical loading increases ($F_{\text{ext}} = 1\,\text{N}$ and $4\,\text{N}$), large deformations occur, and the material progressively migrates toward the upper region to enhance the overall bending stiffness and mitigate geometric instabilities such as local buckling.

As observed in the optimized results, the multi-material distributions follow a physics-driven logic to accommodate the thermo-mechanical environment. As indicated in Fig. 27, the Ogden material is predominantly allocated to the cooler, high-stress regions near the clamped supports to improve structural stiffness. Meanwhile, the SVK and neo-Hookean materials are distributed in the hotter upper regions to accommodate significant thermal strains and large deformations. This spatial material tailoring demonstrates the capability of the proposed method to utilize TDMPs in 3D problems.

To gain deeper insights into the physical responses of the optimized designs, the deformed configuration along with temperature distribution for two representative cases, including Fig. 27(b) and (f), are presented in Fig. 28. It can be observed that under a low mechanical loading and high temperature, the structure utilizes thermal expansion to induce a downward displacement, yielding the aforementioned negative compliance. Conversely, under a high mechanical loading, where large deformation occurs, the material accumulates in the upper region of the design domain. This topological configuration mitigates geometric instabilities such as buckling, thereby enhancing structural stiffness against large deformations.

4.5. Example 5: 3D hand gripper

Having verified the fundamental capabilities of the proposed framework, the final numerical example considers the design of a bio-inspired three-dimensional hand gripper, as shown in Fig. 29(a). This example is intended to demonstrate the practical applicability of the proposed framework. Unlike the previous regular examples, this irregular case represents a practical scenario in which a robotic end-effector manipulates objects within a thermally active environment. This hand gripper model consists of a fixed non-design wrist interface (colored black), a design domain (colored gray), and five fingertips (colored red). The mechanical boundary conditions consist of fully clamping the wrist interface while applying vertical downward external forces F_{ext} at each fingertip. Also, a Dirichlet boundary condition ($\bar{T} = T_0$) is imposed on the fingertips, while the wrist interface acts as a heat sink, creating a thermal gradient across the entire design domain. To ensure a lightweight and porous structure capable of housing internal components such as actuators and motors, the total mass fraction constraint $f_{\max}$ is set to 0.20. Following the voxelization strategy detailed in our previous work [62], the complex geometry is discretized into a voxelized mesh of 51,530 voxel elements, as depicted in Fig. 29(b). The fingertip temperature is fixed to $T_0 = 25\,\text{K}$.

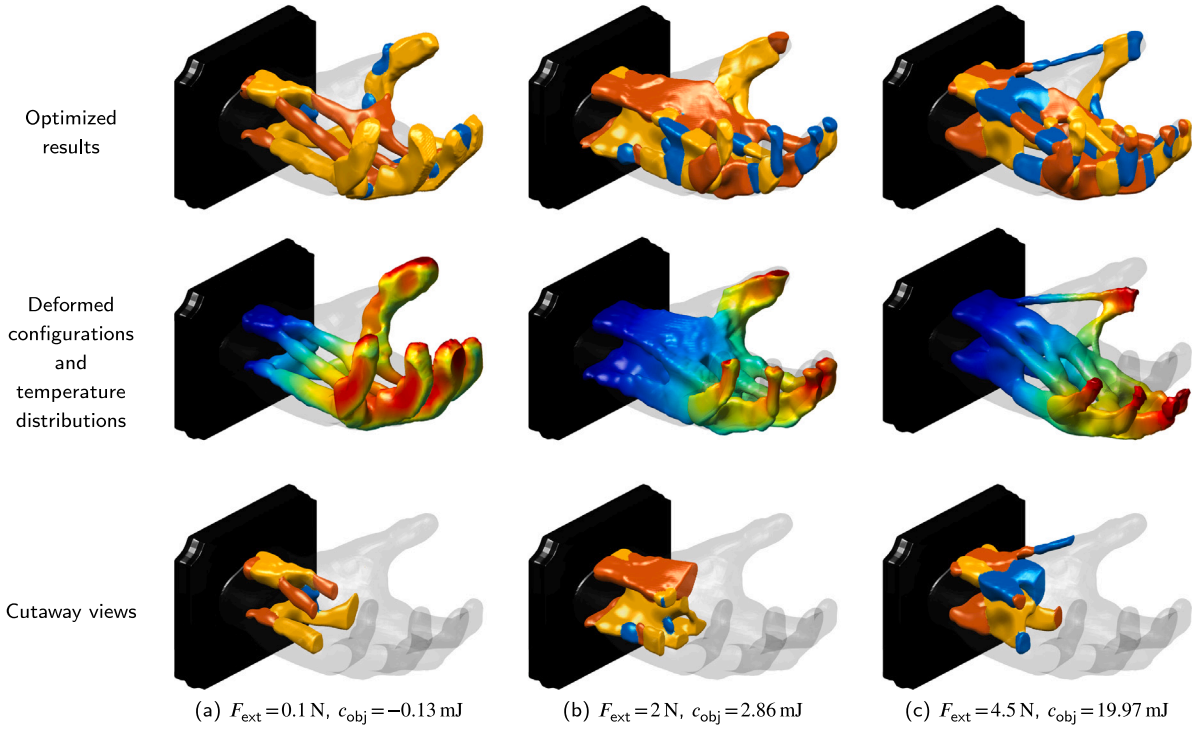


Fig. 30. Optimized results with TDMPs for Example 5. (The blue-, orange-, and yellow-colored materials correspond to the SVK, neo-Hookean, and Ogden.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

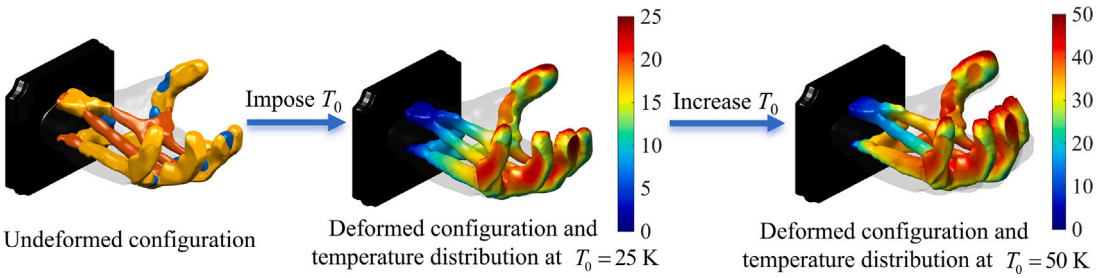


Fig. 31. Thermally induced actuation for Example 5.

The optimized configurations, along with their corresponding deformed configurations and temperature distributions, are presented in Fig. 30. Three-dimensional cutaway views are also provided to expose the internal architectures. Under varying applied loadings ($F_{ext}=0.1$ N, 2 N, and 4.5 N), different topological configurations emerge. Notably, under higher external loads ($F_{ext}=2$ N and $F_{ext}=4.5$ N), the cutaway views reveal the evolution of skeletal internal structures that maximize the moment of inertia to enhance bending resistance. More importantly, the three candidate materials are distributed to satisfy specific multi-physics demands. For instance, under the lowest loading ($F_{ext}=0.1$ N; refer to Fig. 30(a)), the Ogden material is predominantly allocated within the design domain to counteract the mechanical deformation by leveraging its superior thermal expansion coefficient. As the loading increases, the other two materials are placed along the hotter regions of the fingers; refer to Fig. 30(b) and (c). These tailored configurations significantly improve the overall structural performance.

Furthermore, the topological variations observed across different loading cases highlight the necessity of tailoring complex grippers to specific payload magnitudes and actual gripping positions, rather than relying on simplified linear elastic assumptions. Instead of acting merely as a passive load-bearing component, the optimized structure exhibits a pronounced upward deformation driven by internal thermal expansion, effectively functioning as an active, thermo-responsive gripper. This implies that by integrating thermal actuators at the fingertips, the structure can be actively triggered to generate an upward grasping force. To quantitatively demonstrate this capability, the applied temperature is increased to $T_0=50$ K, as illustrated in Fig. 31. This higher thermal gradient induces a larger upward grasping motion compared to the 25 K baseline, specifically amplifying the maximum displacement from 14.55 mm to 32.16 mm. This tunable deformation confirms that the optimized configuration successfully utilizes thermal expansion

to drive larger actuation strokes. This numerical example illustrates the application potential of the proposed method for a complex design domain under the prescribed material functions and loading conditions.

5. Conclusion

In this work, a multi-material topology optimization (MMTO) framework for structures operating under strongly coupled finite-strain thermoelastic conditions with temperature-dependent material properties (TDMPs) is established. Its two methodological contributions are (i) a kinematically scaled interpolation scheme that combines the considered hyperelastic models and improves stability in the investigated large-deformation cases and (ii) a fully coupled adjoint sensitivity analysis for strongly coupled finite-strain thermoelasticity with different objective functions. Unlike conventional approaches based on weakly coupled or linear assumptions, the proposed framework incorporates the deformation-dependent thermal conductivity tensor directly into the TO process, together with the corresponding fully coupled adjoint sensitivity analysis.

To resolve the numerical instabilities associated with large thermo-mechanical deformations, a kinematically scaled material interpolation scheme is proposed. Instead of interpolating material parameters, the proposed scheme scales the displacement gradient directly at the kinematic level. This strategy mitigates element distortion in low- and intermediate-density regions through the induced fictitious strain energy. Consequently, a unified nonlinear multi-material (UNM) interpolation scheme is established, enabling the integration of the considered hyperelastic constitutive models, including the modified Saint Venant–Kirchhoff, compressible neo-Hookean, and Ogden models, within an MMTO framework.

Under the prescribed material laws and loading conditions, the numerical examples exhibit structural bifurcation (Section 4.1.3), boundary detachment (Section 4.3.1), and load-direction-dependent layouts (Section 4.3.2). Comparisons within the specified constitutive settings indicate that accounting for deformation-dependent heat conduction and TDMPs can affect the computed responses and optimized designs. The additional exponential-integral example illustrates the applicability of the proposed framework to an alternative prescribed thermal-expansion law. Furthermore, the design of a three-dimensional thermally actuated hand gripper demonstrates the potential applicability of the proposed framework under the assumed material functions.

The high-expansion studies should be regarded as algorithmic demonstrations based on prescribed constitutive functions. Quantitative assessment of the sensitivity to alternative thermal-expansion laws and material-specific characterization for engineering applications remain necessary before broader physical conclusions can be drawn. The present formulation can handle specified isotropic thermal stretches and their temperature derivatives, while anisotropic thermal expansion would require extending the present spherical thermal deformation gradient. Thermal buckling was observed in some slender regions during optimization; however, structural stability constraints are not included in the present formulation and therefore constitute an important direction for future work. The effects of interfacial strength and the computational scalability of the proposed framework also require further investigation.

CRediT authorship contribution statement

Zhaoyou Sun: Writing – original draft, Visualization, Software, Methodology, Funding acquisition. **Jike Han:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Shinji Nishiwaki:** Writing – review & editing, Supervision, Funding acquisition. **Yangjun Luo:** Writing – review & editing, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementals for Section 2.3

The specific terms in Eqs. (19) and (20) are derived as follows:

$$\mathbb{C}_{\text{tan}} = \frac{\partial \boldsymbol{S}}{\partial \boldsymbol{E}} = \frac{\partial(\vartheta^{-2}\boldsymbol{S}^{\text{m}})}{\partial \boldsymbol{E}} = \vartheta^{-2}\frac{\partial \boldsymbol{S}^{\text{m}}}{\partial \boldsymbol{E}^{\text{m}}} : \frac{\partial \boldsymbol{E}^{\text{m}}}{\partial \boldsymbol{E}} = \vartheta^{-4}\mathbb{C}^{\text{m}}, \tag{A.1}$$

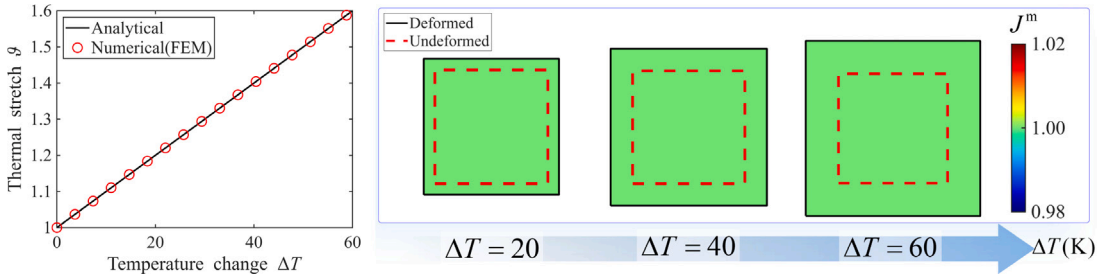
$$\frac{\partial \boldsymbol{S}}{\partial T} = \frac{\partial(\vartheta^{-2})}{\partial T}\boldsymbol{S}^{\text{m}} + \vartheta^{-2}\left(\frac{\partial \boldsymbol{S}^{\text{m}}}{\partial \boldsymbol{E}^{\text{m}}} : \frac{\partial \boldsymbol{E}^{\text{m}}}{\partial T}\right) + \vartheta^{-2}\frac{\partial \boldsymbol{S}^{\text{m}}}{\partial T}, \tag{A.2}$$

$$\frac{\partial \boldsymbol{q}_0}{\partial T} = -\frac{\partial k(T)}{\partial T}J\boldsymbol{C}^{-1}\nabla_{\boldsymbol{X}}T, \quad \frac{\partial \boldsymbol{q}_0}{\partial \nabla T} = -k(T)J\boldsymbol{C}^{-1}, \tag{A.3}$$



(a) Free thermal expansion with unconstrained boundaries (b) Constrained thermal expansion with fixed boundaries

Fig. 32. Problem setup for the single-element tests.



(a) Evolution of the thermal stretch (b) Deformed configurations and stress distributions

Fig. 33. Verification of the free thermal expansion with unconstrained boundaries.

$$\frac{\partial \boldsymbol{q}_0}{\partial \boldsymbol{F}} = -k(T)J\nabla_{\boldsymbol{X}}T\cdot\left(\boldsymbol{C}^{-1}\otimes\boldsymbol{F}^{-\mathrm{T}}-\boldsymbol{F}^{-1}\,\overline{\otimes}\,\boldsymbol{C}^{-1}-\boldsymbol{C}^{-1}\,\underline{\otimes}\,\boldsymbol{F}^{-1}\right), \tag{A.4}$$

where $\mathbb{C}^{\mathrm{m}} = \partial_{\boldsymbol{E}^{\mathrm{m}}}\boldsymbol{S}^{\mathrm{m}}$ represents the material tangent tensor with respect to the intermediate configuration, and the scalar factor ϑ^{-4} arises from the thermal deformation and represents the mapping of the tangent tensor from the intermediate configuration to the reference configuration. Notably, the third term of the right-hand side in Eq. (A.2) corresponds to the derivative of the temperature-dependent material properties (TDMPs), such as the bulk and shear moduli.

Meanwhile, the sub-matrices for Eq. (21) are derived as follows:

$$\boldsymbol{\mathcal{K}}_{\boldsymbol{uu}} = \bigcup_{e=1}^{n_{\mathrm{e}}}\left[\int_{\Omega_0^e}\left((\boldsymbol{B}^{\mathrm{m}})^{\mathrm{T}}(\vartheta^{-4}\mathbb{C}^{\mathrm{m}})\boldsymbol{B}^{\mathrm{m}}+\left(\boldsymbol{B}^{\mathrm{geo}}\right)^{\mathrm{T}}\boldsymbol{S}\boldsymbol{B}^{\mathrm{geo}}\right)\mathrm{d}V\right], \tag{A.5}$$

$$\boldsymbol{\mathcal{K}}_{\boldsymbol{u}T} = \bigcup_{e=1}^{n_{\mathrm{e}}}\left[\int_{\Omega_0^e}(\boldsymbol{B}^{\mathrm{m}})^{\mathrm{T}}\frac{\partial \boldsymbol{S}}{\partial T}\boldsymbol{N}^{\mathrm{th}}\mathrm{d}V\right], \tag{A.6}$$

$$\boldsymbol{\mathcal{K}}_{TT} = \bigcup_{e=1}^{n_{\mathrm{e}}}\left[-\int_{\Omega_0^e}\left((\boldsymbol{B}^{\mathrm{th}})^{\mathrm{T}}\frac{\partial \boldsymbol{q}_0}{\partial T}\boldsymbol{N}^{\mathrm{th}}+(\boldsymbol{B}^{\mathrm{th}})^{\mathrm{T}}\frac{\partial \boldsymbol{q}_0}{\partial \nabla T}\boldsymbol{B}^{\mathrm{th}}\right)\mathrm{d}V\right], \tag{A.7}$$

$$\boldsymbol{\mathcal{K}}_{T\boldsymbol{u}} = \bigcup_{e=1}^{n_{\mathrm{e}}}\left[-\int_{\Omega_0^e}(\boldsymbol{B}^{\mathrm{th}})^{\mathrm{T}}\frac{\partial \boldsymbol{q}_0}{\partial \boldsymbol{F}}\boldsymbol{B}^{\mathrm{geo}}\mathrm{d}V\right], \tag{A.8}$$

where $\boldsymbol{B}^{\mathrm{geo}}$ denotes the discrete geometric gradient operator.

Appendix B. Numerical verification and performance comparison of computational methods

In this appendix, the derived formulation and the associated solution schemes are verified. First, the numerical implementation is compared with analytical solutions to confirm its kinematic consistency. Subsequently, the computational efficiency of the proposed monolithic solver is assessed through comparisons with conventional sequential and staggered approaches.

B.1. Benchmark verification compared with analytical solutions

To verify the theoretical consistency and implementation correctness of the proposed method, the numerical results are compared with benchmark solutions under uniform temperature distributions [53,63]. Two single-element benchmark problems, namely free thermal expansion and fully constrained thermal expansion, are considered to compare the numerical results with the corresponding analytical solutions. The boundary conditions and problem setups for these tests are illustrated in Fig. 32.

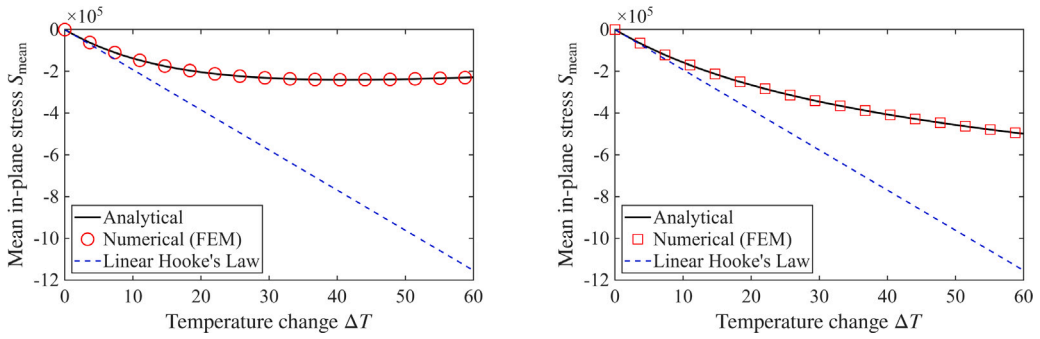


Fig. 34. Comparison of the computed mean in-plane stresses with the analytical solutions under fully constrained thermal expansion.

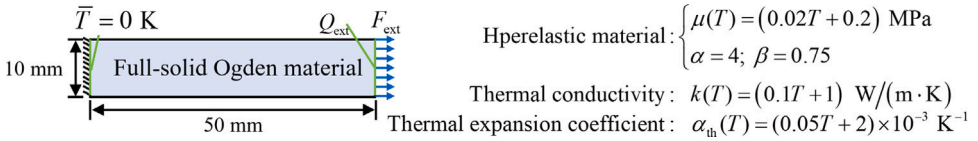


Fig. 35. Problem setup for the comparison of different computational methods.

In the first case (Fig. 32(a)), all boundaries are unconstrained under a uniform temperature change ΔT. The total deformation gradient is therefore given by $\boldsymbol{F} = \boldsymbol{F}^{\text{th}} = \vartheta \boldsymbol{I}$. According to the multiplicative decomposition, the elastic deformation gradient simplifies to $\boldsymbol{F}^{\text{m}} = \boldsymbol{I}$, resulting in a stress-free state. To demonstrate this kinematic consistency, the numerically computed thermal stretch and stress are plotted over the temperature range of 0–60 K using the modified SVK model. As depicted in Fig. 33(a), the numerical principal stretch reproduces the imposed thermal expansion, matching the definition $\vartheta = 1 + \alpha_{\text{th}} \Delta T$. Simultaneously, as shown in Fig. 33(b), the computed elastic volume ratio J^{m} remains essentially equal to 1 throughout the entire temperature range. These results verify the correctness of the multiplicative kinematics, confirming that no spurious elastic deformation is induced during free thermal expansion.

In the second case (Fig. 32(b)), all boundary displacements are fully constrained, enforcing $\boldsymbol{F} = \boldsymbol{I}$. Here, the material parameters are set to Young's modulus $E = 10^6$ Pa, Poisson's ratio $\nu = 0.3$, and a constant thermal expansion coefficient $\alpha_{\text{th}} = 0.01\,\text{K}^{-1}$. The temperature change ΔT is increased incrementally from 0 to 60 K. Under the plane strain assumption, the out-of-plane thermal expansion is restricted to zero to isolate the in-plane thermo-mechanical coupling. This results in a purely elastic compression state with $\boldsymbol{F}^{\text{m}} = \vartheta^{-1} \boldsymbol{I}$ and an elastic volumetric ratio $J^{\text{m}} = \vartheta^{-2}$. Consequently, such an isotropic deformation yields an equibiaxial in-plane stress state, where the mean in-plane stress is defined as $S_{\text{mean}} = (S_{11} + S_{22})/2$. The analytical solution of S_{mean} can thus be derived from the constitutive equations (Eqs. (22) and (25)). For the modified SVK model, the relationship is expressed as

$$S_{\text{mean}} = (\mu + \lambda)\left(\vartheta^{-4} - \vartheta^{-2}\right), \tag{B.1}$$

whereas for the compressible neo-Hookean model, replacing the bulk and shear moduli with Lamé parameters yields

$$S_{\text{mean}} = \left[\frac{\lambda}{2}\left(\vartheta^{-4} - 1\right) + \frac{\mu}{3}\left(\vartheta^{-4} - \vartheta^{4/3} + \vartheta^{-2/3} - 1\right)\right]. \tag{B.2}$$

The computed mean in-plane stress values S_{mean} are depicted in Fig. 34, in which an agreement is observed between the numerical and analytical solutions. Furthermore, to verify the degeneration to the linear regime, the responses are compared with the classical linear thermoelastic Hooke's law, which yields $S_{\text{mean,lin}} = -2(\mu + \lambda)\alpha_{\text{th}} \Delta T$ under a fully constrained plane strain condition. For a small temperature change (e.g., $\Delta T \leq 0.1$), two nonlinear solutions reduce to this linear expression. As illustrated by the dashed lines in Fig. 34, the nonlinear curves agree closely with the linear one in the small temperature change regime, exhibiting a relative error below 1%. These results further confirm the correctness of the adopted numerical implementation.

B.2. Comparison of different computational methods

As previously noted, the interaction between deformation and heat conduction induces strong coupling effects. In this appendix, the following three computational approaches are compared:

- Sequential Coupling (SC):
  A weakly coupled approach, in which the heat conduction is solved on the undeformed configuration, followed by the mechanical equilibrium analysis. The geometric effect of deformation on heat transfer is neglected.

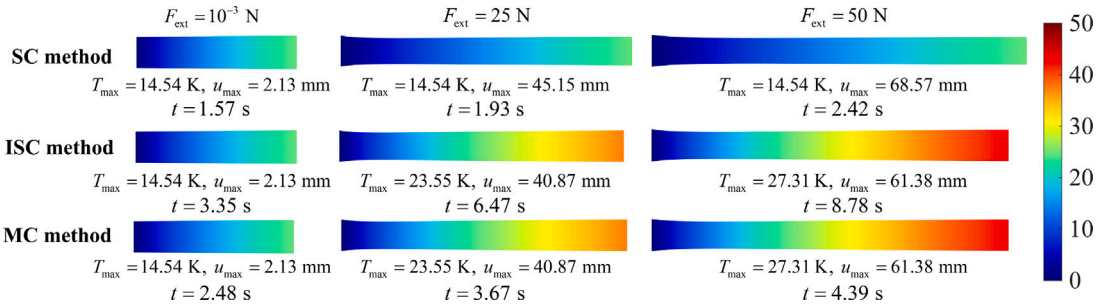


Fig. 36. Comparison between the SC, ISC, and MC methods. (The deformed shapes are colored by the temperature distribution.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- Iterative Sequential Coupling (ISC):
  A staggered approach that alternates between thermal and mechanical Newton–Raphson solvers until convergence is achieved. Although simple to implement, it requires a higher computational cost.
- Monolithic Coupling (MC):
  The employed approach in numerical examples, in which the Newton–Raphson solver includes all coupling terms ($\boldsymbol{\mathcal{K}}_{uT}$ and $\boldsymbol{\mathcal{K}}_{Tu}$) within a consistent tangent matrix, enabling good convergence and facilitating adjoint analysis.

As detailed in Fig. 35, a 2D cantilever beam (50 mm × 10 mm) is clamped at the left edge ($\bar{T} = 0, \bar{\boldsymbol{u}} = \boldsymbol{0}$). At the right edge, an external mechanical traction F_{ext} and a thermal loading $Q_{\text{ext}} = 10$ W are applied simultaneously. The material is modeled using an Ogden constitutive model with the TDMPs (hyperelastic parameters $\mu(T)$, thermal conductivity $k(T)$, and thermal expansion coefficient $\alpha_{\text{th}}(T)$). The tests are conducted under varying loading levels ($F_{\text{ext}} = 10^{-3}$ N, 25 N, 50 N), with results summarized in Fig. 36. Under small deformations ($F_{\text{ext}} = 10^{-3}$ N), all methods yield identical results ($T_{\max} = 14.54\,\text{K}, u_{\max} = 2.13$ mm), indicating that the one-way coupling assumption in the SC method is sufficient when geometric nonlinearities are negligible. Under large deformations ($F_{\text{ext}} = 25$ N, 50 N), clear discrepancies appear. The SC method does not capture the coupling effects, and tends to underestimate the temperature field, thermal expansion, and displacements (e.g., $u_{\max} = 45.15$ mm at $F_{\text{ext}} = 50$ N).

Conversely, both ISC and the MC capture the bidirectional interactions, where large deformations alter heat conduction paths, which in turn affects thermal expansion and material stiffness. While both ISC and MC produce consistent results, MC is computationally more efficient. Benefiting from exact linearization, MC achieves fast convergence, requiring only 4.39 s at $F_{\text{ext}} = 50$ N compared to 8.78 s for ISC. This demonstrates that the MC method provides a better balance between accuracy and computational cost for strongly coupled thermoelastic problems.

Appendix C. Sensitivity analysis

To employ MMA as the optimizer, the optimization model in Eq. (48) should quantify the sensitivities of the objective function and the total mass constraint with respect to the design variable vector $\boldsymbol{\eta}_i$ ($i = 1, 2, \ldots, n_{\text{m}}$). The detailed derivation is outlined below. According to the differential chain rule, the design sensitivities can be expressed as

$$\begin{cases} \dfrac{\mathrm{d}f_{\text{obj}}}{\mathrm{d}\boldsymbol{\eta}_i} = \displaystyle\sum_{e=1}^{n_{\text{e}}} \left(\sum_{j=1}^{n_{\text{m}}} \frac{\mathrm{d}f_{\text{obj}}}{\mathrm{d}w_j(\boldsymbol{X}_e)} \frac{\partial w_j(\boldsymbol{X}_e)}{\partial \rho_i(\boldsymbol{X}_e)} \frac{\mathrm{d}\rho_i(\boldsymbol{X}_e)}{\mathrm{d}\varphi_i(\boldsymbol{X}_e)} \frac{\mathrm{d}\varphi_i(\boldsymbol{X}_e)}{\mathrm{d}\boldsymbol{\eta}_i} \right), \\ \dfrac{\mathrm{d}M}{\mathrm{d}\boldsymbol{\eta}_i} = \displaystyle\sum_{e=1}^{n_{\text{e}}} \left(M_i v(\boldsymbol{X}_e) \frac{\mathrm{d}\rho_i(\boldsymbol{X}_e)}{\mathrm{d}\varphi_i(\boldsymbol{X}_e)} \frac{\mathrm{d}\varphi_i(\boldsymbol{X}_e)}{\mathrm{d}\boldsymbol{\eta}_i} \right). \end{cases} \tag{C.1}$$

where $\boldsymbol{X}_e$ denotes the physical coordinate of the eth element, n_{e} is the total number of elements in the design domain, and $v(\boldsymbol{X}_e)$ represents the corresponding elemental volume.

C.1. Sensitivity analysis of the material weights and associated functions

The derivative $\partial_{\rho_k(\boldsymbol{X}_e)} w_j(\boldsymbol{X}_e)$ can be derived based on the DMO-type interpolation in Eq. (43) as

$$\frac{\partial w_j(\boldsymbol{X}_e)}{\partial \rho_k(\boldsymbol{X}_e)} = \begin{cases} p\left(\rho_j(\boldsymbol{X}_e)\right)^{p-1} \prod_{r=1,\ r\neq j}^{n_{\text{m}}} \left(1 - \left(\rho_r(\boldsymbol{X}_e)\right)^p\right) & \text{if } k = j, \\ -p\left(\rho_j(\boldsymbol{X}_e)\right)^{p} \left(\rho_k(\boldsymbol{X}_e)\right)^{p-1} \prod_{r=1,\ r\neq j,k}^{n_{\text{m}}} \left(1 - \left(\rho_r(\boldsymbol{X}_e)\right)^p\right) & \text{if } k \neq j. \end{cases} \tag{C.2}$$

Also, the derivatives of the sigmoid-based Heaviside projection function and the EOLE-based material field function are explicitly given as

$$\begin{aligned} &\frac{\mathrm{d}\rho_i(\boldsymbol{X}_e)}{\mathrm{d}\varphi_i(\boldsymbol{X}_e)} = \beta_{\text{sig}} \rho_i(\boldsymbol{X}_e)\left(1 - \rho_i(\boldsymbol{X}_e)\right), \\ &\frac{\mathrm{d}\varphi_i(\boldsymbol{X}_e)}{\mathrm{d}\boldsymbol{\eta}_i} = \boldsymbol{\Xi}^{-1/2} \boldsymbol{\Phi}^{\mathrm{T}} \boldsymbol{m}_{\text{div}}(\boldsymbol{X}_e). \end{aligned} \tag{C.3}$$

Notably, the sensitivities of the total mass constraint with respect to the MFSE coefficients $\boldsymbol{\eta}_i$ can be calculated utilizing the above derivations.

C.2. Sensitivity analysis of the objective function

Following Eq. (C.1), evaluating the sensitivity of the objective function requires the derivative $\mathrm{d}_{w_i} f_{\mathrm{obj}}$. Recalling that the objective function implicitly depends on the state variables, $f_{\mathrm{obj}} = f_{\mathrm{obj}}(\boldsymbol{w}, \boldsymbol{U}, \boldsymbol{T})$, the augmented Lagrangian functional $\hat{f}_{\mathrm{obj}}$ is constructed as

$$\hat{f}_{\mathrm{obj}} = f_{\mathrm{obj}}(\boldsymbol{w}, \boldsymbol{U}, \boldsymbol{T}) + \lambda_{\boldsymbol{u}}^{\mathrm{T}} \boldsymbol{\mathcal{R}}^{\mathrm{m}}(\boldsymbol{w}, \boldsymbol{U}, \boldsymbol{T}) + \lambda_{\boldsymbol{T}}^{\mathrm{T}} \boldsymbol{\mathcal{R}}^{\mathrm{th}}(\boldsymbol{w}, \boldsymbol{U}, \boldsymbol{T}), \tag{C.4}$$

where $\lambda = [\lambda_{\boldsymbol{u}}, \lambda_T]^{\mathrm{T}}$ denotes the vector of adjoint variables (Lagrange multipliers) associated with the mechanical and thermal residuals. Since the equilibrium state enforces $\boldsymbol{\mathcal{R}} = [\boldsymbol{\mathcal{R}}^{\mathrm{m}}, \boldsymbol{\mathcal{R}}^{\mathrm{th}}]^{\mathrm{T}} = \mathbf{0}$, the augmented functional $\hat{f}_{\mathrm{obj}}$ is inherently identical to the original objective function f_{obj}.

By collecting the state variables into the global vector $\boldsymbol{H} = [\boldsymbol{U}, \boldsymbol{T}]^{\mathrm{T}}$ and differentiating the augmented Lagrangian functional $\hat{f}_{\mathrm{obj}}$ with respect to the design variables w_i via the chain rule, it yields

$$\frac{\mathrm{d}\hat{f}_{\mathrm{obj}}}{\mathrm{d}w_i} = \left(\frac{\partial f_{\mathrm{obj}}}{\partial w_i} + \lambda^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}}{\partial w_i} \right) + \left(\frac{\partial f_{\mathrm{obj}}}{\partial \boldsymbol{H}} + \lambda^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}}{\partial \boldsymbol{H}} \right) \frac{\mathrm{d}\boldsymbol{H}}{\mathrm{d}w_i}. \tag{C.5}$$

To circumvent the computationally prohibitive evaluation of the state sensitivities $\mathrm{d}_{w_i} \boldsymbol{H}$, the adjoint vector λ is chosen to enforce the second bracketed term to vanish:

$$\frac{\partial f_{\mathrm{obj}}}{\partial \boldsymbol{H}} + \lambda^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}}{\partial \boldsymbol{H}} = \mathbf{0} \quad \Rightarrow \quad \left(\frac{\partial \boldsymbol{\mathcal{R}}}{\partial \boldsymbol{H}} \right)^{\mathrm{T}} \lambda = - \left(\frac{\partial f_{\mathrm{obj}}}{\partial \boldsymbol{H}} \right)^{\mathrm{T}}. \tag{C.6}$$

Recognizing that $\partial_{\boldsymbol{H}} \boldsymbol{\mathcal{R}}$ is the global tangent stiffness matrix $\boldsymbol{\mathcal{K}}_{\mathrm{tan}}$, the adjoint system is governed by its transpose $\boldsymbol{\mathcal{K}}_{\mathrm{tan}}^{\mathrm{T}}$, which incorporates the transposed off-diagonal thermoelastic coupling terms as

$$\boldsymbol{\mathcal{K}}_{\mathrm{tan}}^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{\mathcal{K}}_{\boldsymbol{uu}}^{\mathrm{T}} & \boldsymbol{\mathcal{K}}_{\boldsymbol{Tu}}^{\mathrm{T}} \\ \boldsymbol{\mathcal{K}}_{\boldsymbol{uT}}^{\mathrm{T}} & \boldsymbol{\mathcal{K}}_{\boldsymbol{TT}}^{\mathrm{T}} \end{bmatrix}. \tag{C.7}$$

For the end compliance considered in this work ($c_{\mathrm{obj}} = \boldsymbol{F}_{\mathrm{ext}}^{\mathrm{T}} \boldsymbol{U}$), its partial derivatives with respect to the state variables become $(\partial_{\boldsymbol{U}} c_{\mathrm{obj}})^{\mathrm{T}} = \boldsymbol{F}_{\mathrm{ext}}$ and $(\partial_{\boldsymbol{T}} c_{\mathrm{obj}})^{\mathrm{T}} = \mathbf{0}$. Consequently, the assembled adjoint system becomes

$$\begin{bmatrix} \boldsymbol{\mathcal{K}}_{\boldsymbol{uu}}^{\mathrm{T}} & \boldsymbol{\mathcal{K}}_{\boldsymbol{Tu}}^{\mathrm{T}} \\ \boldsymbol{\mathcal{K}}_{\boldsymbol{uT}}^{\mathrm{T}} & \boldsymbol{\mathcal{K}}_{\boldsymbol{TT}}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \lambda_{\boldsymbol{u}} \\ \lambda_T \end{bmatrix} = - \begin{bmatrix} \boldsymbol{F}_{\mathrm{ext}} \\ \mathbf{0} \end{bmatrix}, \tag{C.8}$$

where the adjoint equation only needs to be solved once per optimization iteration; consequently, the additional computational effort for sensitivity analysis is negligible compared to the forward nonlinear thermoelastic analysis.

Since the external force $\boldsymbol{F}_{\mathrm{ext}}$ is design-independent, the explicit derivative of the end compliance with respect to the material weight is zero (i.e., $\partial_{w_i} c_{\mathrm{obj}} = 0$). Once the adjoint vector $\lambda = [\lambda_{\boldsymbol{u}}, \lambda_T]^{\mathrm{T}}$ is obtained, the final sensitivity of the objective function with respect to the material weight w_i is computed as

$$\frac{\mathrm{d}c_{\mathrm{obj}}}{\mathrm{d}w_i} = \lambda_{\boldsymbol{u}}^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}^{\mathrm{m}}}{\partial w_i} + \lambda_{\boldsymbol{T}}^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}^{\mathrm{th}}}{\partial w_i}, \tag{C.9}$$

where the partial derivatives of the residuals can be derived from the interpolation scheme presented in Eqs. (45) and (47). The proposed framework can be directly applied to other thermo-mechanical objective functions by replacing the relevant derivatives in the adjoint formulation. For the total stored strain energy $\Pi_{\mathrm{obj}} = \int_{\Omega_0} \psi \, \mathrm{d}V$ in Eq. (50), the partial derivative with respect to the mechanical state variable is equal to the internal mechanical force vector:

$$(\partial_{\boldsymbol{U}} \Pi_{\mathrm{obj}})^{\mathrm{T}} = \boldsymbol{F}_{\mathrm{int}}^{\mathrm{m}}, \tag{C.10}$$

which reduces to the external force vector $\boldsymbol{F}_{\mathrm{ext}}$ at equilibrium. Thus, the mechanical component of the adjoint load is identical to that for the end compliance. In contrast, because the stored strain energy depends on the temperature through both the thermal stretch ϑ and the TDMPs, its derivative with respect to the temperature field is non-zero:

$$(\partial_{\boldsymbol{T}} \Pi_{\mathrm{obj}})^{\mathrm{T}} = \int_{\Omega_0} \left(\frac{\partial \psi}{\partial \vartheta} \frac{\partial \vartheta}{\partial \boldsymbol{T}} + \frac{\partial \psi}{\partial \boldsymbol{g}} \cdot \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{T}} \right) \mathrm{d}V \neq \mathbf{0}, \tag{C.11}$$

where ϑ is the thermal stretch defined in Eq. (3) and $\boldsymbol{g}(T)$ collects the TDMP functions. Hence, the assembled adjoint system for the stored energy-type objective function yields

$$\begin{bmatrix} \boldsymbol{\mathcal{K}}_{\boldsymbol{uu}}^{\mathrm{T}} & \boldsymbol{\mathcal{K}}_{\boldsymbol{Tu}}^{\mathrm{T}} \\ \boldsymbol{\mathcal{K}}_{\boldsymbol{uT}}^{\mathrm{T}} & \boldsymbol{\mathcal{K}}_{\boldsymbol{TT}}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \lambda_{\boldsymbol{u}} \\ \lambda_T \end{bmatrix} = - \begin{bmatrix} \boldsymbol{F}_{\mathrm{int}}^{\mathrm{m}} \\ \partial_{\boldsymbol{T}} \Pi_{\mathrm{obj}} \end{bmatrix}. \tag{C.12}$$

Moreover, unlike the end compliance case, the stored energy case depends on the material weights through the interpolated strain energy density in Eq. (44), so that the derivative $\partial_{w_i} \Pi_{\mathrm{obj}}$ is non-zero. The resulting sensitivity of the stored energy-type objective function is thus

$$\frac{\mathrm{d}\Pi_{\mathrm{obj}}}{\mathrm{d}w_i} = \partial_{w_i} \Pi_{\mathrm{obj}} + \lambda_{\boldsymbol{u}}^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}^{\mathrm{m}}}{\partial w_i} + \lambda_{\boldsymbol{T}}^{\mathrm{T}} \frac{\partial \boldsymbol{\mathcal{R}}^{\mathrm{th}}}{\partial w_i}. \tag{C.13}$$

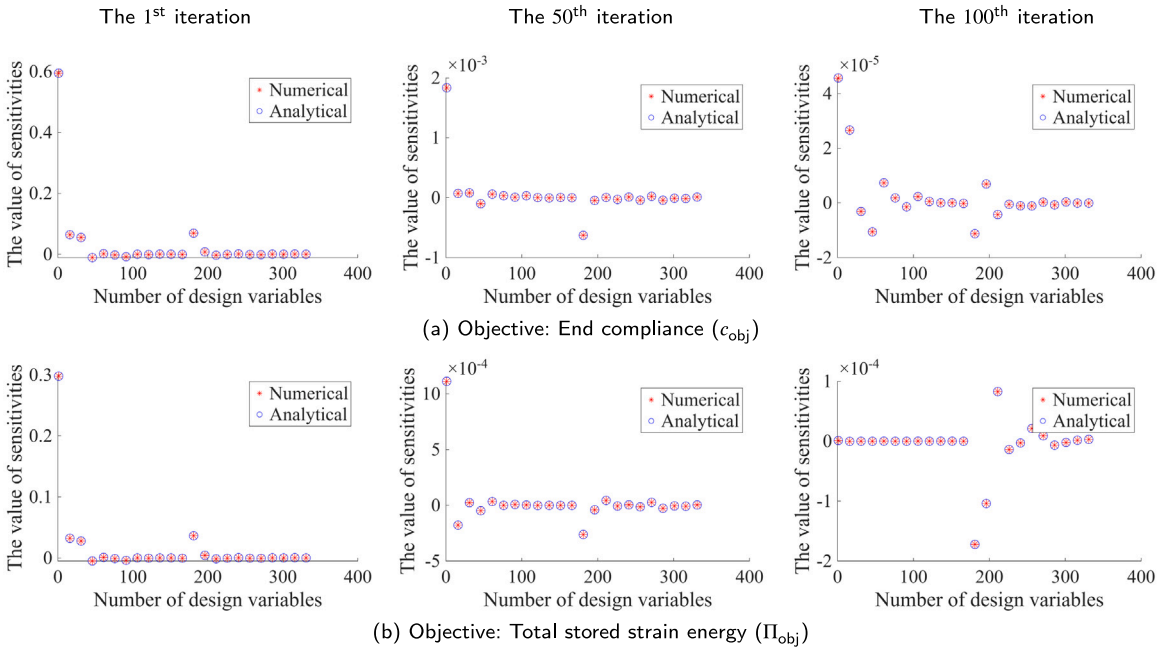


Fig. 37. Verification of the design sensitivities using the finite difference method across different iterations for both objective functions.

C.3. Verification of sensitivity

To verify the correctness of the derived analytical sensitivities and their numerical implementation, verification is presented using the finite difference method. Without loss of generality, Case 2 in Section 4.2.3 is adopted to verify the sensitivity of end compliance- and stored energy-type objective functions, while all other parameters are kept consistent. The finite difference (FD) approximation is employed as a numerical reference to evaluate the derived sensitivities, expressed as

$$\frac{\mathrm{d}f_{\mathrm{obj}}}{\mathrm{d}\eta_{i,j}} \approx \frac{\Delta f_{\mathrm{obj}}}{\Delta\eta} = \frac{f_{\mathrm{obj}}(\eta_{i,j}+\Delta\eta) - f_{\mathrm{obj}}(\eta_{i,j})}{\Delta\eta}, \tag{C.14}$$

where $\Delta\eta = 10^{-6}$ is the perturbation step size applied to the design variable to obtain the corresponding objective function variation. To verify the accuracy throughout the optimization process, three distinct iterations, i.e., the 1st, 50th, and 100th, are selected for comparison.

The verification results are illustrated in Fig. 37. To enhance visual clarity, the results are plotted for every 15th design variable. The comparison result demonstrates an agreement between the analytical sensitivities and the FD approximations. Furthermore, the near-zero design sensitivities observed at the 100th iteration indicate that the optimization process has satisfied the stationary conditions and successfully converged to a local minimum.

Appendix D. Mechanism of numerical stability induced by kinematic scaling

The kinematically scaled material interpolation scheme (introduced in Section 3.2) improves computational stability in intermediate-density regions. Assuming that the thermal deformation $\boldsymbol{F}^{\mathrm{th}}$ is neglected for simplicity, the underlying mechanism can be understood through the polar decomposition theorem. That is, any non-singular elastic deformation gradient tensor $\boldsymbol{F}^{\mathrm{m}}$ can be uniquely decomposed into an orthogonal rotation tensor $\boldsymbol{R}$ and a symmetric positive-definite stretch tensor $\boldsymbol{U}_R$ as

$$\boldsymbol{F}^{\mathrm{m}} = \boldsymbol{R}\boldsymbol{U}_R. \tag{D.1}$$

For a pure rigid body rotation, no actual stretching or compression occurs, and thus $\boldsymbol{U}_R = \boldsymbol{I}$. Consequently, $\boldsymbol{F}^{\mathrm{m}} = \boldsymbol{R}$. Substituting this expression into the standard elastic Green–Lagrange strain tensor $\boldsymbol{E}^{\mathrm{m}}$ demonstrates its physical objectivity (frame indifference), since the following condition holds:

$$\boldsymbol{E}^{\mathrm{m}} = \frac{1}{2}((\boldsymbol{F}^{\mathrm{m}})^{\mathrm{T}}\boldsymbol{F}^{\mathrm{m}} - \boldsymbol{I}) = \frac{1}{2}(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R} - \boldsymbol{I}) = \boldsymbol{0}. \tag{D.2}$$

However, this objectivity is intentionally relaxed in the kinematically scaled scheme for intermediate-density regions. Let $\alpha = \gamma\rho^p$ be the effective kinematic scaling factor corresponding to an intermediate density. The modified elastic deformation gradient is defined as $\boldsymbol{F}^{\mathrm{m}}_{\rho} = \boldsymbol{I} + \alpha\left(\boldsymbol{F}^{\mathrm{m}} - \boldsymbol{I}\right)$. Substituting the pure rotation condition, $\boldsymbol{F}^{\mathrm{m}} = \boldsymbol{R}$, into this expression gives

$$\boldsymbol{F}^{\mathrm{m}}_{\rho} = \boldsymbol{I} + \alpha(\boldsymbol{R} - \boldsymbol{I}) = (1-\alpha)\boldsymbol{I} + \alpha\boldsymbol{R}. \tag{D.3}$$

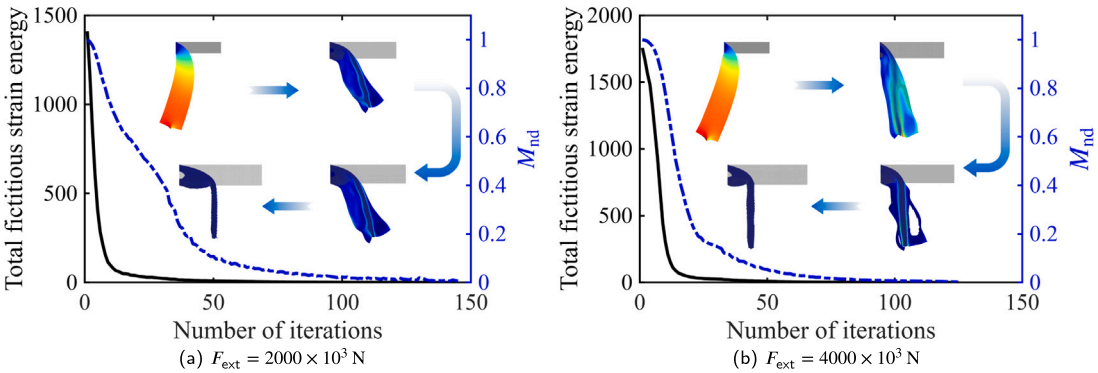


Fig. 38. Evolution histories of the total fictitious strain energy and the non-discreteness index M_{nd}, accompanied by the spatial distributions of the equivalent fictitious strain during the optimization process.

Also, the corresponding modified elastic right Cauchy–Green deformation tensor $\boldsymbol{C}_\rho^{\mathrm{m}}$ is then evaluated as

$$\begin{aligned}\boldsymbol{C}_\rho^{\mathrm{m}} &= (\boldsymbol{F}_\rho^{\mathrm{m}})^{\mathrm{T}}\boldsymbol{F}_\rho^{\mathrm{m}} = [(1-\alpha)\boldsymbol{I} + \alpha\boldsymbol{R}^{\mathrm{T}}][(1-\alpha)\boldsymbol{I} + \alpha\boldsymbol{R}] \\ &= (1-\alpha)^2\boldsymbol{I} + \alpha^2\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R} + \alpha(1-\alpha)(\boldsymbol{R} + \boldsymbol{R}^{\mathrm{T}}).\end{aligned} \tag{D.4}$$

Recalling the orthogonality condition $\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R} = \boldsymbol{I}$, Eq. (D.4) reduces to

$$\boldsymbol{C}_\rho^{\mathrm{m}} = [(1-\alpha)^2 + \alpha^2]\boldsymbol{I} + \alpha(1-\alpha)(\boldsymbol{R} + \boldsymbol{R}^{\mathrm{T}}). \tag{D.5}$$

Consequently, the modified elastic Green–Lagrange strain $\boldsymbol{E}_\rho^{\mathrm{m}}$ induced by pure rigid body rotation becomes

$$\begin{aligned}\boldsymbol{E}_\rho^{\mathrm{m}} &= \frac{1}{2}(\boldsymbol{C}_\rho^{\mathrm{m}} - \boldsymbol{I}) = \frac{1}{2}\left[-2\alpha(1-\alpha)\boldsymbol{I} + \alpha(1-\alpha)(\boldsymbol{R} + \boldsymbol{R}^{\mathrm{T}})\right] \\ &= \alpha(1-\alpha)\left[\frac{1}{2}(\boldsymbol{R} + \boldsymbol{R}^{\mathrm{T}}) - \boldsymbol{I}\right].\end{aligned} \tag{D.6}$$

This equation demonstrates that, for any non-trivial rigid body rotation ($\boldsymbol{R} \neq \boldsymbol{I}$) and intermediate scaling factor α, the modified elastic strain becomes non-zero ($\boldsymbol{E}_\rho^{\mathrm{e}} \neq \boldsymbol{0}$). In other words, the kinematic scaling transforms rigid body rotations into fictitious strains. During the Newton–Raphson iterations, the resulting fictitious strain energy introduces a positive-definite artificial tangent stiffness that suppresses excessive element rotations and prevents mesh distortion. The magnitude of this fictitious strain is governed by the parabolic factor $\alpha(1-\alpha)$. Since this factor vanishes at the two limits, $\alpha = 0$ and $\alpha = 1$, the non-physical effects disappear once the topology converges to a binary design. Therefore, the proposed formulation enhances numerical stability in intermediate-density regions without compromising the physical consistency of the final optimized structure.

To quantitatively verify the self-mitigating nature of the fictitious strain energy, the evolution of the total fictitious strain energy is compared with that of the structural discreteness throughout the optimization process. The structural discreteness of the topological configuration is evaluated using the non-discreteness index (M_{nd}) proposed by Sigmund [64], which is defined as

$$M_{\mathrm{nd}} = \frac{1}{n_{\mathrm{e}}}\sum_{e=1}^{n_{\mathrm{e}}} 4\rho_e(1-\rho_e), \tag{D.7}$$

where ρ_e denotes the physical density of the eth element, and n_{e} denotes the total number of elements. A value of $M_{\mathrm{nd}} = 0$ indicates a fully binary design consisting only of solid and void regions, whereas $M_{\mathrm{nd}} = 1$ corresponds to a uniform distribution of intermediate density with $\rho_e = 0.5$. Meanwhile, to quantify the magnitude of the local artificial penalty induced by the kinematic scaling scheme, the Frobenius norm of $\boldsymbol{E}_\rho^{\mathrm{e}}$ in Eq. (D.6) is evaluated. For a two-dimensional case, the rotation matrix $\boldsymbol{R}$ is fully characterized by a single rotation angle θ. Accordingly, the Frobenius norm can be expressed analytically as

$$\|\boldsymbol{E}_\rho^{\mathrm{m}}\|_F = \sqrt{\mathrm{tr}((\boldsymbol{E}_\rho^{\mathrm{m}})^{\mathrm{T}}\boldsymbol{E}_\rho^{\mathrm{m}})} = \sqrt{2}\alpha(1-\alpha)(1-\cos\theta). \tag{D.8}$$

This expression shows that the local fictitious strain is governed by both the density-dependent parabolic factor $\alpha(1-\alpha)$ and the rotation-dependent kinematic factor $(1-\cos\theta)$.

Using the numerical results shown in Fig. 6, the evolution histories of the total fictitious strain energy (evaluated as the sum of the squared Frobenius norms) and the M_{nd} index are plotted in Fig. 38. As can be seen, the total fictitious strain energy initially takes a relatively large value owing to the presence of intermediate densities in the early stages of optimization. However, as the topological layout binarizes and M_{nd} approaches zero, the fictitious strain energy decreases continuously throughout the optimization process. Furthermore, the inset plots of the equivalent fictitious strain show that the introduced strain energy is concentrated in highly distorted intermediate-density regions during the early iterations and gradually vanishes in the final optimized design. These observations demonstrate that the proposed interpolation scheme enhances the numerical stability of the optimization process without compromising the physical consistency of the final optimized structure.

Data availability

Data will be made available on request.

References

[1] O. Uyanna, H. Najafi, A novel solution for inverse heat conduction problem in one-dimensional medium with moving boundary and temperature-dependent material properties, Int. J. Heat Mass Transfer 182 (2022) 122023, http://dx.doi.org/10.1016/j.ijheatmasstransfer.2021.122023.

[2] F. Retailleau, V. Allheily, L. Merlat, J.-F. Henry, J. Randrianalisoa, Temperature dependent radiative properties of semi-transparent fiberglass-epoxy composite materials from 20 °C to 200 °C, Int. J. Heat Mass Transfer 184 (2022) 122319, http://dx.doi.org/10.1016/j.ijheatmasstransfer.2021.122319.

[3] D. He, D. Huang, L. Wu, Y. Xu, Investigation on thermal failure of functionally graded materials using fully coupled thermo-mechanical peridynamics, Compos. Struct. 305 (2023) 116454, http://dx.doi.org/10.1016/j.compstruct.2022.116454.

[4] M. Baghani, R. Naghdabadi, J. Arghavani, S. Sohrabpour, A thermodynamically-consistent 3D constitutive model for shape memory polymers, Int. J. Plast. 35 (2012) 13–30, http://dx.doi.org/10.1016/j.ijplas.2012.01.007.

[5] Y. Cui, Z. Yu, P. Lan, A novel method of thermo-mechanical coupled analysis based on the unified description, Mech. Mach. Theory 134 (2019) 376–392, http://dx.doi.org/10.1016/j.mechmachtheory.2019.01.001.

[6] J. Carter, J. Booker, Finite element analysis of coupled thermoelasticity, Comput. Struct. 31 (1989) 73–80, http://dx.doi.org/10.1016/0045-7949(89)90169-7.

[7] O. Sigmund, K. Maute, Topology optimization approaches, Struct. Multidiscip. Optim. 48 (2013) 1031–1055, http://dx.doi.org/10.1007/s00158-013-0978-6.

[8] J. Zhu, H. Zhou, C. Wang, L. Zhou, S. Yuan, W. Zhang, A review of topology optimization for additive manufacturing: Status and challenges, Chin. J. Aeronaut. 34 (2021) 91–110, http://dx.doi.org/10.1016/j.cja.2020.09.020.

[9] L. Meng, W. Zhang, D. Quan, G. Shi, L. Tang, Y. Hou, P. Breitkopf, J. Zhu, T. Gao, From topology optimization design to additive manufacturing: Today's success and tomorrow's roadmap, Arch. Comput. Methods Eng. 27 (2020) 805–830, http://dx.doi.org/10.1007/s11831-019-09331-1.

[10] S. Xue, X. Ma, D. Liu, Z. Huo, P. Hao, B. Wang, Thermoelastic topology optimization for stiffened thin-walled structures under design-dependent thermal loading problems, Comput. Methods Appl. Mech. Engrg. 432 (2024) 117344, http://dx.doi.org/10.1016/j.cma.2024.117344.

[11] S. Liu, X. Hong, K. Fu, Z. Zhang, Y. Li, Topology optimization of 3D-printed continuous fiber-reinforced composites considering buckling constraint, Eng. Struct. 353 (2026) 122309, http://dx.doi.org/10.1016/j.engstruct.2026.122309.

[12] X. Gao, Y. Li, H. Ma, G. Chen, Improving the overall performance of continuum structures: A topology optimization model considering stiffness, strength and stability, Comput. Methods Appl. Mech. Engrg. 359 (2020) 112660, http://dx.doi.org/10.1016/j.cma.2019.112660.

[13] S. Zhou, R. Tao, Q. Sun, CFGLSs: Conformal filling gradient lattice structures designed by multiscale isogeometric topology optimization for 3d swept volume, Comput. Methods Appl. Mech. Engrg. 442 (2025) 118023, http://dx.doi.org/10.1016/j.cma.2025.118023.

[14] T. Gao, P. Xu, W. Zhang, Topology optimization of thermo-elastic structures with multiple materials under mass constraint, Comput. Struct. 173 (2016) 150–160, http://dx.doi.org/10.1016/j.compstruc.2016.06.002.

[15] C. Wu, J. Fang, Q. Li, Multi-material topology optimization for thermal buckling criteria, Comput. Methods Appl. Mech. Engrg. 346 (2019) 1136–1155, http://dx.doi.org/10.1016/j.cma.2018.08.015.

[16] L.M. Anaya-Jaimes, W.M. Vicente, R. Pavanello, Metamaterials design with a desired thermal expansion using a multi-material BESO method, Struct. Multidiscip. Optim. 65 (2022) 355, http://dx.doi.org/10.1007/s00158-022-03448-x.

[17] L. Tang, T. Gao, L. Song, L. Meng, C. Zhang, W. Zhang, Topology optimization of nonlinear heat conduction problems involving large temperature gradient, Comput. Methods Appl. Mech. Engrg. 357 (2019) 112600, http://dx.doi.org/10.1016/j.cma.2019.112600.

[18] L. Tang, T. Gao, W. Zhang, J. Zeng, W. Zhang, Topology optimization of thermo-elastic structures with temperature-dependent material properties under large temperature gradient, Internat. J. Numer. Methods Engrg. 124 (19) (2023) 4224–4253, http://dx.doi.org/10.1002/nme.7304.

[19] J. Zheng, X. Rong, C. Jiang, Thermoelastic topology optimization for structures with temperature-dependent material properties, Sci. China Technol. Sci. 66 (2023) 3488–3503, http://dx.doi.org/10.1007/s11431-023-2458-6.

[20] Z. Sun, J. Han, S. Nishiwaki, Y. Luo, Multi-material topology optimization for nonlinear heat conduction with temperature-dependent properties, Internat. J. Engrg. Sci. 226 (2026) 104585, http://dx.doi.org/10.1016/j.ijengsci.2026.104585.

[21] E. Sanders, M. Aguiló, G. Paulino, Multi-material continuum topology optimization with arbitrary volume and mass constraints, Comput. Methods Appl. Mech. Engrg. 340 (2018) 798–823, http://dx.doi.org/10.1016/j.cma.2018.01.032.

[22] A. Bandyopadhyay, B. Heer, Additive manufacturing of multi-material structures, Mater. Sci. Eng.: R: Rep. 129 (2018) 1–16, http://dx.doi.org/10.1016/j.mser.2018.04.001.

[23] J. Han, K. Furuta, T. Kondoh, K. Izui, S. Nishiwaki, K. Terada, Topology optimization with a finite strain nonlocal damage model using the continuous adjoint method, Comput. Methods Appl. Mech. Engrg. 432 (2024) 117333, http://dx.doi.org/10.1016/j.cma.2024.117333.

[24] S. Tisato, G. Vera, Q. Song, N. Nekoonam, D. Helmer, Additive manufacturing of multi-material and hollow structures by embedded extrusion-volumetric printing, Nat. Commun. 16 (2025) 6730, http://dx.doi.org/10.1038/s41467-025-62057-6.

[25] J. Han, Y. Yamakawa, K. Izui, S. Nishiwaki, K. Terada, Multi-material topology optimization based on finite strain subloading surface nonlocal elastoplasticity, Comput. Methods Appl. Mech. Engrg. 442 (2025) 118038, http://dx.doi.org/10.1016/j.cma.2025.118038.

[26] Z. Du, Y. Guo, C. Liu, W. Zhang, R. Xue, Y. Guo, S. Tang, X. Guo, Structural topology optimization of three-dimensional multi-material composite structures with finite deformation, Compos. Struct. 328 (2024) 117692, http://dx.doi.org/10.1016/j.compstruct.2023.117692.

[27] T.T. Banh, D. Lee, Efficient topology optimization for geometrically nonlinear multi-material systems under design-dependent pressure loading, Eng. Comput. 41 (2) (2025) 1155–1189, http://dx.doi.org/10.1007/s00366-024-02083-y.

[28] B. Goh, Z. Du, H. Chung, Explicit topology optimization of large deforming hyperelastic composite structures, Struct. Multidiscip. Optim. 67 (2024) 154, http://dx.doi.org/10.1007/s00158-024-03860-5.

[29] Y. Jia, Z. Du, Y. Guo, X. Wang, X. Liu, X. Guo, Novel optimization-based method for three-dimensional stms considering thermoelastic effects, J. Struct. Eng. 152 (3) (2026) 04026005, http://dx.doi.org/10.1061/JSENDH.STENG-15631.

[30] R.W. Ogden, Large deformation isotropic elasticity – on the correlation of theory and experiment for incompressible rubberlike solids, Proc. R. Soc. A 326 (1567) (1972) 565–584, http://dx.doi.org/10.1098/rspa.1972.0026.

[31] T.J. Pence, K. Gou, On compressible versions of the incompressible neo-hookean material, Math. Mech. Solids 20 (2) (2015) 157–182, http://dx.doi.org/10.1177/1081286514544258.

[32] H. Dal, K. Açıkgöz, Y. Badienia, On the performance of isotropic hyperelastic constitutive models for rubber-like materials: A state of the art review, Appl. Mech. Rev. 73 (2) (2021) 020802, http://dx.doi.org/10.1115/1.4050978.

[33] O. Sigmund, S. Torquato, Design of materials with extreme thermal expansion using a three-phase topology optimization method, J. Mech. Phys. Solids 45 (1997) 1037–1067, http://dx.doi.org/10.1016/S0022-5096(96)00114-7.

[34] W. Zuo, K. Saitou, Multi-material topology optimization using ordered simp interpolation, Struct. Multidiscip. Optim. 55 (2017) 477–491, http://dx.doi.org/10.1007/s00158-016-1513-3.

[35] Z. Sun, W. Liu, P. Liu, Y. Luo, A new multiscale concurrent topology optimization method based on the ordered interpolation and mfse model, Comput. Methods Appl. Mech. Engrg. 430 (2024) 117230, http://dx.doi.org/10.1016/j.cma.2024.117230.
[36] J. Han, K. Furuta, T. Kondoh, K. Izui, S. Nishiwaki, K. Terada, Topology optimization for nonlocal elastoplasticity at finite strain, Comput. Methods Appl. Mech. Engrg. 435 (2025) 117678, http://dx.doi.org/10.1016/j.cma.2024.117678.
[37] J. Han, K. Izui, S. Nishiwaki, Extended discrete material optimization: A generalized framework for multi-material topology optimization of nonlinear material constitutive models, Comput. Methods Appl. Mech. Engrg. 451 (2026) 118682, http://dx.doi.org/10.1016/j.cma.2025.118682.
[38] M. Nguyen, V. Hoang, D. Lee, Stress-based multi-material topology optimization framework for self-weight and thermal loads involving dynamic constraints, Struct. Multidiscip. Optim. 69 (2026) 37, http://dx.doi.org/10.1007/s00158-026-04247-4.
[39] B. Goh, H.A. Kim, H. Chung, Investigation of thermoelastic compliances considering finite strain, Comput. Methods Appl. Mech. Engrg. 410 (2023) 116011, http://dx.doi.org/10.1016/j.cma.2023.116011.
[40] G. Granlund, M. Wallin, O. Günther-Hanssen, D. Tortorelli, S. Watts, Topology optimization of compliant mechanisms under transient thermal conditions, Comput. Methods Appl. Mech. Engrg. 418 (2024) 116478, http://dx.doi.org/10.1016/j.cma.2023.116478.
[41] T. Bruns, D. Tortorelli, An element removal and reintroduction strategy for the topology optimization of structures and compliant mechanisms, Internat. J. Numer. Methods Engrg. 57 (2003) 1413–1430, http://dx.doi.org/10.1002/nme.783.
[42] Y. Luo, M. Wang, Z. Kang, Topology optimization of geometrically nonlinear structures based on an additive hyperelasticity technique, Comput. Methods Appl. Mech. Engrg. 286 (2015) 422–441, http://dx.doi.org/10.1016/j.cma.2014.12.023.
[43] Q. Chen, X. Zhang, B. Zhu, A 213-line topology optimization code for geometrically nonlinear structures, Struct. Multidiscip. Optim. 59 (2019) 1863–1879, http://dx.doi.org/10.1007/s00158-018-2138-5.
[44] L. Song, F. Wang, T. Gao, W. Zhang, A neo-yeoh hyperelastic interpolation model for stable multi-material topology optimization under geometric nonlinearity, Comput. Methods Appl. Mech. Engrg. 448 (2026) 118467, http://dx.doi.org/10.1016/j.cma.2025.118467.
[45] F. Wang, B. Lazarov, O. Sigmund, J. Jensen, Interpolation scheme for fictitious domain techniques and topology optimization of finite strain elastic problems, Comput. Methods Appl. Mech. Engrg. 276 (2014) 453–472, http://dx.doi.org/10.1016/j.cma.2014.03.021.
[46] M. Wallin, A. Dalklint, D. Tortorelli, Topology optimization of bistable elastic structures - an application to logic gates, Comput. Methods Appl. Mech. Engrg. 383 (2021) 113912, http://dx.doi.org/10.1016/j.cma.2021.113912.
[47] P. Dunning, Stability constraints for geometrically nonlinear topology optimization, Struct. Multidiscip. Optim. 66 (2023) 253, http://dx.doi.org/10.1007/s00158-023-03712-8.
[48] J. Zhu, Y. Li, F. Wang, W. Zhang, Shape preserving design of thermo-elastic structures considering geometrical nonlinearity, Struct. Multidiscip. Optim. 61 (2020) 1787–1804, http://dx.doi.org/10.1007/s00158-020-02532-4.
[49] H. Chung, O. Amir, H. Kim, Level-set topology optimization considering nonlinear thermoelasticity, Comput. Methods Appl. Mech. Engrg. 361 (2020) 112735, http://dx.doi.org/10.1016/j.cma.2019.112735.
[50] B. Yuan, H. Ye, J. Li, N. Wei, Y. Sui, Topology optimization of geometrically nonlinear structures under thermal–mechanical coupling, Acta Mech. Solida Sin. 36 (2023) 22–33, http://dx.doi.org/10.1007/s10338-022-00342-3.
[51] J. Stegmann, E. Lund, Discrete material optimization of general composite shell structures, Internat. J. Numer. Methods Engrg. 62 (2005) 2009–2027, http://dx.doi.org/10.1002/nme.1259.
[52] Y. Luo, J. Bao, A material-field series-expansion method for topology optimization of continuum structures, Comput. Struct. 225 (2019) 106122, http://dx.doi.org/10.1016/j.compstruc.2019.106122.
[53] G.A. Holzapfel, Nonlinear Solid Mechanics: A Continuum Approach for Engineering, John Wiley & Sons, Chichester, 2000.
[54] J.D. James, J.A. Spittle, S.G.R. Brown, R.W. Evans, A review of measurement techniques for the thermal expansion coefficient of metals and alloys at elevated temperatures, Meas. Sci. Technol. 12 (3) (2001) R1–R15, http://dx.doi.org/10.1088/0957-0233/12/3/201.
[55] X. Zhang, H. Chi, G. Paulino, Adaptive multi-material topology optimization with hyperelastic materials under large deformations: A virtual element approach, Comput. Methods Appl. Mech. Engrg. 370 (2020) 112976, http://dx.doi.org/10.1016/j.cma.2020.112976.
[56] C. Li, A.D. Kiureghian, Optimal discretization of random fields, J. Eng. Mech. 119 (6) (1993) 1136–1154, http://dx.doi.org/10.1061/(ASCE)0733-9399(1993)119:6(1136).
[57] J. Han, Z. Sun, Y. Luo, S. Nishiwaki, Material interpolation in finite strain topology optimization: A comprehensive assessment and a generalized energy-strain-combined interpolation scheme, Struct. Multidiscip. Optim. 69 (2026) 122, http://dx.doi.org/10.1007/s00158-026-04332-8.
[58] O. Giraldo-Londoño, G. Paulino, Polystress: A matlab implementation for local stress-constrained topology optimization using the augmented Lagrangian method, Struct. Multidiscip. Optim. 63 (2021) 2065–2097, http://dx.doi.org/10.1007/s00158-020-02760-8.
[59] O. Sigmund, Design of multiphysics actuators using topology optimization—Part I: One-material structures, Comput. Methods Appl. Mech. Engrg. 190 (49–50) (2001) 6577–6604, http://dx.doi.org/10.1016/S0045-7825(01)00251-1.
[60] O. Sigmund, Design of multiphysics actuators using topology optimization—Part II: Two-material structures, Comput. Methods Appl. Mech. Engrg. 190 (49–50) (2001) 6605–6627, http://dx.doi.org/10.1016/S0045-7825(01)00252-3.
[61] Z. Liu, J.G. Korvink, Using artificial reaction force to design compliant mechanism with multiple equality displacement constraints, Finite Elem. Anal. Des. 45 (2009) 555–568, http://dx.doi.org/10.1016/j.finel.2009.03.005.
[62] Z. Sun, Q. Duan, Y. Luo, A movable partition region strategy for multiscale concurrent topology optimization with multiple connectable microstructures, Comput. Methods Appl. Mech. Engrg. 445 (2025) 118202, http://dx.doi.org/10.1016/j.cma.2025.118202.
[63] C. Miehe, Entropic thermoelasticity at finite strains, aspects of the formulation and numerical implementation, Comput. Methods Appl. Mech. Engrg. 120 (3–4) (1995) 243–269, http://dx.doi.org/10.1016/0045-7825(94)00057-T.
[64] O. Sigmund, Morphology-based black and white filters for topology optimization, Struct. Multidiscip. Optim. 33 (2007) 401–424, http://dx.doi.org/10.1007/s00158-006-0087-x.